Quantum cellular automata on lattice phase space

A quantum cellular automaton is a discrete-time dynamics of an infinite lattice of quantum systems that commutes with the lattice translations and moves observables by at most a fixed distance per step [7]. These notes are the mathematical account of the Lean library QCALean. They begin with the Gaussian case, following Chapter 4 of [1], in which each step acts on the Weyl operators through a linear map of the classical phase space.

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About these notes

Coined terms. These terms are coined for these notes and defined where they first appear: the lifted form (Definition 1.6); the twisted criterion (Theorem 1.7); cell data (Definition 3.22); the regime (Definition 4.8); the limit symbol and the limit cells (Definition 5.4); lattice channels (Definition 7.1); pull-back cells (Definition 7.5). The section on the notation of Krüger's thesis, after the Notation, matches the thesis's symbols with those used here.

Reading conventions. C*-algebras are unital and complex. Inner products are linear in the second argument. The CCR algebra and its relabellings, states on it, the cell index and the symplectic matrix of a cell are those of the library BosonicLean; nets of algebras are those of the library LinearAQFT.

A statement marked Lean ✓ has a checked proof in QCALean or in the library BosonicLean on which it builds, whose correspondence with the statement here has been independently reviewed; the declaration index at the end names the checked declarations. Every Example carries exactly one status: Lean ✓ as above, script ✓ when a check script verifies it exhaustively, or paper when it is checked on paper only. Definitions, Remarks and paragraphs of motivation carry no marks. A sentence tagged formalization records a choice made for the formalization that a proof on paper would not need.

Notation

Each symbol has one meaning throughout, shared with LinearAQFT and BosonicLean where they have one, with one exception: N denotes a neighbourhood of a rule, and $N(k)$, always with its argument, the matrix of the example system below.

Letters that serve only within one proof, example or remark, such as scalars, indices and auxiliary matrices, are introduced where they appear and may differ from the table. Second meanings over a longer stretch are these: μ is a complex combination of vectors in the discussion after Definition 3.11 and in Subsection 3.5; S is a symplectic matrix in Chapters 1–4 and a symmetric invariant matrix in the part of Chapter 4 on invariant forms; K is a second rule and the twisted kernel in Chapters 2 and 3; T_O is the matrix of Corollary 3.32; and in Chapter 6 the letters V, σ and L denote a general real vector space, an alternating form on it and a linear map of it, of which the lattice data are one instance.

G, x	the lattice (an abelian group; $\mathbb{Z}$ for chains) and a point of it
n	the number of modes per cell
Ω	the symplectic matrix of a cell, $\left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, positions first
V, u, w	finitely supported phase space $G \to \mathbb{R}^{2n}$, its vectors
σ	the symplectic form $\sigma(u,w) = \sum_x u(x)^{\mathsf{T}} \Omega\, w(x)$
$W(u)$, $\mathrm{CCR}(V,\sigma)$	a Weyl operator, the C*-algebra they generate
O, N, $\mathcal{A}(O)$	a region, a neighbourhood (a finite set of sites), the local algebra of O
κ_{a*}, α_a	translation by a on V, $(\kappa_{a*}u)(x) = u(x-a)$, and on $\mathrm{CCR}(V,\sigma)$
L, α_L	a rule (translation-invariant linear map of V), its automorphism when symplectic
ℓ_m, $\hat{L}(z)$	the coefficients of a rule, $(Lu)(x) = \sum_m \ell_m u(x-m)$; its symbol $\sum_m \ell_m z^m$
λ	the coupling of a nearest-neighbour rule
F	the leapfrog step of the lattice Klein–Gordon field
ω, γ, d	a state, its covariance form on V, its displacement
χ_ω	the characteristic function $\chi_\omega(u) = \omega(W(u))$
$\frac{1}{4}$	the normalization $\chi_\omega(u) = \exp(\mathrm{i}d(u) - \frac{1}{4}\gamma(u,u))$ of a Gaussian state; the vacuum has $\gamma = 1$
$e_{x,i}$, $\mathrm{loc}(O)$	the site vector (i-th unit vector at x), the vectors supported in O
n_O, Ω_M	the number $n\lvert O\rvert$ of modes of a region, the symplectic matrix of M modes, positions first
$R_O(\beta)$	the matrix of a bilinear form β on the site vectors of O, positions first
γ_B	a covariance matrix of finitely many modes, positions first, $\gamma_B = 2\times$(symmetrized second moments)
$\mathrm{cov}_O(\gamma)$	the covariance matrix $\Omega_{n_O}^{\mathsf{T}} R_O(\gamma)\, \Omega_{n_O}$ of γ on O; a matrix γ_B is admissible if $\gamma_B - \mathrm{i}\Omega_{n_O} \geq 0$
γ_x, $\hat{\gamma}(k)$	the cells of a translation-invariant covariance form, its symbol $\sum_x \gamma_x e^{-\mathrm{i}kx}$
$\tau(k)$, $\theta(k)$	half the trace of the example system's symbol, $\cos\varphi + \lambda\cos k \sin\varphi$, and the rotation angle $\arccos\tau(k)$
$N(k)$, P_k	the real matrix $(\cos\theta(k)1 - \hat{L}(k))/\sin\theta(k)$ with $N(k)^2 = -1$, and the projector $\frac{1}{2}(1 + \mathrm{i}N(k))$
$\hat{\varepsilon}(k)$, $\eta(k)$	the invariant form of the example system, and the factor of a stationary state, $\hat{\gamma} = \eta\hat{\varepsilon}$
$C(k)$, $\gamma_{\infty,x}$, γ_∞	the limit symbol $\frac{1}{2}(\hat{\gamma} + N^{\mathsf{T}}\hat{\gamma}N)$, its Fourier coefficients (the limit cells), and their covariance form
f_{crit}	the critical coupling $(1 - \lvert\cos\varphi\rvert)/\lvert\sin\varphi\rvert$
T	the map of observables of an automaton or of a Gaussian channel
ν	the noise form of a Gaussian channel
μ, Δ_T	the displacement of a Gaussian channel T, its dissipation form $\nu + \mathrm{i}(\sigma - \sigma(L\cdot, L\cdot))$
ζ	the amplitude transmission $0 < \zeta < 1$ of the attenuator, the running example of Chapter 6, and of the lossy chain of Chapter 7
(X, Y)	the channel matrices of a Gaussian channel on finitely many modes, $\gamma_B \mapsto X\gamma_B X^{\mathsf{T}} + Y$
S_{11}, S_{21}	the system and ancilla parts of a dilation S, $S(u,0) = (S_{11}u, S_{21}u)$
N'	the separation set in the factorization condition of a linear map
N_L	the range $\{m : \ell_m \neq 0\}$ of a rule L
$(L_*\nu)_j$, $(L_*\sigma)_j$	the pull-back cells (also $L_*\gamma$ in Chapters 4 and 5) $\sum_{m,m'} \ell_m^{\mathsf{T}} \nu_{j+m-m'} \ell_{m'}$ of cells (ν_x), and those of the on-site cells $\Omega\delta_{x,0}$ of σ
n'	the number of ancilla modes per site of a dilation

Notation of Krüger's thesis

Formulas of [1] are quoted in its own letters; elsewhere these notes use the letters above.

Object	[1]	here
phase space, vectors	Ξ; ξ, η	V; u, w
one-mode symplectic matrix	σ_0, σ_s	Ω
modes per cell	f	n
region	Λ	O
translation	τ_Δ, $(\tau_\Delta \xi)_x = \xi_{x+\Delta}$	κ_{a*} with $a = -\Delta$
rule, global rule	Γ, T	L, α_L
rotation angle, noise factor	$\alpha(k)$, $g(k)$	$\theta(k)$, $\eta(k)$
coefficients, symbol	Γ_x, $\hat{\Gamma}(k)$	ℓ_x (same index, (4.8)), $\hat{L}(z)$ with $z = e^{-\mathrm{i}k}$
coupling	f	λ
noise form	g	ν

Forms and matrices: a reading guide

Most texts on Gaussian states and channels work with matrices: covariance matrices of finitely many modes and channel matrices (X, Y). The thesis [1], like these notes, works on the lattice with forms on labels, which it writes through the matrices $\Gamma(x)$ and $g(x)$, the coefficients and cells below. These notes describe all objects by bilinear forms on the phase space V of the whole chain and by their coefficients, the cells (not to be confused with the cell of a site, its $2n$ degrees of freedom). The reason is that a Weyl operator $W(u)$ is labelled by a vector u of V, so that a Gaussian state or a channel is most directly a prescription on labels, with no region and no basis chosen. The two descriptions are equivalent, and the dictionary below is all that is needed to pass from one to the other.

Vectors and forms. A vector $u \in V$ assigns to every site x a column $u(x) \in \mathbb{R}^{2n}$, positions first, and only finitely many are nonzero. A bilinear form β on V takes two such vectors to a number. If β is translation invariant, it is determined by the matrices β_x, its *cells*, which record the correlation between the site x and the site 0:

$$\beta(u, w) = \sum_{x,y} u(x)^{\mathsf{T}} \beta_{x-y}\, w(y), \qquad (\beta_x)_{ij} = \beta(e_{x,i}, e_{0,j}).$$

Restricted to finitely many sites, β is the block matrix $[\beta_{x-y}]$, constant along its diagonals. The noise matrices $g(x)$ of the thesis are cells.

Covariance matrices. A Gaussian state is given by a covariance *form* γ, through its characteristic function $\omega(W(u)) = \exp(\mathrm{i}d(u) - \frac{1}{4}\gamma(u, u))$. Its covariance *matrix* on a finite region, the familiar γ_B with entries twice the symmetrized second moments of the centred coordinates (the vacuum has $\gamma_B = 1$), is

$$\gamma_B = \Omega^{\mathsf{T}} \gamma\, \Omega \qquad \text{(Definition 3.15)},$$

where γ stands for the matrix of the form on the region, positions first. The congruence by Ω exchanges positions and momenta, because the position slot of a label multiplies the momentum operator in $W(u)$: for one mode, with position and momentum operators Q and P, $W(u) = \exp(\mathrm{i}(u_p Q - u_q P))$. Thus the entry $(\gamma)_{11}$ of the form is the momentum entry of γ_B, and the state condition $\gamma + \mathrm{i}\sigma \geq 0$ on forms is equivalent to the familiar $\gamma_B - \mathrm{i}\Omega \geq 0$ on matrices (Proposition 3.17).

Rules and symbols. A translation-invariant linear map of V, a *rule*, has coefficients ℓ_m, the thesis's $\Gamma(m)$, and acts by $(Lu)(x) = \sum_m \ell_m u(x - m)$. Its *symbol* is the matrix of Laurent polynomials $\hat{L}(z) = \sum_m \ell_m z^m$, with z standing for the shift by one site; composing rules multiplies

symbols (Theorem 1.3). Substituting $z = e^{-ik}$ gives the Fourier symbol $\hat{L}(k)$, a $2n \times 2n$ matrix at each wave number k, and the cells of a form give $\hat{\beta}(k) = \sum_x \beta_x e^{-ikx}$ in the same way.

Dynamics. A rule or a channel moves labels, $W(u) \mapsto W(Lu)$ up to a factor, and therefore acts on forms by pull-back, $\beta \mapsto \beta(L\cdot, L\cdot)$. In cells and in Fourier symbols,

$$(L_*\beta)_j = \sum_{m,m'} \ell_m^{\mathsf{T}} \beta_{j+m-m'} \ell_{m'}, \qquad \widehat{L_*\beta}(k) = \hat{L}(k)^\dagger \, \hat{\beta}(k) \, \hat{L}(k),$$

so that on the chain every step is a congruence at each wave number (Theorem 3.34). Two consequences recur. An on-site rule acting at every site by a symplectic matrix S acts on covariance matrices through its inverse, $\gamma_B \mapsto S^{-1}\gamma_B S^{-\mathsf{T}}$ at each site (Corollary 3.32), since labels and moments transform oppositely; and pull-backs compose in the reverse order, $(L_2 L_1)_* = L_{1*} L_{2*}$.

Channels. A Gaussian channel is a linear map L of labels, a noise form ν and a displacement μ, with $T(W(u)) = e^{i\mu(u) - \nu(u,u)/4} \, W(Lu)$. It sends the covariance form γ to $\gamma(L\cdot, L\cdot) + \nu$, which in matrices is the familiar $\gamma_B \mapsto X\gamma_B X^{\mathsf{T}} + Y$ with $X = \Omega^{\mathsf{T}} L^{\mathsf{T}} \Omega$ and $Y = \Omega^{\mathsf{T}} \nu \, \Omega$ (Proposition 6.12); the condition $\nu + i(\sigma - \sigma(L\cdot, L\cdot)) \geq 0$ becomes $Y - i\Omega + iX\Omega X^{\mathsf{T}} \geq 0$. On the chain, the noise has cells ν_x; the two notions of locality are read off from where the cells vanish, and the condition from the Fourier symbols at each k (Chapter 7).

Roadmap

These notes are also the blueprint of the Lean library, and their chapters are ordered by logical dependence. Dashed boxes are chapters not yet written.

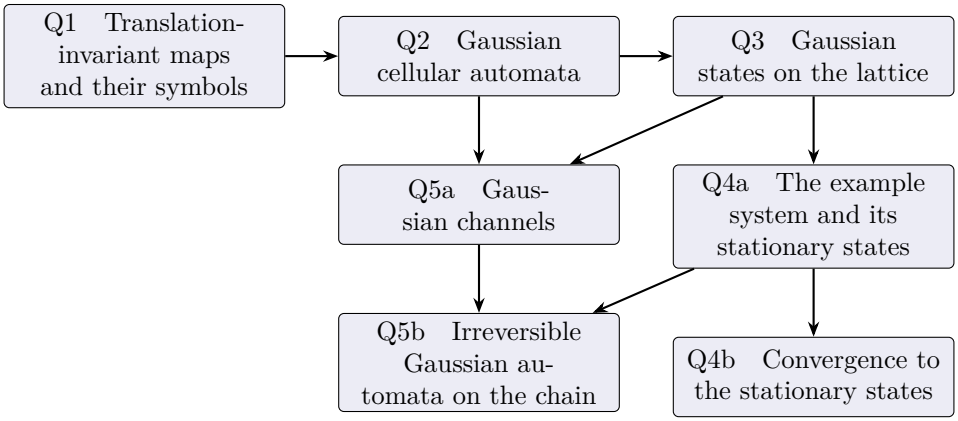


Figure 1: Milestones and their dependence. Each box links to its chapter.

1 Translation-invariant maps and their symbols

A Gaussian cellular automaton acts on Weyl operators through a linear map L of the classical phase space of a lattice. Three properties are required of L: it preserves the symplectic form, so that it defines an automorphism of the CCR algebra; it commutes with the lattice translations; and it moves each vector by a bounded distance.

This chapter shows that translation invariance turns L into a matrix over a ring of Laurent polynomials, its *symbol*, and that preservation of the form becomes a matrix identity for the symbol. For one mode per site and nearest-neighbour coupling the identity is solved completely. The running example is one time step of the lattice Klein–Gordon field.

The first three subsections work over a commutative ring R and an abelian group G, since the same algebra describes Clifford cellular automata over finite fields [8]. From Subsection 1.4 on, $R = \mathbb{R}$ and $G = \mathbb{Z}$.

1.1 Lattice phase space

The *cell* is a finite set ι indexing the degrees of freedom at one site; for n bosonic modes, $\iota = \{q_1, \dots, q_n, p_1, \dots, p_n\}$. The group algebra $R[G]$ is the ring of finite sums

$$c = \sum_{a \in G} c_a \, z^a, \qquad c_a \in R, \qquad z^a z^b = z^{a+b}.$$

For $G = \mathbb{Z}$ these are the Laurent polynomials in z. The coefficient at the origin is written $\varepsilon(c) = c_0$.

Definition 1.1 (Lattice phase space)**.** The *lattice phase space* is $V = R[G]^\iota$, the vectors $u = (u_i)_{i \in \iota}$ with $u_i \in R[G]$. The *value* of u at $x \in G$ is $u(x) = \big((u_i)_x\big)_{i \in \iota} \in R^\iota$, and the *support* of u is the finite set $\operatorname{supp} u = \{x \in G : u(x) \neq 0\}$. Translation by $a \in G$ is multiplication by z^a, written $\kappa_{a*} u = z^a u$.

Reading u through its values identifies V with the finitely supported functions $G \to R^\iota$. Multiplication by z^a shifts the values forward:

$$(\kappa_{a*} u)(x) = u(x - a).$$

For example, with one mode and $G = \mathbb{Z}$, the vector with position 1 at the site 0 and momentum 2 at the site 3 is $u = (1,\, 2z^3)$, and $\kappa_{1*} u = (z,\, 2z^4)$ has the same values at the sites 1 and 4.

1.2 Translation-invariant maps and symbols

Definition 1.2. An R-linear map $L \colon V \to V$ is *translation invariant* if $L\kappa_{a*} = \kappa_{a*} L$ for every $a \in G$.

Theorem 1.3 (Symbols)**.** Lean ✓

(a) An R-linear map $L \colon V \to V$ is translation invariant if and only if it is $R[G]$-linear.

(b) The $R[G]$-linear maps $V \to V$ are the maps $u \mapsto \hat{L}u$ with $\hat{L} \in M_\iota(R[G])$, and $L \mapsto \hat{L}$ is an isomorphism of $R[G]$-algebras. In particular the symbol of a composite $L_2 L_1$ is the product $\hat{L}_2 \hat{L}_1$.

(c) Write $\hat{L} = \sum_m \ell_m z^m$ with coefficient matrices $\ell_m \in M_\iota(R)$, finitely many nonzero. Then

$$(Lu)(x) = \sum_{m \in G} \ell_m \, u(x - m).$$

The matrix $\hat{L}$ is the symbol *of L.*

Proof.

(a) Multiplication by z^a is κ_{a*}, so an $R[G]$-linear map is translation invariant. Conversely, if L is translation invariant and $c = \sum_a c_a z^a$, then $L(cu) = \sum_a c_a L(z^a u) = \sum_a c_a z^a Lu = c\,Lu$.

(b) V is a free $R[G]$-module with basis $(e_i)_{i\in\iota}$. The endomorphisms of a free module of finite rank over a commutative ring are the square matrices, and composition is the matrix product.

(c) $(Lu)_i = \sum_j \hat{L}_{ij} u_j$, and the coefficient at x of a product in $R[G]$ is $(\hat{L}_{ij}u_j)_x = \sum_m (\hat{L}_{ij})_m (u_j)_{x-m}$.

□

By (c), the coefficient ℓ_m carries the value at $x - m$ to the value at x. The simplest example is the shift $(Lu)(x) = u(x-1)$, with $\ell_1 = 1$ and symbol $z \cdot 1$.

Example 1.4 (The leapfrog rule)**.** Lean ✓ The lattice Klein–Gordon equation for $\varphi\colon \mathbb{Z}^2 \to \mathbb{R}$, with points (t, x) and a parameter $m^2 \in \mathbb{R}$, is

$$\varphi(t+1,x) + \varphi(t-1,x) - \varphi(t,x+1) - \varphi(t,x-1) + m^2\varphi(t,x) = 0.$$

Given the values on two consecutive time slices,

$$d_0(x) = \varphi(t,x), \qquad d_1(x) = \varphi(t+1,x),$$

the equation at time $t+1$ determines the next slice:

$$d_2(x) = \varphi(t+2,x) = -d_0(x) + d_1(x+1) + d_1(x-1) - m^2 d_1(x).$$

One time step is the map $F\colon (d_0, d_1) \mapsto (d_1, d_2)$. Take one mode per site with the cell $u(x) = (d_0(x), d_1(x))$. Comparing

$$(Fu)(x) = \begin{pmatrix} d_1(x) \\ -d_0(x) - m^2 d_1(x) + d_1(x+1) + d_1(x-1) \end{pmatrix}$$

with Theorem 1.3(c), the term $d_1(x+1) = u(x-(-1))_2$ comes from ℓ_{-1} and $d_1(x-1)$ from ℓ_{+1}, so

$$\ell_0 = \begin{pmatrix} 0 & 1 \\ -1 & -m^2 \end{pmatrix}, \qquad \ell_{-1} = \ell_{+1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \hat{F} = \begin{pmatrix} 0 & 1 \\ -1 & z + z^{-1} - m^2 \end{pmatrix}.$$

Since only finitely many ℓ_m are nonzero, a translation-invariant map moves every vector by a bounded distance.

Corollary 1.5 (Finite range)**.** Lean ✓ *Let $N = \bigcup_{i,j} \operatorname{supp} \hat{L}_{ij}$. Then $\operatorname{supp} Lu \subseteq \operatorname{supp} u + N$ for every $u \in V$.*

Proof. If $(Lu)(x) \neq 0$, then by Theorem 1.3(c) some $\ell_m u(x-m)$ is nonzero, so $m \in N$ and $x - m \in \operatorname{supp} u$. □

For the leapfrog, $N = \{-1, 0, 1\}$: one step changes the Cauchy data only within distance one of their support.

1.3 Forms preserved by symbols

A form on one cell, given by a matrix $B \in M_\iota(R)$, defines a form on V by summing over sites. Writing it through symbols needs the *involution* of $R[G]$, the R-algebra automorphism

$$c \mapsto c^\star, \qquad (z^a)^\star = z^{-a}.$$

For a matrix M over $R[G]$, $M^{\star\mathsf{T}}$ is the transpose of the matrix with entries $M^\star_{ij}$. Since $R[G]$ is commutative, $(MN)^{\star\mathsf{T}} = N^{\star\mathsf{T}} M^{\star\mathsf{T}}$.

Definition 1.6 (Lifted form)**.** For $B \in M_\iota(R)$, the *lifted form* on V is

$$\tilde{B}(u,w) = \sum_{x \in G} u(x)^{\mathsf{T}} B\, w(x) = \varepsilon\Big(\sum_{i,j \in \iota} u_i^\star B_{ij} w_j \Big).$$

The matrix B is *alternating* if $v^{\mathsf{T}} B v = 0$ for every $v \in R^\iota$.

The two expressions agree because the coefficient of $u_i^\star w_j$ at the origin is $\sum_x (u_i)_x (w_j)_x$. If B is alternating, so is $\tilde{B}$. For n bosonic modes and $B = \Omega = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, the lifted form $\sigma = \tilde{\Omega}$ is the symplectic form of the Notation table. (Alternating is stronger than $B^{\mathsf{T}} = -B$ only when 2 is not invertible, as for Clifford automata over $\mathbb{Z}/2$.)

Theorem 1.7 (Twisted criterion)**.** Lean ✓ *Let $B \in M_\iota(R)$ and let L be translation invariant with symbol $\hat{L}$. Then $\tilde{B}(Lu, Lw) = \tilde{B}(u,w)$ for all $u, w \in V$ if and only if*

$$\hat{L}^{\star\mathsf{T}} B\, \hat{L} = B.$$

Proof. For $P \in M_\iota(R[G])$ put $\langle u, Pw \rangle = \varepsilon\big(\sum_{i,j} u_i^\star P_{ij} w_j\big)$, so that $\tilde{B}(u,w) = \langle u, Bw \rangle$. The involution is a ring homomorphism fixing R, so $(\hat{L}u)^\star_i = \sum_k \hat{L}^\star_{ik} u^\star_k$, and therefore $\tilde{B}(Lu, Lw) = \langle u, \hat{L}^{\star\mathsf{T}} B \hat{L} w \rangle$. Invariance thus says $\langle u, Qw \rangle = 0$ for all u, w, where $Q = \hat{L}^{\star\mathsf{T}} B \hat{L} - B$. If $Q = 0$ this holds. Conversely, $u = e_i$ and $w = z^k e_j$ give $\langle u, Qw \rangle = \varepsilon(Q_{ij} z^k) = (Q_{ij})_{-k}$, so every coefficient of every entry of Q vanishes. □

Without the involution the criterion would fail for rules that move vectors. For the shift $\hat{L} = z \cdot 1$ one has $\hat{L}^{\star\mathsf{T}} B \hat{L} = z^{-1} B z = B$, so the shift preserves every lifted form; the plain transpose would give $z^2 B$. Expanding $\hat{L}^{\star\mathsf{T}} = \sum_m \ell_m^{\mathsf{T}} z^{-m}$ and collecting the coefficient of z^k gives the criterion in coefficients:

$$\sum_m \ell_m^{\mathsf{T}} B\, \ell_{m+k} = \delta_{k0} B \qquad \text{for every } k \in G. \tag{1.1}$$

Example 1.8 (On-site rules)**.** Lean ✓ Let $\hat{L} = S$ be constant, with n modes per cell and S a real $2n \times 2n$ matrix. Then $\hat{L}^{\star\mathsf{T}} = S^{\mathsf{T}}$, so L preserves σ if and only if $S^{\mathsf{T}} \Omega S = \Omega$. This is equivalent to $S \Omega S^{\mathsf{T}} = \Omega$: if $S^{\mathsf{T}} \Omega S = \Omega$, then S is invertible with $S^{\mathsf{T}} = \Omega S^{-1} \Omega^{-1}$, and

$$S \Omega S^{\mathsf{T}} = S \Omega^2 S^{-1} \Omega^{-1} = -\Omega^{-1} = \Omega,$$

using $\Omega^2 = -1$; the converse is the same computation for S^{T}.

Under the condition $S \Omega S^{\mathsf{T}} = \Omega$ the map $\gamma \mapsto S \gamma S^{\mathsf{T}}$ preserves the uncertainty relation $\gamma - \mathrm{i}\Omega \geq 0$ of real symmetric covariance matrices.

Example 1.9 (Coefficient form of the criterion)**.** Lean ✓ For n modes and $\hat{L} = \sum_m z^m \ell_m$, put $\ell_m^+ = \Omega^{-1} \ell_m^{\mathsf{T}} \Omega$, the symplectic adjoint. Multiplying (1.1) with $B = \Omega$ on the left by Ω^{-1}, $\hat{L}$ is symplectic if and only if

$$\sum_m \ell_m^+ \ell_{m+k} = \delta_{k0} 1 \qquad \text{for every } k \in \mathbb{Z}.$$

Theorem 1.10 (Inverse). Lean ✓ *Let B be invertible and $\hat{L}^{\star\mathsf{T}} B \hat{L} = B$. Then L is bijective. Its inverse is translation invariant with symbol $B^{-1}\hat{L}^{\star\mathsf{T}}B$, preserves $\tilde{B}$, and has finite range.*

Proof. $(B^{-1}\hat{L}^{\star\mathsf{T}}B)\hat{L} = B^{-1}B = 1$, so $B^{-1}\hat{L}^{\star\mathsf{T}}B$ is a left inverse. A square matrix over a commutative ring with a left inverse has invertible determinant, since $\det(PM) = \det P \det M = 1$, so the left inverse is two-sided. The inverse $\hat{N}$ satisfies $\hat{N}^{\star\mathsf{T}}B\hat{N} = \hat{N}^{\star\mathsf{T}}(\hat{L}^{\star\mathsf{T}}B\hat{L})\hat{N} = (\hat{L}\hat{N})^{\star\mathsf{T}}B(\hat{L}\hat{N}) = B$, and it has finite range by Corollary 1.5. □

On an infinite-dimensional phase space, a map preserving a nondegenerate form is injective but need not be onto. For translation-invariant maps the theorem gives an inverse, and the inverse again has finite range.

Corollary 1.11 (The group of rules). Lean ✓ *For invertible B, the symbols $\hat{L}$ with $\hat{L}^{\star\mathsf{T}}B\hat{L} = B$ form a group under multiplication.*

Proof. They are invertible by Theorem 1.10, which also shows that the inverse satisfies the identity. The identity matrix satisfies it, and if $\hat{L}$ and $\hat{N}$ do, then $(\hat{L}\hat{N})^{\star\mathsf{T}}B\hat{L}\hat{N} = \hat{N}^{\star\mathsf{T}}(\hat{L}^{\star\mathsf{T}}B\hat{L})\hat{N} = B$. □

A rule preserving σ satisfies $\sigma(Lu, Lw) = \sigma(u, w)$ for all u, w, which is exactly what is needed for $W(u) \mapsto W(Lu)$ to extend to an automorphism of $\mathrm{CCR}(V, \sigma)$.

1.4 Nearest-neighbour rules on one mode

From now on $R = \mathbb{R}$, $G = \mathbb{Z}$, and there is one mode per site, $\iota = \{q, p\}$, with

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

A symbol is *symplectic* if $\hat{L}^{\star\mathsf{T}}\Omega\hat{L} = \Omega$, and *nearest-neighbour* if

$$\hat{L} = \ell_- z^{-1} + \ell_0 + \ell_+ z, \qquad \ell_-, \ell_0, \ell_+ \in M_2(\mathbb{R}),$$

so that $(Lu)(x) = \ell_- u(x+1) + \ell_0 u(x) + \ell_+ u(x-1)$. The leapfrog is of this form. For 2×2 matrices the symplectic condition is governed by the adjugate

$$\operatorname{adj}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \qquad A \operatorname{adj} A = \operatorname{adj}(A) A = (\det A)\, 1.$$

Lemma 1.12 (2×2 identities). Lean ✓ *For $A, A' \in M_2(\mathbb{R})$:*

(a) $A^{\mathsf{T}}\Omega A' = \Omega \operatorname{adj}(A) A'$, *and in particular* $A^{\mathsf{T}}\Omega A = (\det A)\,\Omega$;

(b) adj *is additive,* $\operatorname{adj}(AA') = \operatorname{adj}(A')\operatorname{adj}(A)$ *and* $\operatorname{adj}\operatorname{adj} A = A$;

(c) $\det(A + A') = \det A + \det A' + \operatorname{tr}(\operatorname{adj}(A) A')$;

(d) if $\operatorname{adj} A = A$, *then A is a multiple of* 1.

Proof.

(a) For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, both $A^{\mathsf{T}}\Omega$ and $\Omega\operatorname{adj}(A)$ equal $\left(\begin{smallmatrix} -c & a \\ -d & b \end{smallmatrix}\right)$; multiply on the right by A'. For $A' = A$, use $\operatorname{adj}(A)A = (\det A)1$.

(b) The entries of $\operatorname{adj} A$ are linear in those of A; the other two identities are checked on entries.

(c) Checked on entries.

(d) $\operatorname{adj} A = A$ says $-A_{12} = A_{12}$, $-A_{21} = A_{21}$ and $A_{22} = A_{11}$.

□

Part (d) uses $2 \neq 0$; over $\mathbb{Z}/2$ every matrix with equal diagonal entries equals its adjugate. The classification below is therefore proved over $\mathbb{R}$ (any field of characteristic other than 2 would do).

Proposition 1.13 (Coefficient equations). Lean ✓ *A nearest-neighbour symbol is symplectic if and only if the coefficients of $z^2, z^{-2}, z, z^{-1}, z^0$ in $\hat{L}^{\star\mathsf{T}}\Omega\hat{L} - \Omega$ vanish:*

$$(C_2)\ \ \ell_-^{\mathsf{T}}\Omega\ell_+ = 0, \qquad (C_{-2})\ \ \ell_+^{\mathsf{T}}\Omega\ell_- = 0, \qquad (C_1)\ \ \ell_-^{\mathsf{T}}\Omega\ell_0 + \ell_0^{\mathsf{T}}\Omega\ell_+ = 0,$$
$$(C_{-1})\ \ \ell_0^{\mathsf{T}}\Omega\ell_- + \ell_+^{\mathsf{T}}\Omega\ell_0 = 0, \qquad (C_0)\ \ \ell_-^{\mathsf{T}}\Omega\ell_- + \ell_0^{\mathsf{T}}\Omega\ell_0 + \ell_+^{\mathsf{T}}\Omega\ell_+ = \Omega.$$

Proof. $\hat{L}^{\star\mathsf{T}} = \ell_-^{\mathsf{T}} z + \ell_0^{\mathsf{T}} + \ell_+^{\mathsf{T}} z^{-1}$; multiply out and collect powers of z. □

By Lemma 1.12(a), $(C_{\pm 2})$ say $\operatorname{adj}(\ell_\mp)\ell_\pm = 0$, and (C_0) says $\det \ell_- + \det \ell_0 + \det \ell_+ = 1$. Two families of solutions can be read off:

- the *pure shifts* Sz^{-1} and Sz with $\det S = 1$, where $\ell_0 = 0$;
- $\ell_0 \in SL_2(\mathbb{R})$ and $\ell_+ = \ell_- = \ell$ with $\det \ell = 0$; then $(C_{\pm 2})$ and (C_0) hold, and $(C_{\pm 1})$ is one remaining condition on ℓ.

The next proposition shows that there are no others.

Proposition 1.14 (Classification of nearest-neighbour rules). Lean ✓ *A nearest-neighbour symbol is symplectic if and only if one of the following holds:*

(i) $\ell_0 = 0$*, and either* $\ell_+ = 0$ *and* $\det \ell_- = 1$*, or* $\ell_- = 0$ *and* $\det \ell_+ = 1$*;*

(ii) $\det \ell_0 = 1$*,* $\ell_+ = \ell_-$*,* $\det \ell_+ = 0$*, and* $\ell_+^{\mathsf{T}}\Omega\ell_0 + \ell_0^{\mathsf{T}}\Omega\ell_+ = 0$*.*

Proof. The two families above show that (i) and (ii) are symplectic; in (ii), (C_{-1}) is (C_1) with the two terms exchanged. Conversely, assume the equations (C).

ℓ_- invertible. (C_2) gives $\ell_+ = 0$. Then (C_1) reads $\operatorname{adj}(\ell_-)\ell_0 = 0$, so $\ell_0 = 0$, and (C_0) gives $\det \ell_- = 1$: case (i).

ℓ_+ invertible. The same argument with (C_{-2}) and (C_{-1}) gives the other half of case (i).

ℓ_- and ℓ_+ both singular. (C_0) gives $\det \ell_0 = 1$, so $\ell_0 \operatorname{adj} \ell_0 = 1$. Put $D = \ell_- - \ell_+$. Subtracting (C_1) from (C_{-1}), in the adjugate form of Lemma 1.12(a), gives $\operatorname{adj}(\ell_0)D = \operatorname{adj}(D)\ell_0$. For $K = \operatorname{adj}(\ell_0)D$, Lemma 1.12(b) and this identity give

$$\operatorname{adj} K = \operatorname{adj}(D)\,\ell_0 = \operatorname{adj}(\ell_0)D = K,$$

so $K = c1$ for some $c \in \mathbb{R}$ by Lemma 1.12(d), and $D = \ell_0 K = c\,\ell_0$. Hence $\det D = c^2$. On the other hand, Lemma 1.12(c) with $A = \ell_-$, $A' = -\ell_+$ and (C_2) gives $\det D = \det \ell_- + \det \ell_+ - \operatorname{tr}(\operatorname{adj}(\ell_-)\ell_+) = 0$. So $c = 0$ and $\ell_+ = \ell_-$: case (ii). □

In case (ii) the last condition is one scalar equation. The identity $\operatorname{adj}(A)A' + \operatorname{adj}(A')A = \operatorname{tr}(\operatorname{adj}(A)A')\,1$, checked on entries, and Lemma 1.12(a) give $\ell^{\mathsf{T}}\Omega\ell_0 + \ell_0^{\mathsf{T}}\Omega\ell = \operatorname{tr}(\operatorname{adj}(\ell_0)\ell)\,\Omega$, so the condition is $\operatorname{tr}(\operatorname{adj}(\ell_0)\ell) = 0$.

Remark 1.15 (Pure shifts). Case (i) contains the translation $z^{\pm 1} \cdot 1$, which moves the state of every cell one site along the chain. It is a symplectic nearest-neighbour rule with one mode per site and $\ell_0 = 0$, and it is not of the form (ii).

In case (ii), $\det \ell = 0$, so $\ell = va^{\mathsf{T}}$ for some vectors v, a, with $v \neq 0$ if $\ell \neq 0$. A change of basis in the cell reduces to $v = e_1 = (1, 0)$, the position direction: for $S \in SL_2(\mathbb{R})$, the conjugation $\hat{L} \mapsto S\hat{L}S^{-1}$ maps symplectic symbols to symplectic symbols (Corollary 1.11 with Example 1.8), replaces v by Sv, and replaces ℓ_0 by $S\ell_0 S^{-1}$. For $v = e_1$ the last condition fixes a.

Proposition 1.16 (Coupling in a fixed direction). Lean ✓ *Let* $\det \ell_0 = 1$ *and let* r *be the second row of* ℓ_0*. For* $a \in \mathbb{R}^2$*, the symbol with* $\ell_+ = \ell_- = e_1 a^{\mathsf{T}}$*, which satisfies all conditions of case (ii) except the last, is symplectic if and only if* $a = \lambda r$ *for some* $\lambda \in \mathbb{R}$*.*

Proof. Write $\ell = e_1 a^{\mathsf{T}}$. Since $e_1^{\mathsf{T}}\Omega = e_2^{\mathsf{T}}$ and $\Omega e_1 = -e_2$,

$$\ell^{\mathsf{T}}\Omega\ell_0 = a\, e_2^{\mathsf{T}}\ell_0 = a r^{\mathsf{T}}, \qquad \ell_0^{\mathsf{T}}\Omega\ell = -\ell_0^{\mathsf{T}} e_2 a^{\mathsf{T}} = -r a^{\mathsf{T}}.$$

So the last condition of case (ii) reads $a r^{\mathsf{T}} = r a^{\mathsf{T}}$, which holds if and only if $a = \lambda r$, since $r \neq 0$. The other conditions of case (ii) hold because $\det(e_1 a^{\mathsf{T}}) = 0$. □

The number λ is the *coupling* of the rule.

Example 1.17 (The leapfrog step is symplectic). Lean ✓ For every real m^2, the leapfrog symbol $\hat{F}$ of Example 1.4 and

$$\hat{F}^{-1} = \begin{pmatrix} z + z^{-1} - m^2 & -1 \\ 1 & 0 \end{pmatrix}$$

are symplectic, and $\hat{F}\hat{F}^{-1} = 1$.

Both lie in case (ii) of Proposition 1.14: for $\hat{F}$, $\det \ell_0 = 1$, $\det \ell_\pm = 0$, and the last condition holds; the same computation applies to $\hat{F}^{-1}$. In the Cauchy data (d_0, d_1), the form $\sigma(u, u') = \sum_x \left(d_0(x) d_1'(x) - d_1(x) d_0'(x)\right)$ is the conserved Wronskian of the lattice Klein–Gordon equation, so d_0 plays the role of position and d_1 of momentum. The inverse $\hat{F}^{-1}$ has $\ell_\pm = e_1 e_1^{\mathsf{T}}$; it is Proposition 1.16 with $r = e_1$ and $\lambda = 1$. For $\hat{F}$ itself, $\ell_\pm = e_2 e_2^{\mathsf{T}}$ points in the momentum direction, so Proposition 1.16 applies to it only after the change of basis described before that proposition.

1.5 Fourier modes

Evaluating a symbol on the unit circle gives its Fourier transform: the formula of Theorem 1.3(c), applied to a plane wave $u(x) = e^{\mathrm{i}kx} v$ with $v \in \mathbb{C}^\iota$, gives $(Lu)(x) = e^{\mathrm{i}kx} \hat{L}(e^{-\mathrm{i}k}) v$.

Lemma 1.18 (Evaluation on the circle). Lean ✓ *For* $k \in \mathbb{R}$*, evaluation at* $z = e^{-\mathrm{i}k}$ *is a ring homomorphism from the Laurent polynomials* $\mathbb{R}[z, z^{-1}]$ *to* $\mathbb{C}$*, with* $p(e^{-\mathrm{i}k}) = \sum_m p_m e^{-\mathrm{i}km}$*, and it carries the involution to complex conjugation:* $p^\star(e^{-\mathrm{i}k}) = \overline{p(e^{-\mathrm{i}k})}$*.*

Proof. $e^{-\mathrm{i}k}$ is a unit of $\mathbb{C}$, and a Laurent polynomial ring maps to any ring by sending z to a unit. On monomials, $(z^m)^\star = z^{-m} \mapsto e^{\mathrm{i}km} = \overline{e^{-\mathrm{i}km}}$, and the coefficients are real. □

The *Fourier symbol* is $\hat{L}(k) = \hat{L}(e^{-\mathrm{i}k}) \in M_\iota(\mathbb{C})$, evaluated entrywise. By the lemma, $\hat{L}^{\star\mathsf{T}}$ evaluates to the conjugate transpose $\hat{L}(k)^\dagger$.

Example 1.19 (Fourier modes). Lean ✓ If $\hat{L}$ is symplectic, then

$$\hat{L}(k)^\dagger\, \Omega\, \hat{L}(k) = \Omega \qquad \text{for every } k \in \mathbb{R}.$$

For the leapfrog, $\operatorname{tr} \hat{F}(k) = 2\cos k - m^2$.

The conjugate transpose cannot be replaced by the transpose: for the shift $\hat{L} = z \cdot 1$, $\hat{L}(k)^{\mathsf{T}}\Omega\hat{L}(k) = e^{-2\mathrm{i}k}\Omega$. The converse also holds: if $\hat{L}(k)^\dagger\Omega\hat{L}(k) = \Omega$ for all k, each entry of $\hat{L}^{\star\mathsf{T}}\Omega\hat{L} - \Omega$ is a Laurent polynomial vanishing at infinitely many points, hence zero.

Example 1.20 (Rotation with symmetric coupling). Lean ✓ Let ℓ_0 be the rotation by φ, with second row $r = (\sin\varphi, \cos\varphi)$, and $\ell_\pm = \lambda\, e_1 r^{\mathsf{T}}$ as in Proposition 1.16:

$$\ell_0 = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad \ell_\pm = \lambda \begin{pmatrix} \sin\varphi & \cos\varphi \\ 0 & 0 \end{pmatrix}.$$

The rule is symplectic for all φ and λ, and

$$\operatorname{tr} \hat{L}(k) = 2\cos\varphi + 2\lambda \cos k \sin\varphi.$$

The trace comes from $\hat{L}(k) = \ell_0 + 2\cos k\,\ell_\pm$. Since $\det \hat{L}(k) = 1$, writing $\operatorname{tr} \hat{L}(k) = 2\cos\theta(k)$ gives the dispersion relation $\cos\theta(k) = \cos\varphi + \lambda \cos k \sin\varphi$.

1.6 Notes and references

The material of this chapter is the phase-space part of Chapter 4 of [1]; the Notation section at the front lists the correspondence of symbols.

1. *Conventions.* In [1] a rule is written $(\Gamma\xi)_x = \sum_z \Gamma_z \xi_{x-z}$ (4.8), so $\Gamma_z = \ell_z$, and its Fourier transform $\hat{\Gamma}(k) = \sum_x \Gamma_x e^{-\mathrm{i}kx}$ (4.14)–(4.15) is $\hat{L}(e^{-\mathrm{i}k})$. Translations there act by $(\tau_\Delta \xi)_x = \xi_{x+\Delta}$ (p. 78), which is κ_{a*} with $a = -\Delta$.

2. *Corresponding equations.* Theorem 1.3(b),(c) are (4.13) and (4.8) of [1]; Proposition 1.13 is (4.11); Example 1.9 is (4.10), with Γ^+ the symplectic adjoint (2.1); Proposition 1.16 is (4.12), in the basis of footnote 6 (p. 80), with the coupling written f; Example 1.20 is (4.18)–(4.19); Example 1.19 is the statement on p. 82 that $\hat{\Gamma}(k)$ "induces a symplectic transformation on every single mode k".

3. *Range.* In [1, p. 92] the condition $\Gamma_x = 0$ for $x \notin N$ is imposed on a rule. By Theorem 1.3(c) it holds for every translation-invariant map, with N as in Corollary 1.5.

4. *Invertibility.* [1, p. 78] states that Γ is invertible "as a symplectic transformation"; Theorem 1.10 proves this for translation-invariant rules.

5. *Pure shifts.* Proposition 4.4 of [1] derives from $(C_{\pm 2})$ that "Γ_- and Γ_+ project onto the same, one-dimensional subspace" (p. 80), which assumes that both are nonzero, and so omits case (i) of Proposition 1.14. The Remark after it, "the left- and right-shifter mentioned as examples in the introduction cannot be realized with one mode per site" (p. 80; the shifters are listed on p. 70), is contradicted by Remark 1.15.

6. *Direction of time.* For the lattice Klein–Gordon field, the forward rule in the convention of [1] corresponds to F^{-1}; this is made precise in Proposition 2.16.

2 Gaussian cellular automata

Chapter 1 studied rules acting on the phase space itself; this chapter lets them act instead on the observables. A symplectic rule L defines an automorphism α_L of the CCR algebra by $\alpha_L(W(u)) = W(Lu)$, and this automorphism is local, commutes with the lattice translations, and composes, so that it is a quantum cellular automaton in the sense made precise below. Conversely, any automorphism sending Weyl operators to multiples of Weyl operators arises from a symplectic linear bijection of V together with a character, and the bijection is a rule whenever the automorphism also commutes with the translations. The leapfrog again serves as the running example, and at the end of the chapter it is identified with one time step of the lattice Klein–Gordon field.

Throughout this chapter G is an abelian group, the cell carries n modes, and V denotes the lattice phase space of Definition 1.1 over $\mathbb{R}$, equipped with the symplectic form $\sigma = \tilde{\Omega}$. The algebra $\mathrm{CCR}(V, \sigma)$ is the C*-algebra generated by the Weyl operators $W(u)$ on $\ell^2(V)$, which act by

$$(W(u)f)(v) = e^{\mathrm{i}\sigma(u,v)/2} f(v+u),$$

and consequently satisfy

$$W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)/2}\, W(u+w), \qquad W(u)^* = W(-u).$$

Here $\ell^2(V)$ is the space of square-summable functions on the set V, with the point masses δ_v as orthonormal basis; it serves only to construct the algebra, and it is not the Fock space of the oscillators, since $t \mapsto W(tu)$ is not continuous on it and it carries no Gaussian state (Remark 3.10). The algebra, and not this representation, is what the states and channels of later chapters act on.

2.1 The lattice CCR net

Local observables are attached to regions of the lattice, understood as subsets $O \subseteq G$: observables belonging to disjoint regions should commute, and the observables of a region belong as well to every larger region containing it. Since the lattice carries no time direction, there is no time-slice condition beyond this isotony.
Every linear bijection of V that preserves σ relabels the Weyl operators, and both the translations and the rules of the previous chapter are bijections of this kind.

Lemma 2.1 (Relabelling)**.** Lean ✓ *Let $P\colon V \to V$ be a linear bijection with $\sigma(Pu, Pw) = \sigma(u, w)$. The unitary $(U_P\psi)(v) = \psi(P^{-1}v)$ on $\ell^2(V)$ satisfies $U_P W(u) U_P^* = W(Pu)$, so $\alpha_P = U_P \cdot U_P^*$ is a $*$-automorphism of $\mathrm{CCR}(V, \sigma)$ with $\alpha_P(W(u)) = W(Pu)$, and $\alpha_{PP'} = \alpha_P \alpha_{P'}$.*

Proof. U_P is unitary because P is a bijection, and $(U_P^*\psi)(v) = \psi(Pv)$. Hence

$$(U_P W(u) U_P^* \psi)(v) = e^{\mathrm{i}\sigma(u, P^{-1}v)/2}\, \psi(v+Pu) = e^{\mathrm{i}\sigma(Pu,v)/2}\, \psi(v+Pu) = (W(Pu)\psi)(v).$$

Conjugation by U_P is a $*$-automorphism of the bounded operators, isometric, and maps the Weyl operators onto the Weyl operators; so it maps $\mathrm{CCR}(V, \sigma)$, the closure of their span, onto itself. Finally $U_{PP'} = U_P U_{P'}$. Two automorphisms that agree on the Weyl operators agree on their span, and so on its closure; so α_P is the only automorphism with $\alpha_P(W(u)) = W(Pu)$. □

Definition 2.2 (The lattice CCR net)**.** For $O \subseteq G$, $\mathcal{A}(O)$ is the closed $*$-subalgebra of $\mathrm{CCR}(V, \sigma)$ generated by the $W(u)$ with $\operatorname{supp} u \subseteq O$. The translation by $a \in G$ acts by $\alpha_a = \alpha_{\kappa_{a*}}$, so $\alpha_a(W(u)) = W(\kappa_{a*}u)$.

Theorem 2.3. Lean ✓ *The net $O \mapsto \mathcal{A}(O)$ is isotone, generates $\mathrm{CCR}(V, \sigma)$, and satisfies causality: elements of $\mathcal{A}(O_1)$ and $\mathcal{A}(O_2)$ commute when $O_1 \cap O_2 = \emptyset$. The map $a \mapsto \alpha_a$ is a group homomorphism into the automorphisms of $\mathrm{CCR}(V, \sigma)$, and $\alpha_a \mathcal{A}(O) = \mathcal{A}(O + a)$.*

Proof. Isotony holds because $\mathcal{A}(O)$ has fewer generators than $\mathcal{A}(O')$ whenever $O \subseteq O'$, while $\mathcal{A}(G) = \mathrm{CCR}(V, \sigma)$ by definition. For causality, if $\operatorname{supp} u \cap \operatorname{supp} w = \emptyset$ then every term of $\sigma(u, w) = \sum_x u(x)^{\mathsf{T}} \Omega w(x)$ vanishes, so that $W(u)$ and $W(w)$ commute; since the commutant of the generators of $\mathcal{A}(O_2)$ is a closed $*$-algebra, being closed under $*$ as a set, and contains the generators of $\mathcal{A}(O_1)$, it contains $\mathcal{A}(O_1)$ itself, and by the same argument the commutant of $\mathcal{A}(O_1)$ contains $\mathcal{A}(O_2)$. For the last statement, translations preserve σ, so α_a exists by Lemma 2.1, and $\alpha_{a+b} = \alpha_a \alpha_b$ because $\kappa_{(a+b)*} = \kappa_{a*}\kappa_{b*}$; since $\operatorname{supp} \kappa_{a*} u = \operatorname{supp} u + a$, α_a carries the generators of $\mathcal{A}(O)$ onto those of $\mathcal{A}(O + a)$, and being an isometric $*$-automorphism it therefore carries $\mathcal{A}(O)$ onto $\mathcal{A}(O + a)$. □

2.2 The automorphism of a rule

Since a symplectic rule L is a bijection of V preserving σ (Theorem 1.10), Lemma 2.1 already furnishes its automorphism α_L.

Proposition 2.4. Lean ✓ *For symplectic rules L and K:*

(a) $\alpha_L(W(u)) = W(Lu)$*;*

(b) $\alpha_{LK} = \alpha_L \alpha_K$ *and* $\alpha_1 = \mathrm{id}$*;*

(c) $\alpha_L \alpha_a = \alpha_a \alpha_L$ *for every* $a \in G$*;*

(d) $\alpha_L^{-1} = \alpha_{L^{-1}}$*, where* L^{-1} *has symbol* $\Omega^{-1} \hat{L}^{\star\mathsf{T}} \Omega$*.*

Proof. (a) Lemma 2.1.

(b) Lemma 2.1.

(c) L commutes with κ_{a*} (Theorem 1.3(a)), so both sides send $W(u)$ to $W(L\kappa_{a*}u)$.

(d) By (b) and Theorem 1.10.

□

Theorem 2.5 (Locality)**.** Lean ✓ *Let N be the range of the symbol of L. Then $\alpha_L \mathcal{A}(O) \subseteq \mathcal{A}(O + N)$ for every $O \subseteq G$.*

Proof. If $\operatorname{supp} u \subseteq O$, then $\operatorname{supp} Lu \subseteq O + N$ (Corollary 1.5), so α_L maps the generators of $\mathcal{A}(O)$ into $\mathcal{A}(O + N)$. A $*$-homomorphism maps the $*$-algebra generated by a set onto the $*$-algebra generated by its image; a $*$-automorphism of a C*-algebra is isometric, so it maps closures into closures, and $\mathcal{A}(O + N)$ is closed. □

2.3 Quantum cellular automata

Definition 2.6 (Reversible quantum cellular automaton)**.** Let $\mathcal{A}$ be a net over the subsets of G in a C*-algebra A, with a group homomorphism $a \mapsto \alpha_a$ into the automorphisms of A such that $\alpha_a \mathcal{A}(O) = \mathcal{A}(O + a)$. A *reversible quantum cellular automaton* is a $*$-automorphism T of A together with a finite set $N \subseteq G$, its *neighbourhood*, such that $T\alpha_a = \alpha_a T$ for all a and $T\mathcal{A}(O) \subseteq \mathcal{A}(O + N)$ for all O.

The locality condition $T\mathcal{A}(O) \subseteq \mathcal{A}(O + N)$ requires that a single step change the observables of a region only within the neighbourhood of that region.

Proposition 2.7 (Composition)**.** Lean ✓ *If (T_1, N_1) and (T_2, N_2) are reversible quantum cellular automata on the same net, so is $(T_2 T_1, N_1 + N_2)$.*

Proof. $T_2 T_1$ commutes with every α_a, and $T_2 T_1 \mathcal{A}(O) \subseteq T_2 \mathcal{A}(O + N_1) \subseteq \mathcal{A}(O + N_1 + N_2)$. □

Theorem 2.8. Lean ✓ *For a symplectic rule L with range N, (α_L, N) is a reversible quantum cellular automaton on the lattice CCR net.*

Proof. Proposition 2.4(c) and Theorem 2.5; N is finite because the entries of the symbol are finitely supported. The inverse $\alpha_{L^{-1}}$ is likewise a reversible quantum cellular automaton, with neighbourhood $-N$, since the entries of $\Omega^{-1}\hat{L}^{\star\mathsf{T}}\Omega$ are, up to sign, exactly those of $\hat{L}^{\star\mathsf{T}}$. □

Example 2.9 (The leapfrog)**.** Lean ✓ The leapfrog F of Example 1.4 and its inverse give reversible quantum cellular automata α_F and $\alpha_{F^{-1}}$ with neighbourhood $\{-1, 0, 1\}$, and $\alpha_F^{-1} = \alpha_{F^{-1}}$. In the Cauchy data $u(x) = (d_0(x), d_1(x))$, $\alpha_F(W(u)) = W(Fu)$ with $(Fu)(x) = \big(d_1(x),\ -d_0(x) + d_1(x+1) + d_1(x-1) - m^2 d_1(x)\big)$.

Example 2.10 (Shifts)**.** Lean ✓ For $a \in G$, the symbol $z^a \cdot 1$ gives the reversible quantum cellular automaton with neighbourhood $\{a\}$ whose automorphism is the translation α_a itself.

Example 2.11 (On-site rules)**.** Lean ✓ For $G = \mathbb{Z}$ and a real $2n \times 2n$ matrix S with $S\Omega S^{\mathsf{T}} = \Omega$, the constant symbol S gives a reversible quantum cellular automaton with neighbourhood $\{0\}$: every site evolves separately by S. For one mode this covers the rotations $R_\varphi = \left(\begin{smallmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{smallmatrix}\right)$, the free evolution of uncoupled oscillators, and the squeezings $\mathrm{diag}(e^r, e^{-r})$, which iterated squeeze every site further.

Example 2.12 (Rotation with symmetric coupling)**.** Lean ✓ The rule of Example 1.20 gives a reversible quantum cellular automaton with neighbourhood $\{-1, 0, 1\}$ for all φ and λ. A neighbourhood need not be the smallest possible one, as is seen already at $\lambda = 0$, where the range of the symbol reduces to $\{0\}$.

2.4 The local rule

A rule is entirely determined by its action on the degrees of freedom of a single site. For $i \in \iota$, let $u_i \in V$ denote the vector taking the value e_i at the site 0 and vanishing elsewhere, so that the operators $W(tu_i)$, $t \in \mathbb{R}$, are precisely the Weyl operators of the site 0.

Proposition 2.13. Lean ✓

(a) A translation-invariant linear map is determined by the vectors Lu_i, $i \in \iota$.

(b) A translation-invariant linear map L is symplectic if and only if

$$\sigma(Lu_i, \kappa_{x*} Lu_j) = \delta_{x0}\, \Omega_{ij} \qquad \text{for all } i, j \in \iota,\ x \in G.$$

Proof. (a) Lu_i is the i-th column of the symbol (the vector whose value at $x \in G$ is $\hat{L}_{ji}$, read as a function on G), and the symbol in turn determines L by Theorem 1.3.

(b) $\sigma(Lu_i, \kappa_{x*} Lu_j)$ is the coefficient of $(\hat{L}^{\star\mathsf{T}}\Omega\hat{L})_{ij}$ at $-x$, as in the proof of Theorem 1.7; the condition says that this symbol is the constant Ω.

□

In terms of the automaton, part (b) says that the images $\alpha_L(W(tu_i)) = W(tLu_i)$ of the one-site Weyl operators, together with their translates, satisfy the same commutation relations as the one-site Weyl operators themselves.

2.5 Automorphisms of Weyl type

The automorphisms of $\mathrm{CCR}(V,\sigma)$ that send every Weyl operator to a multiple of a Weyl operator are characterized by means of a norm estimate. Let $\delta_0 \in \ell^2(V)$ be the unit vector at $0 \in V$; the functional $\varphi(a) = (a\delta_0)(0)$ satisfies $|\varphi(a)| \le \|a\|$, and $\varphi(W(u)) = 1$ if $u = 0$ and 0 otherwise, since $(W(u)\delta_0)(0) = \delta_0(u)$.

Lemma 2.14 (Separation)**.** Lean ✓ *If y lies in the closed linear span of $\{W(v) : v \neq u\}$, then $\|W(u) - y\| \ge 1$. In particular, $\mu W(u) = \nu W(w)$ with $\nu \neq 0$ implies $u = w$ and $\mu = \nu$.*

Proof. For $v \neq u$, $W(u)^*W(v)$ is a multiple of $W(v-u)$, so $\varphi(W(u)^*y') = 0$ for y' in the span and, by continuity, in its closure, while $\varphi(W(u)^*W(u)) = 1$. Hence $1 = |\varphi(W(u)^*(W(u) - y))| \le \|W(u) - y\|$. For the second statement apply $\varphi(W(w)^* \cdot)$ to both sides. □

Theorem 2.15. Lean ✓ *Let T be a $*$-automorphism of $\mathrm{CCR}(V,\sigma)$, $Q\colon V \to V$ a map and $c\colon V \to \mathbb{C}$ with $T(W(u)) = c(u)W(Qu)$ for all u. Then Q is a linear bijection with $\sigma(Qu,Qw) = \sigma(u,w)$, and c is a character: $c(u+w) = c(u)c(w)$, $|c(u)| = 1$.*

Proof. $c(u) \neq 0$ since T is injective, and $|c(u)| = 1$ since T is isometric and Weyl operators have norm 1.

Additivity. Applying T to $W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)/2}W(u+w)$ gives

$$c(u)c(w)\, e^{-\mathrm{i}\sigma(Qu,Qw)/2}\, W(Qu+Qw) = e^{-\mathrm{i}\sigma(u,w)/2}\, c(u+w)\, W(Q(u+w)),$$

so $Q(u+w) = Qu + Qw$ and the coefficients agree, by Lemma 2.14.

Symplecticity. Exchanging u and w and dividing gives $e^{-\mathrm{i}\sigma(Qu,Qw)} = e^{-\mathrm{i}\sigma(u,w)}$, so $\sigma(u,w) - \sigma(Qu,Qw) = 2\pi k(u,w)$ with $k(u,w) \in \mathbb{Z}$. By additivity, $m\,Q(u/m) = Qu$ for $m \in \mathbb{N}$, hence $k(u,w) = m\,k(u/m,w)$; so $k(u,w)$ is divisible by every m, and $k = 0$.

Surjectivity. If $v' \notin QV$, then T maps the span of the Weyl operators into the span of $\{W(v) : v \neq v'\}$. The Weyl operators span a dense subspace of $\mathrm{CCR}(V,\sigma)$ (their span is already a $*$-algebra), and T is continuous, so $W(v') = T(T^{-1}W(v'))$ lies in the closure of that span, contradicting Lemma 2.14.

Linearity. For $t \in \mathbb{R}$, $\sigma(Q(tu) - tQu, Qw) = \sigma(tu,w) - t\sigma(u,w) = 0$ for every w; since Q is onto and σ is nondegenerate (if $\sigma(u,w) = 0$ for all w, pairing with the one-site vectors $z^x e_j$ gives $\Omega^{\mathsf{T}}u(x) = 0$ for every x, so $u = 0$), $Q(tu) = tQu$. If $Qu = 0$ then $\sigma(u,w) = 0$ for all w, so $u = 0$.

Character. With symplecticity, the coefficient identity of the first step reads $c(u)c(w) = c(u+w)$. □

Nontrivial phases c do occur in Theorem 2.15. For any character c with $|c| = 1$, the unitary $(U\psi)(v) = \overline{c(v)}\,\psi(v)$ satisfies $UW(u)U^* = c(u)W(u)$, so that α_L followed by conjugation with U is an automorphism of the form of the theorem, with $Q = L$ and phase $c \circ L$. When T also commutes with the translations, comparing $T\alpha_a(W(u))$ with $\alpha_a T(W(u))$ shows that Q commutes with κ_{a*}, so that Q is itself a rule and $c \circ \kappa_{a*} = c$. Such translation-invariant characters do exist, for instance $c(u) = \exp\left(\mathrm{i}\sum_x \theta^{\mathsf{T}}u(x)\right)$ with $\theta \in \mathbb{R}^{2n}$, and conjugation with the corresponding U is a reversible quantum cellular automaton with neighbourhood $\{0\}$. The translation-covariant automata of Weyl type are therefore exactly the automorphisms α_L followed by such a conjugation.

2.6 The leapfrog and the direction of time

The Cauchy data (d_0, d_1) of the lattice Klein–Gordon field form exactly the one-mode lattice phase space over $\mathbb{Z}$, through $u(x) = (d_0(x), d_1(x))$, and their conserved form $\sum_x \left(d_0(x)d_1'(x) - d_1(x)d_0'(x)\right)$ is σ. The leapfrog step F of Example 1.4 advances the data from the slices $t, t+1$ to $t+1, t+2$, and it remains to determine which automaton represents time evolution itself.

Proposition 2.16. Lean ✓ *Let Λ_t send a solution φ of the lattice Klein–Gordon equation with finitely supported Cauchy data to its Cauchy data $(\varphi(t,\cdot), \varphi(t+1,\cdot))$, and let τ translate solutions forward in time by one step, $(\tau\varphi)(t,x) = \varphi(t-1,x)$. Then $\Lambda_t\tau = F^{-1}\Lambda_t$.*

Proof. $\Lambda_t \tau \varphi = (\varphi(t-1,\cdot), \varphi(t,\cdot)) = \Lambda_{t-1}\varphi$, and $\Lambda_t = F\Lambda_{t-1}$ by the construction of F in Example 1.4. Hence $\Lambda_{t-1} = F^{-1}\Lambda_t$. □

Translating a solution forward in time shifts its Cauchy data at a fixed time backwards, and on observables the consequence of this shift depends on how $W(u)$ is to be read. If $W(u)$ is thought of as $e^{\mathrm{i}\sigma(u,\Phi_t)}$, with Φ_t the Cauchy data of the quantum field at time t (a heuristic reading here, since $\ell^2(V)$ carries no field operators and $t \mapsto W(tu)$ is not continuous), then the field moves as $\Phi_{t+1} = F\Phi_t$, so that $\sigma(u, \Phi_{t+1}) = \sigma(F^{-1}u, \Phi_t)$; under this reading the forward time step is therefore the automaton $\alpha_{F^{-1}}$ of Example 2.9, rather than α_F.

2.7 Notes and references

1. *Definition.* Definition 2.6 corresponds to Definition 4.1 of [1, p. 73] for reversible automata, with the quasi-local algebra there replaced by a net; the condition $0 \in N$ imposed in the thesis is not needed here. Irreversible automata, that is, quantum channels, are treated in Chapter 6.

2. *Composition and local rule.* Proposition 2.7 corresponds to Corollary 4.3 of the thesis. Proposition 2.13(b) is the Gaussian analogue of Lemma 4.2(ii) rather than a special case of it: that lemma, like the results of [7], concerns quasi-local algebras built from finite-dimensional cells, whereas each cell here carries the Weyl algebra of $\mathbb{R}^{2n}$.

3. *Weyl-type automorphisms.* After (4.7), [1, p. 78] states that any Γ for which $T(W(\xi)) = W(\Gamma\xi)$ defines a homomorphism "would have to be linear ... and fulfill $\sigma(\Gamma\xi, \Gamma\eta) = \sigma(\xi, \eta)$". Theorem 2.15 proves this claim and additionally records the character c, which the thesis omits. Lemma 2.1, together with the uniqueness of the CCR algebra for nondegenerate σ (established in the proof of Theorem 2.15), of which $\mathrm{CCR}(V, \sigma)$ is a concrete realization, is due to Manuceau [10], Slawny [12], and Manuceau, Sirugue, Testard and Verbeure [11]; see [9, §5.2].

4. *Examples.* The thesis lists the free evolution, the left- and right-shifter, continued squeezing, and symplectic rotations as examples of Gaussian automata (p. 70); Example 2.12 corresponds to its system (4.18).

5. *Direction of time.* In the convention $T(W(\xi)) = W(\Gamma\xi)$ used in the thesis, the rule governing the forward time step of the lattice Klein–Gordon field is F^{-1}.

3 Gaussian states on the lattice

The automata of Chapter 2 act on observables, and a dynamics of observables becomes a dynamics of physical situations only once it acts on states. This chapter introduces the states of the lattice CCR algebra through their characteristic functions, singles out the Gaussian states, whose characteristic functions are exponentials of quadratic forms, and derives the condition that the quadratic form of a Gaussian state must satisfy. Because a Gaussian state is determined by a covariance form on the whole infinite lattice, whereas the classical theory of Gaussian states concerns covariance matrices of finitely many modes, a large part of the chapter is devoted to showing that the two descriptions agree: restricted to a finite region, a covariance form is a covariance matrix of finitely many modes, and the condition on the form becomes the familiar matrix inequality on every region. On the chain the condition can also be read off pointwise from a Fourier transform, which leads to a matrix condition that saturates the state condition at every wave number, and to an explicit law for the evolution of the covariance under a Gaussian automaton.

Two product states serve as examples throughout, the vacuum and the squeezed states, and the running example of the previous chapters returns at the end, where one step of the leapfrog is applied to the vacuum and the correlations it creates between neighbouring sites are computed exactly.

As in Chapter 2, G is an abelian group, the cell carries n modes with coordinates ordered positions first, $u(x) = (q_1, \dots, q_n, p_1, \dots, p_n)$, and $\sigma(u, w) = \sum_x u(x)^{\mathsf{T}} \Omega\, w(x)$ with $\Omega = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$, so that $\Omega^{\mathsf{T}} = -\Omega = \Omega^{-1}$. For a site x and a cell index i we write $e_{x,i} \in V$ for the vector whose value at x is the i-th unit vector and which vanishes elsewhere, so that $\sigma(e_{x,i}, e_{y,j}) = \delta_{xy}\Omega_{ij}$. Since products and adjoints of Weyl operators are multiples of Weyl operators, the linear span of the Weyl operators is a $*$-subalgebra of $\mathrm{CCR}(V, \sigma)$, and it is dense by the definition of $\mathrm{CCR}(V, \sigma)$ as its closure. For a complex vector or matrix, † denotes the conjugate transpose, and $A \geq 0$ means that the Hermitian matrix A is positive semidefinite.

3.1 States and characteristic functions

A state assigns to each observable its expectation value, and the only requirements on such an assignment are linearity, normalization and positivity on the squares a^*a, which are the observables with non-negative spectrum.

Definition 3.1 (State)**.** A *state* on $\mathrm{CCR}(V, \sigma)$ is a continuous $\mathbb{C}$-linear functional ω with $\omega(1) = 1$ and $\omega(a^*a) \geq 0$ for every a, where, for a complex number z, $z \geq 0$ means that z is real and non-negative.

Positivity is required of the complex numbers $\omega(a^*a)$, so that no order on the algebra itself needs to be introduced. Continuity follows from positivity on every C*-algebra, but it is included in the definition because it is the property the arguments below use directly.
Since the Weyl operators span a dense subalgebra, a continuous functional is known once its values on them are known, and these values form a function on the phase space. The products $W(u_j)^*W(u_k)$ are again multiples of Weyl operators, and positivity of the state on combinations of such products becomes a positivity property of this function.

Definition 3.2 (Characteristic function)**.** The *characteristic function* of a state ω is $\chi_\omega(u) = \omega(W(u))$. For a function $\chi\colon V \to \mathbb{C}$ and a family $(u_j)_{j \in I}$ in V indexed by a finite set I, the *twisted kernel* is the matrix

$$K_{jk} = e^{\mathrm{i}\sigma(u_j, u_k)/2}\, \chi(u_k - u_j) \qquad (j, k \in I).$$

Proposition 3.3 (Twisted positivity)**.** Lean ✓ *Let ω be a state with characteristic function χ.*

(a) *For every finite family* $(u_j)_{j\in I}$ *and all* $c \in \mathbb{C}^I$, $\sum_{j,k} \bar{c}_j c_k \, e^{\mathrm{i}\sigma(u_j,u_k)/2}\chi(u_k - u_j) \geq 0$, *so that the twisted kernel of* χ *on every finite family is positive semidefinite.*

(b) $\chi(0) = 1$, $\chi(-u) = \overline{\chi(u)}$ *and* $|\chi(u)| \leq 1$ *for every* $u \in V$.

Proof.

(a) By the Weyl relations, $W(u_j)^* W(u_k) = W(-u_j)W(u_k) = e^{\mathrm{i}\sigma(u_j,u_k)/2} W(u_k - u_j)$, so that for $X = \sum_k c_k W(u_k)$ the product X^*X equals $\sum_{j,k} \bar{c}_j c_k \, e^{\mathrm{i}\sigma(u_j,u_k)/2} W(u_k - u_j)$, and the sum in question is $\omega(X^*X) \geq 0$. A complex matrix K whose quadratic form $c^\dagger K c$ is real for every c is Hermitian, because the quadratic form of $K - K^\dagger$, which is $c^\dagger K c - \overline{c^\dagger K c}$, then vanishes identically, and a complex matrix with vanishing quadratic form is zero by polarization. The twisted kernel is therefore Hermitian with non-negative quadratic form, that is, positive semidefinite.

(b) First, $\chi(0) = \omega(1) = 1$. For the family $(0, u)$ the phases are trivial, because $\sigma(0,u) = \sigma(u,0) = 0$, and the twisted kernel is $\left(\begin{smallmatrix} 1 & \chi(u) \\ \chi(-u) & 1 \end{smallmatrix}\right)$. By (a) it is Hermitian, which gives $\chi(-u) = \overline{\chi(u)}$, and its determinant $1 - |\chi(u)|^2$ is non-negative.

□

Remark 3.4 (Twisted positive-definiteness). Part (a) says that χ is *σ-positive-definite*. For $\sigma = 0$ this is the positive-definiteness of Bochner's theorem, by which a continuous positive-definite function on a finite-dimensional real vector space with $\chi(0) = 1$ is the characteristic function of a probability measure, and the twisted condition is its analogue for the canonical commutation relations. The analogue of Bochner's theorem itself, for finitely many degrees of freedom, is the quantum Bochner theorem of Kastler [13] and of Loupias and Miracle-Sole [14]: a continuous σ-positive-definite function on a finite-dimensional symplectic space with $\chi(0) = 1$ is the characteristic function of a density operator in the Schrödinger representation, and its symplectic Fourier transform is the Wigner function of that density operator, just as the Fourier transform of χ is the probability density in the commutative case (statement recalled, not checked against the two sources). Both hypotheses of that theorem belong to finitely many degrees of freedom, continuity of χ and a Schrödinger representation in which the density operator lives. On the infinite lattice there is no such distinguished representation, and a state need not be given by a density operator in any fixed representation, as Remark 3.10 shows for the representation on $\ell^2(V)$. The result that replaces the quantum Bochner theorem there is the characterization of states by their characteristic functions quoted in that remark, which requires no continuity of χ. This Fourier transform acts on the phase-space variable u and is not used below; it is to be distinguished from the lattice Fourier transform $x \mapsto k$ of Subsection 3.5, which acts on the site variable of a vector.

Two properties of characteristic functions suffice for the evolution of states: a state is determined by its characteristic function, and pulling a state back along an automorphism that relabels the Weyl operators composes the characteristic function with the relabelling.

Proposition 3.5 (Pullback along automorphisms)**.** Lean ✓

(a) *Two states with the same characteristic function are equal.*

(b) *If* α *is a* $*$*-automorphism of* $\mathrm{CCR}(V,\sigma)$ *and* ω *a state, then* $\omega \circ \alpha$ *is a state. If in addition* $\alpha(W(u)) = W(Qu)$ *for a map* $Q\colon V \to V$, *then* $\chi_{\omega\circ\alpha} = \chi_\omega \circ Q$.

Proof.

(a) Two continuous linear functionals that agree on the Weyl operators agree on their linear span, and hence on its closure, which is the whole algebra.

(b) A $*$-automorphism of a C*-algebra is isometric, so that $\omega \circ \alpha$ is continuous as well as linear; it is normalized because $\alpha(1) = 1$, and positive because $\omega(\alpha(a^*a)) = \omega(\alpha(a)^*\alpha(a)) \geq 0$. Finally $\chi_{\omega\circ\alpha}(u) = \omega(W(Qu)) = \chi_\omega(Qu)$.

□

Corollary 3.6 (Translation invariance)**.** Lean ✓ *A state ω satisfies $\omega \circ \alpha_a = \omega$ for every $a \in G$ if and only if $\chi_\omega \circ \kappa_{a*} = \chi_\omega$ for every $a \in G$.*

Proof. By Proposition 3.5(b) with $Q = \kappa_{a*}$, the state $\omega \circ \alpha_a$ has the characteristic function $\chi_\omega \circ \kappa_{a*}$, and by Proposition 3.5(a) it equals ω exactly when the two characteristic functions agree. □

3.2 Gaussian states

In classical probability a Gaussian measure is one whose characteristic function is the exponential of a quadratic polynomial, and the quantum definition takes over this description word for word; on the CCR algebra these are the quasi-free states of Manuceau and Verbeure [15]. The factor $\frac{1}{4}$ in the exponent is the normalization in which the vacuum of a single oscillator has covariance 1, as Example 3.20 below shows; it agrees with the finite-mode convention in which the covariance matrix is twice the matrix of symmetrized second moments.

Definition 3.7 (Gaussian state)**.** A *Gaussian state* is a state ω together with a symmetric bilinear form γ on V, its *covariance form*, and a linear form d on V, its *displacement*, such that

$$\chi_\omega(u) = \exp\bigl(\mathrm{i}\, d(u) - \tfrac{1}{4}\gamma(u,u)\bigr) \qquad \text{for all } u \in V.$$

Every result below concerns a Gaussian state given together with its data (γ, d). That a state with prescribed data exists is not proved in these notes, and no result below depends on it; the examples of the chapter are accordingly covariance forms satisfying the state condition of Definition 3.11, which by Theorem 3.13 is necessary for a Gaussian state, and Remark 3.10 explains why existence cannot simply be read off from the construction of the algebra.

Remark 3.8 (The data are determined by the state)*.* Although the definition lists γ and d as data, a state admits at most one pair of them. Indeed, $|\chi_\omega(u)| = \exp(-\frac{1}{4}\gamma(u,u))$ determines $\gamma(u,u)$ for every u, and hence the symmetric form γ by polarization; then $\chi_\omega(tu)/|\chi_\omega(tu)| = e^{\mathrm{i}t\, d(u)}$ for all $t \in \mathbb{R}$, and two real numbers r, r' with $e^{\mathrm{i}tr} = e^{\mathrm{i}tr'}$ for every real t are equal, as one sees by taking $|t(r - r')| < 2\pi$ with $t \neq 0$, so that $d(u)$ is determined as well.

Remark 3.9 (Normalizations)*.* In the normalization of these notes $\chi_\omega(u) = \exp(\mathrm{i}d(u) - \frac{1}{4}\gamma(u,u))$, and for one mode the corresponding covariance matrix γ_B is twice the matrix of symmetrized second moments, as the paragraph before Example 3.21 shows. Other normalizations of the literature write the exponent with a different constant, for instance as $-\frac{1}{2}\gamma'(u,u)$, and then $\gamma = 2\gamma'$; such a rescaling of γ by a constant makes conditions such as $\sigma(u,w)^2 \leq \gamma(u,u)\gamma(w,w)$ acquire factors, here $\frac{1}{4}\sigma(u,w)^2 \leq \gamma'(u,u)\gamma'(w,w)$, without changing their content. The sign of $\mathrm{i}\Omega$ is likewise a matter of convention: for a real symmetric matrix γ_B of M modes the matrices $\gamma_B - \mathrm{i}\Omega_M$ and $\gamma_B + \mathrm{i}\Omega_M$, with Ω_M the symplectic matrix of Section 3.4, are complex conjugates of each other, and the complex conjugate of a positive semidefinite matrix is positive semidefinite, so that $\gamma_B - \mathrm{i}\Omega_M \geq 0$ and $\gamma_B + \mathrm{i}\Omega_M \geq 0$ are equivalent and either may be called the finite-mode condition.

Remark 3.10 (Existence and regularity)*.* The representation of $\mathrm{CCR}(V,\sigma)$ on $\ell^2(V)$ by which the algebra was constructed contains no Gaussian state. For $u \neq 0$ and $t \neq 0$, the vector $W(tu)\delta_v$ is a multiple of δ_{v-tu}, which is orthogonal to δ_v, so that $\|W(tu)\delta_v - \delta_v\| = \sqrt{2}$ and $t \mapsto W(tu)$ is not strongly continuous, and the representation is therefore not regular. Let $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ be a density operator on $\ell^2(V)$ and $u \neq 0$. Each ψ_i has countable support

$\operatorname{supp}\psi_i$, and $\langle\psi_i, W(tu)\psi_i\rangle = \sum_v \overline{\psi_i(v)}\, e^{\mathrm{i}\sigma(tu,v)/2}\psi_i(v+tu)$ vanishes unless $tu \in \operatorname{supp}\psi_i - \operatorname{supp}\psi_i$, which happens for only countably many t. Hence $\operatorname{tr}(\rho\, W(tu)) = 0$ for all but countably many $t \in \mathbb{R}$, whereas a Gaussian characteristic function vanishes nowhere, and no normal state of this representation is Gaussian.

That a Gaussian function with a covariance form satisfying the state condition nevertheless defines a state follows from the theorem that a function χ on V with $\chi(0) = 1$ and positive semidefinite twisted kernels is the characteristic function of a unique state on the CCR algebra [11], [9, §5.2]. Its proof passes through the GNS construction on the group algebra of V twisted by σ and through the uniqueness of the CCR algebra, and it is not reproduced here. The uniqueness of the CCR algebra, and with it the theorem, requires the form σ to be nondegenerate, and this is the one place where nondegeneracy enters the existence question: if $\sigma(u, w) = 0$ for every $w \in V$, then pairing with $w = e_{x,i}$ gives $u(x)^{\mathsf{T}}\Omega = 0$, hence $u(x) = 0$, at every site, so that $u = 0$. Continuity of χ, by contrast, is not needed. The elementary half, that the state condition makes the Gaussian twisted kernels positive semidefinite, is Remark 3.14.

3.3 The state condition and its necessity

For finitely many modes, a real symmetric matrix γ_B is the covariance matrix of a Gaussian state exactly when $\gamma_B - \mathrm{i}\Omega_M \geq 0$ (equivalently $\gamma_B + \mathrm{i}\Omega_M \geq 0$), a form of the uncertainty principle. On the lattice there is no matrix, but the condition can be imposed on every finite family of vectors, and on a real vector space it can be rewritten without complex numbers in two ways, the second of which makes it evident that the sign in front of $\mathrm{i}\sigma$ plays no role.

Definition 3.11 (State condition)**.** Let γ be a symmetric and σ an alternating bilinear form on a real vector space V. The pair (γ, σ) satisfies

(a) the *state condition* if for every finite family $(u_j)_{j\in I}$ in V the complex matrix $\big(\gamma(u_j, u_k) + \mathrm{i}\sigma(u_j, u_k)\big)_{j,k}$ is positive semidefinite;

(b) the state condition *in pair form* if $\gamma(a, a) + \gamma(b, b) \geq 2\sigma(a, b)$ for all $a, b \in V$;

(c) the state condition *in real-pair form* if $\gamma(u, u) \geq 0$ and $\sigma(u, w)^2 \leq \gamma(u, u)\,\gamma(w, w)$ for all $u, w \in V$.

Condition (a) can equally be read on complex vectors: if $\gamma_{\mathbb{C}}$ and $\sigma_{\mathbb{C}}$ denote the sesquilinear extensions to $V_{\mathbb{C}} = V \otimes \mathbb{C}$, then for $\mu = \sum_k c_k u_k$ the quadratic form of the matrix in (a) at c is $\gamma_{\mathbb{C}}(\mu, \mu) + \mathrm{i}\sigma_{\mathbb{C}}(\mu, \mu)$, and every $\mu = a + \mathrm{i}b$ arises in this way from the family (a, b) with $c = (1, \mathrm{i})$.

Proposition 3.12 (Equivalent forms)**.** Lean ✓ *For symmetric γ and alternating σ, the three conditions of Definition 3.11 are equivalent, and they are also equivalent to condition (a) with $\gamma - \mathrm{i}\sigma$ in place of $\gamma + \mathrm{i}\sigma$.*

Proof. The matrix in (a) is Hermitian, since γ is symmetric and σ antisymmetric, and we first compute its quadratic form. For the family (a, b) and $c = (1, \mathrm{i})$, the diagonal contributes $\gamma(a, a) + \gamma(b, b)$, since $\sigma(a, a) = \sigma(b, b) = 0$, while the off-diagonal terms contribute $\mathrm{i}\big(\gamma(a, b) - \gamma(b, a)\big) = 0$ from γ and $\mathrm{i}\cdot\mathrm{i}\big(\sigma(a, b) - \sigma(b, a)\big) = -2\sigma(a, b)$ from σ. For a general family and general coefficients, bilinearity reduces the computation to this case with $a = \sum_k (\operatorname{Re} c_k)u_k$ and $b = \sum_k (\operatorname{Im} c_k)u_k$, and therefore

$$\sum_{j,k} \bar{c}_j c_k\, (\gamma + \mathrm{i}\sigma)(u_j, u_k) = \gamma(a, a) + \gamma(b, b) - 2\sigma(a, b). \tag{3.1}$$

The quadratic forms in (a) are thus non-negative for all families and coefficients exactly when (b) holds, which proves (a)⇔(b).

Assume (b). Taking $b = 0$ gives $\gamma(a, a) \geq 0$. Given u, w and $s > 0$, (b) applied to $a = su$ and $b = \pm w/s$ gives $s^2\gamma(u, u) + s^{-2}\gamma(w, w) \geq 2|\sigma(u, w)|$. When $\gamma(u, u)$ and $\gamma(w, w)$ are both positive, the choice $s^2 = \sqrt{\gamma(w, w)/\gamma(u, u)}$ yields $\sqrt{\gamma(u, u)\gamma(w, w)} \geq |\sigma(u, w)|$, which is (c); when

$\gamma(u,u)=0$, the inequality $s^{-2}\gamma(w,w)\geq 2|\sigma(u,w)|$ holds for every $s>0$ and forces $\sigma(u,w)=0$ as $s\to\infty$, so that (c) holds in the form $0\leq 0$, and the case $\gamma(w,w)=0$ is the same with $s\to 0$. Conversely, (c) and the inequality between the geometric and the arithmetic mean give $2\sigma(a,b)\leq 2\sqrt{\gamma(a,a)\gamma(b,b)}\leq\gamma(a,a)+\gamma(b,b)$, which is (b).

Condition (c) does not change when σ is replaced by $-\sigma$, and condition (a) for $-\sigma$ is condition (a) with $\gamma-\mathrm{i}\sigma$, which proves the last statement. □

The necessity of the state condition is the statement that positivity of a Gaussian state forces the uncertainty relation on its covariance form. The argument below uses positivity only on elements of the form $\sum_k b_k\delta_k^{-1}(W(su_k)-1)$ and lets s tend to zero; the subtraction of the identity removes the constant term of the exponential, which would otherwise dominate, and the limit is controlled by an explicit bound, so that no derivatives are needed.

Theorem 3.13 (Necessity of the state condition)**.** Lean ✓ *The covariance form γ of every Gaussian state satisfies the state condition with σ, in each of the three equivalent forms of Definition 3.11.*

Proof. Let (ω,γ,d) be a Gaussian state, $u_1,\dots,u_J\in V$ and $b\in\mathbb{C}^J$, and put $H_{jk}=\frac{1}{2}(\gamma(u_j,u_k)+\mathrm{i}\sigma(u_j,u_k))$. The matrix H is Hermitian, so that $h=\sum_{j,k}\bar{b}_jb_kH_{jk}$ is real, and it has to be shown that $h\geq 0$.

For $s>0$ consider the family $v_0=0$ and $v_k=su_k$ for $1\leq k\leq J$, and put $\delta_0=1$ and $\delta_k=\exp(\mathrm{i}s\,d(u_k)-\frac{1}{4}s^2\gamma(u_k,u_k))$. Expanding $\gamma(v_k-v_j,v_k-v_j)=\gamma(v_j,v_j)+\gamma(v_k,v_k)-2\gamma(v_j,v_k)$ by symmetry and using the linearity of d, the twisted kernel of χ_ω on this family factorizes as

$$K_{jk}=e^{\mathrm{i}\sigma(v_j,v_k)/2}\exp\bigl(\mathrm{i}d(v_k-v_j)-\tfrac{1}{4}\gamma(v_k-v_j,v_k-v_j)\bigr)=\overline{\delta_j}\,\delta_k\,e^{s^2H_{jk}},$$

where $H_{0k}=H_{j0}=0$. We apply Proposition 3.3(a) with the coefficients $c_0=-B$, where $B=\sum_{k\geq1}b_k$, and $c_k=b_k/\delta_k$ for $k\geq1$, which is possible because $\delta_k\neq0$. Then $c_k\delta_k=b_k$ for $k\geq1$ and $c_0\delta_0=-B$, and the terms of the quadratic form with $j=k=0$, with $j=0<k$, with $k=0<j$ and with $j,k\geq1$ contribute $|B|^2$, $-|B|^2$, $-|B|^2$ and $\sum_{j,k\geq1}\bar{b}_jb_ke^{s^2H_{jk}}$ respectively. Since $\sum_{j,k\geq1}\bar{b}_jb_k=|B|^2$, this gives, with $\tau=s^2$,

$$0\leq\sum_{j,k=1}^{J}\bar{b}_jb_k(e^{\tau H_{jk}}-1)\qquad\text{for every }\tau>0,\tag{3.2}$$

which in terms of the algebra is $\omega(X^*X)\geq0$ for $X=\sum_k b_k\delta_k^{-1}(W(su_k)-1)$.

For a complex number z with $|z|\leq1$ one has $|e^z-1-z|\leq\sum_{r\geq2}|z|^r/r!\leq|z|^2$. Choose $\tau_0>0$ with $\tau_0|H_{jk}|\leq1$ for all j,k, and let $0<\tau\leq\tau_0$. Writing $e^{\tau H_{jk}}-1=\tau H_{jk}+\rho_{jk}$ with $|\rho_{jk}|\leq\tau^2|H_{jk}|^2$ and taking real parts in (3.2), we obtain

$$0\leq\tau h+\tau^2C,\qquad C=\sum_{j,k}|b_j|\,|b_k|\,|H_{jk}|^2,$$

so that $h\geq-\tau C$ for every $\tau\in(0,\tau_0]$, and hence $h\geq0$. Since $2h$ is the quadratic form of condition (a) of Definition 3.11, this proves (a), and the other two forms follow from Proposition 3.12. □

Remark 3.14 (The converse for twisted kernels). Conversely, if γ is symmetric and satisfies the state condition, and d is linear, then the Gaussian function $\chi(u)=\exp(\mathrm{i}d(u)-\frac{1}{4}\gamma(u,u))$ has positive semidefinite twisted kernels. By the factorization in the proof above with $s=1$, the kernel on a family (u_k) is $K=D^\dagger ED$ with $D=\mathrm{diag}(\delta_k)$ and $E_{jk}=e^{H_{jk}}$, and $E=\sum_{r\geq0}H^{\circ r}/r!$ is a convergent sum of entrywise powers of the positive semidefinite matrix H, each of which is positive semidefinite by the Schur product theorem. Together with the theorem cited in Remark 3.10, this yields a Gaussian state for every symmetric γ satisfying the state condition and every linear d.

3.4 Finite regions and the covariance matrix

A finite region $O \subseteq G$ carries $n_O = n|O|$ modes, and a Gaussian state of finitely many modes is described by a real $2n_O \times 2n_O$ covariance matrix γ_B, which is twice the matrix of symmetrized second moments of the canonical coordinates, ordered with all positions before all momenta. Such a matrix is called *admissible* if it is symmetric and $\gamma_B - \mathrm{i}\Omega_{n_O} \geq 0$, where $\Omega_M = \left(\begin{smallmatrix} 0 & I_M \\ -I_M & 0 \end{smallmatrix}\right)$ is the symplectic matrix of M modes in the same order. The covariance form of a lattice state, restricted to a finite region, is exactly such a covariance matrix, and the state condition on the region is exactly admissibility, once two differences of bookkeeping are accounted for: the site coordinates of a region come site by site, whereas the finite-mode convention puts all positions first, and the form γ is evaluated on the labels u of the Weyl operators $W(u)$, as a two-point function is evaluated on test functions, whereas γ_B is a matrix of moments of the canonical coordinates, the two being related by a conjugation with Ω.
We write $\mathrm{loc}(O)$ for the subspace of vectors supported in O, as in Chapter 2, and say that γ satisfies the state condition *on* O if the restrictions of γ and σ to $\mathrm{loc}(O)$ satisfy it; by Proposition 3.12 this may be taken in the real-pair form, $\gamma(u,u) \geq 0$ and $\sigma(u,w)^2 \leq \gamma(u,u)\gamma(w,w)$ for $u, w \in \mathrm{loc}(O)$.

Definition 3.15 (Covariance matrix of a region)**.** Let $O \subseteq G$ be finite and fix an enumeration $r\colon O \times \{1,\dots,n\} \to \{1,\dots,n_O\}$ of its modes. The *site vectors* of O in the finite-mode order are

$$f_{q_{r(x,i)}} = e_{x,i}, \qquad f_{p_{r(x,i)}} = e_{x,n+i} \qquad (x \in O,\ 1 \leq i \leq n),$$

so that the position coordinate of the mode $r(x,i)$ belongs to the i-th position of the site x, and its momentum coordinate to the i-th momentum. For a bilinear form β on V, the *restricted matrix* $R_O(\beta)$ is the real $2n_O \times 2n_O$ matrix with entries $\beta(f_a, f_b)$, rows and columns ordered $q_1,\dots,q_{n_O},p_1,\dots,p_{n_O}$. The *covariance matrix* of γ on O is

$$\mathrm{cov}_O(\gamma) = \Omega_{n_O}^{\mathsf{T}}\, R_O(\gamma)\, \Omega_{n_O}.$$

The matrix $R_O(\gamma)$ is the matrix of γ on the site vectors of O taken site by site, conjugated by the permutation matrix that puts positions first, and the conjugation by Ω_{n_O} is the passage from forms on labels to moments, $\gamma_B = \Omega^{\mathsf{T}}\gamma\,\Omega$, which the paragraph before Example 3.21 derives for one mode. A different enumeration of the modes conjugates $\mathrm{cov}_O(\gamma)$ by $P \oplus P$ for a permutation matrix P of the modes; since $(P\oplus P)\Omega_{n_O}(P\oplus P)^{\mathsf{T}} = \Omega_{n_O}$, this preserves admissibility, and none of the statements below depends on the enumeration.

Lemma 3.16 (Coordinates on a region)**.** Lean ✓ *Let O be finite.*

(a) $R_O(\sigma) = \Omega_{n_O}$, and $\Omega_{n_O}^{\mathsf{T}}\Omega_{n_O} = 1$.

(b) Every $u \in \mathrm{loc}(O)$ is $u = \sum_a \xi_a f_a$ for the coordinate vector $\xi \in \mathbb{R}^{2n_O}$ with ξ_a the corresponding entry $u(x)_i$, and for $w = \sum_a \eta_a f_a$ one has $\beta(u,w) = \xi^{\mathsf{T}} R_O(\beta)\,\eta$ for every bilinear form β.

(c) For $n = 1$ and a one-point region $\{x\}$, $\mathrm{cov}_{\{x\}}(\gamma) = \Omega^{\mathsf{T}} R_{\{x\}}(\gamma)\,\Omega$, where $R_{\{x\}}(\gamma)$ is the matrix of γ on the site vectors $e_{x,1}, e_{x,2}$; and $\Omega^{\mathsf{T}}\,\mathrm{diag}(\alpha,\beta)\,\Omega = \mathrm{diag}(\beta,\alpha)$.

Proof.

(a) Since $\sigma(e_{x,i}, e_{y,j}) = \delta_{xy}\Omega_{ij}$, the only nonzero values of σ on the site vectors are $\sigma(f_{q_r}, f_{p_r}) = 1$ and $\sigma(f_{p_r}, f_{q_r}) = -1$, which are the entries of Ω_{n_O}. The second statement is the block computation $\left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right) = 1$.

(b) The vectors f_a are the vectors $e_{x,i}$ with $x \in O$, each listed once, and these form a basis of $\mathrm{loc}(O)$, so that u is the sum of its entries times these vectors; the formula for β is bilinearity.

(c) For one site and one mode, $f_{q_1} = e_{x,1}$ and $f_{p_1} = e_{x,2}$, and $\Omega_1 = \Omega$. The identity is the product $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & \alpha \\ -\beta & 0 \end{smallmatrix}\right) = \operatorname{diag}(\beta, \alpha)$.

□

Proposition 3.17 (Restriction to a region)**.** Lean ✓ *Let γ be a symmetric bilinear form on V and O a finite region. Then γ satisfies the state condition on O if and only if $\operatorname{cov}_O(\gamma)$ is admissible. In particular, the covariance form of a Gaussian state has an admissible covariance matrix on every finite region.*

Proof. By Lemma 3.16(b), the coordinates identify $\operatorname{loc}(O)$ with $\mathbb{R}^{2n_O}$ and carry γ and σ to the forms of the matrices $R_O(\gamma)$ and $R_O(\sigma) = \Omega_{n_O}$. A complex combination of vectors of $\operatorname{loc}(O)$ has a complex coordinate vector ζ, and by (3.1) the quadratic form of condition (a) of Definition 3.11 at it is $\zeta^\dagger(R_O(\gamma) + \mathrm{i}\Omega_{n_O})\zeta$. The state condition on O is therefore equivalent to $R_O(\gamma) + \mathrm{i}\Omega_{n_O} \geq 0$ and, by the last statement of Proposition 3.12, to $R_O(\gamma) - \mathrm{i}\Omega_{n_O} \geq 0$. Since $R_O(\gamma)$ is symmetric, this says that $R_O(\gamma)$ is admissible.

It remains to pass from $R_O(\gamma)$ to $\operatorname{cov}_O(\gamma)$. If S is an invertible real matrix with $S\Omega_{n_O}S^{\mathsf{T}} = \Omega_{n_O}$, then $S(A - \mathrm{i}\Omega_{n_O})S^{\mathsf{T}} = SAS^{\mathsf{T}} - \mathrm{i}\Omega_{n_O}$, and since a congruence by an invertible matrix preserves positive semidefiniteness in both directions, A is admissible if and only if SAS^{T} is. The matrix $S = \Omega_{n_O}^{\mathsf{T}}$ is of this kind, because $\Omega_{n_O}^{\mathsf{T}}\Omega_{n_O}\Omega_{n_O} = \Omega_{n_O}$ by Lemma 3.16(a), and $SR_O(\gamma)S^{\mathsf{T}} = \operatorname{cov}_O(\gamma)$. The last statement follows from Theorem 3.13, since the state condition on V implies it on $\operatorname{loc}(O)$. □

The conjugation by Ω_{n_O} plays no role for admissibility, since $\Omega_{n_O}^{\mathsf{T}}$ preserves the symplectic matrix; it is there so that $\operatorname{cov}_O(\gamma)$ is the physical covariance matrix of the region and the finite-mode examples below mean the same states on both sides. The step from $\gamma + \mathrm{i}\sigma$ to $\gamma - \mathrm{i}\sigma$ is the complex conjugation of Proposition 3.12.
The familiar operations on covariance matrices of finitely many modes, the passage to a subsystem and the direct sum of independent systems, correspond on the lattice to the passage to a smaller region and to the union of two regions without correlations between them.

Corollary 3.18 (Sub-regions and unions)**.** Lean ✓ *Let γ be a symmetric bilinear form on V.*

(a) If γ satisfies the state condition on a finite region O, it satisfies it on every $O' \subseteq O$.

(b) Let O_1 and O_2 be disjoint finite regions with $\gamma(u, w) = 0$ for $u \in \operatorname{loc}(O_1)$ and $w \in \operatorname{loc}(O_2)$. Then γ satisfies the state condition on $O_1 \cup O_2$ if and only if it satisfies it on O_1 and on O_2.

Proof.

(a) The real-pair form on $\operatorname{loc}(O)$ implies it on the subspace $\operatorname{loc}(O') \subseteq \operatorname{loc}(O)$.

(b) The direction ($\Rightarrow$) is (a). Conversely, every $a \in \operatorname{loc}(O_1 \cup O_2)$ splits as $a = a_1 + a_2$ with $a_i \in \operatorname{loc}(O_i)$, and similarly $b = b_1 + b_2$. The cross terms of γ vanish by hypothesis and by symmetry, and those of σ vanish because $\sigma(u, w) = 0$ whenever u and w have disjoint supports. Hence

$$\gamma(a, a) + \gamma(b, b) - 2\sigma(a, b) = \sum_{i=1,2} \big(\gamma(a_i, a_i) + \gamma(b_i, b_i) - 2\sigma(a_i, b_i)\big),$$

and the pair form on O_1 and on O_2 gives it on $O_1 \cup O_2$; by Proposition 3.12, applied to the restrictions, the pair form is equivalent to the state condition.

□

In terms of covariance matrices, (a) says that $\mathrm{cov}_{O'}(\gamma)$, which is the submatrix of $\mathrm{cov}_O(\gamma)$ on the modes of O' when the enumerations are chosen compatibly, is admissible whenever $\mathrm{cov}_O(\gamma)$ is, and (b) says that under the hypothesis $\mathrm{cov}_O(\gamma)$ is, up to a permutation of modes, the direct sum of $\mathrm{cov}_{O_1}(\gamma)$ and $\mathrm{cov}_{O_2}(\gamma)$, which is admissible exactly when both summands are.

Remark 3.19 (Definiteness). If γ is symmetric and satisfies the state condition on a finite region O, then $\gamma(u,u) > 0$ for every nonzero $u \in \mathrm{loc}(O)$, so that $R_O(\gamma)$ and the congruent matrix $\mathrm{cov}_O(\gamma)$ are positive definite. Indeed, if $\gamma(u,u) = 0$, the real-pair form gives $\sigma(u,w) = 0$ for every $w \in \mathrm{loc}(O)$, and pairing with $w = e_{x,i}$ for $x \in O$ shows that $\Omega^{\mathsf{T}} u(x) = 0$, so that $u(x) = 0$ for every $x \in O$ and $u = 0$.

The two basic examples of the finite-mode theory, the vacuum and the squeezed states, are products over the sites. Each is given here by its covariance form, which by Theorem 3.13 and Proposition 3.17 is what a Gaussian state with these correlations must have.

Example 3.20 (The vacuum)**.** Lean ✓ The *vacuum form* is $\gamma(u,w) = \sum_x u(x)^{\mathsf{T}} w(x)$. It satisfies the state condition on V: since Ω is orthogonal,

$$\gamma(a,a) + \gamma(b,b) - 2\sigma(a,b) = \sum_x \bigl|a(x) - \Omega\, b(x)\bigr|^2 \geq 0,$$

which is the pair form. On every finite region $R_O(\gamma) = 1$, since the site vectors are orthonormal, and hence $\mathrm{cov}_O(\gamma) = \Omega_{n_O}^{\mathsf{T}} \Omega_{n_O} = 1$; for one mode and one site this is the covariance matrix $\mathrm{diag}(1,1)$ of the ground state of the harmonic oscillator in natural units, whose characteristic function is $\exp(-\tfrac{1}{4}|u|^2)$.

Let $n = 1$. On one mode, $W(u) = \exp\bigl(\mathrm{i}(u_p X - u_q P)\bigr)$ realizes the Weyl relations of these notes, and a centred Gaussian state with covariance matrix γ_B has $\chi(u) = \exp\bigl(-\tfrac{1}{4}(\Omega u)^{\mathsf{T}} \gamma_B (\Omega u)\bigr)$, since $\Omega u = (u_p, -u_q)$. Comparing with Definition 3.7 gives $\gamma = \Omega^{\mathsf{T}} \gamma_B \Omega$, or equivalently $\gamma_B = \Omega \gamma \Omega^{\mathsf{T}} = \Omega^{\mathsf{T}} \gamma\, \Omega$, the last because $\Omega^{\mathsf{T}} = -\Omega$. For the vacuum both sides are the identity, and it is the squeezed state that makes the conjugation visible: for $\gamma_B = \mathrm{diag}(\vartheta, 1/\vartheta)$ with $\vartheta > 0$ the characteristic function is $\exp\bigl(-\tfrac{1}{4}(u_q^2/\vartheta + \vartheta u_p^2)\bigr)$, so that $\gamma = \mathrm{diag}(1/\vartheta, \vartheta)$ on one site.

Example 3.21 (Squeezed product states)**.** Lean ✓ Let $n = 1$ and $\vartheta > 0$. The *squeezed form* is

$$\gamma(u,w) = \sum_x u(x)^{\mathsf{T}}\, \mathrm{diag}(1/\vartheta, \vartheta)\, w(x).$$

By Lemma 3.16(c) its covariance matrix on a one-point region is $\mathrm{diag}(\vartheta, 1/\vartheta)$. It satisfies the state condition on V, because for $a(x) = (q,p)$ and $b(x) = (q',p')$ the contribution of the site x to the pair form is

$$\frac{q^2 + q'^2}{\vartheta} + \vartheta(p^2 + p'^2) - 2(qp' - pq') = \Bigl(\frac{q}{\sqrt{\vartheta}} - \sqrt{\vartheta}\, p'\Bigr)^2 + \Bigl(\sqrt{\vartheta}\, p + \frac{q'}{\sqrt{\vartheta}}\Bigr)^2 \geq 0.$$

3.5 The Fourier criterion

From here on $G = \mathbb{Z}$. A translation-invariant covariance form is determined by the correlations between the site 0 and every other site, a family of $2n \times 2n$ matrices, just as a rule is determined by its coefficients. The Fourier transform then diagonalizes the translations, and the state condition, which involves arbitrarily large families of vectors, reduces to a condition on one $2n \times 2n$ Hermitian matrix at each wave number k.

Definition 3.22 (Cell coefficients and symbol)**.** A family $(\gamma_x)_{x \in \mathbb{Z}}$ of real $2n \times 2n$ matrices is *cell data* if it is absolutely summable, $\sum_x \|\gamma_x\| < \infty$, and symmetric, $\gamma_{-x} = \gamma_x^{\mathsf{T}}$. Its *covariance form* and its *symbol* are

$$\gamma(u,w) = \sum_{x,y \in \mathbb{Z}} u(x)^{\mathsf{T}} \gamma_{x-y}\, w(y), \qquad \hat{\gamma}(k) = \sum_{x \in \mathbb{Z}} \gamma_x\, e^{-\mathrm{i}kx} \quad (k \in \mathbb{R}).$$

The family has *range* $\Lambda \in \mathbb{N}$ if $\gamma_x = 0$ for $|x| > \Lambda$.

The sign in the exponent is that of the Fourier symbol of a rule, $\hat{L}(k) = \sum_g \ell_g e^{-\mathrm{i}kg}$ of the paragraph after Lemma 1.18; in this chapter the coefficients of a rule are indexed by $g \in \mathbb{Z}$, because the letter m is needed for the mass in Example 3.35. The cell data of this definition are not required to satisfy the state condition, and admissibility, which concerns covariance matrices of finite regions, is a different notion. The vacuum form of Example 3.20 has the cells $\gamma_0 = 1$ and $\gamma_x = 0$ for $x \neq 0$, so that $\hat{\gamma}(k) = 1$, and the squeezed form has $\gamma_0 = \operatorname{diag}(1/\vartheta, \vartheta)$ and $\gamma_x = 0$ otherwise; both have range 0.

Lemma 3.23 (Cell forms). Lean ✓ *Let (γ_x) be cell data.*

(a) The covariance form is a well-defined symmetric bilinear form on V, invariant under translations.

(b) The series for $\hat{\gamma}(k)$ converges absolutely, the function $\hat{\gamma}$ is continuous, and $\hat{\gamma}(k)$ is Hermitian with $\hat{\gamma}(-k) = \overline{\hat{\gamma}(k)}$.

Proof.

(a) The double sum is finite because u and w are finitely supported, and it is bilinear. It is symmetric because $\gamma(w,u) = \sum_{x,y} u(y)^{\mathsf{T}} \gamma_{x-y}^{\mathsf{T}} w(x) = \sum_{x,y} u(y)^{\mathsf{T}} \gamma_{y-x} w(x) = \gamma(u,w)$, and translation invariance follows from the substitution $x \mapsto x + a$, $y \mapsto y + a$.

(b) The terms are bounded by $\|\gamma_x\|$, so that the Weierstrass test gives absolute and uniform convergence of continuous functions, and hence continuity. Since the γ_x are real, $\hat{\gamma}(k)^\dagger = \sum_x \gamma_x^{\mathsf{T}} e^{\mathrm{i}kx} = \sum_x \gamma_{-x} e^{\mathrm{i}kx} = \hat{\gamma}(k)$ and $\overline{\hat{\gamma}(k)} = \sum_x \gamma_x e^{\mathrm{i}kx} = \hat{\gamma}(-k)$.

□

The convergence in (b) is uniform in k, and each term is 2π-periodic, so that $\hat{\gamma}$ is a continuous function on the circle.
The state condition concerns complex combinations of vectors, that is, finitely supported functions $\mu\colon \mathbb{Z} \to \mathbb{C}^{2n}$, with Fourier transform $\hat{\mu}(k) = \sum_x \mu(x) e^{-\mathrm{i}kx}$. By the discussion after Definition 3.11, (γ, σ) satisfies the state condition if and only if $Q(\mu) \geq 0$ for every such μ, where

$$Q(\mu) = \sum_{x,y \in \mathbb{Z}} \mu(x)^\dagger a_{x-y}\, \mu(y), \qquad a_g = \gamma_g + \mathrm{i}\delta_{g,0}\Omega,$$

and $\sum_g a_g e^{-\mathrm{i}kg} = \hat{\gamma}(k) + \mathrm{i}\Omega$.

Lemma 3.24 (Parseval identity). Lean ✓ *For cell data and every finitely supported $\mu\colon \mathbb{Z} \to \mathbb{C}^{2n}$,*

$$Q(\mu) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{\mu}(k)^\dagger \big(\hat{\gamma}(k) + \mathrm{i}\Omega\big) \hat{\mu}(k)\, dk.$$

Proof. For fixed k,

$$\hat{\mu}(k)^\dagger \big(\hat{\gamma}(k) + \mathrm{i}\Omega\big) \hat{\mu}(k) = \sum_{x,y} \sum_g \mu(x)^\dagger a_g \mu(y)\, e^{\mathrm{i}k(x-y-g)},$$

a finite sum over x, y of series in g that converge uniformly in k, the terms being bounded by $|\mu(x)|\,|\mu(y)|\,\|a_g\|$. The series may therefore be integrated term by term, and since $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\mathrm{i}kj}\, dk = \delta_{j,0}$ for $j \in \mathbb{Z}$, exactly the terms with $g = x - y$ remain, whose sum is $Q(\mu)$. □

Theorem 3.25 (Fourier criterion). Lean ✓ *Let (γ_x) be cell data with covariance form γ.*

(a) If $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$ for every $k \in \mathbb{R}$, then (γ, σ) satisfies the state condition.

(b) If (γ_x) has finite range and (γ, σ) satisfies the state condition, then $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$ for every $k \in \mathbb{R}$.

Proof.

(a) The integrand in Lemma 3.24 is non-negative for every k, hence so is $Q(\mu)$.

(b) Suppose that $v^\dagger(\hat\gamma(k_0)+\mathrm{i}\Omega)v = -\varepsilon < 0$ for some k_0 and $v \in \mathbb{C}^{2n}$, and let Λ be the range. For $N > \Lambda$ let $\mu_N(x) = e^{\mathrm{i}k_0 x}v$ for $0 \le x < N$ and $\mu_N(x) = 0$ otherwise. The number of pairs $(x,y) \in \{0,\dots,N-1\}^2$ with $x-y=g$ is $N-|g|$ for $|g|<N$, and $a_g = 0$ for $|g|>\Lambda$, so that

$$Q(\mu_N) = \sum_{|g|\le\Lambda} (N-|g|)\, e^{-\mathrm{i}k_0 g}\, v^\dagger a_g v = -N\varepsilon - \sum_{|g|\le\Lambda} |g|\, e^{-\mathrm{i}k_0 g}\, v^\dagger a_g v.$$

The last sum is bounded in absolute value by $C = \sum_{|g|\le\Lambda} |g|\,|v^\dagger a_g v|$, which does not depend on N, and for $N > \max(\Lambda, C/\varepsilon)$ we obtain $Q(\mu_N) \le -N\varepsilon + C < 0$, contradicting the state condition.

□

Remark 3.26 (Necessity for all cell data). The hypothesis of finite range in part (b) can be dropped: for all cell data, the state condition holds if and only if $\hat\gamma(k)+\mathrm{i}\Omega \ge 0$ for every $k \in \mathbb{R}$. Only necessity needs proof. Assume the state condition, fix $k_0 \in \mathbb{R}$ and $v \in \mathbb{C}^{2n}$, and define μ_N as in the proof of (b) for every $N \ge 1$. The state condition gives $Q(\mu_N) \ge 0$ for every N, and counting the pairs (x,y) with $x-y=g$ as there, now without any restriction on g,

$$\frac{Q(\mu_N)}{N} = \sum_{|g|<N} \Big(1 - \frac{|g|}{N}\Big) e^{-\mathrm{i}k_0 g}\, v^\dagger a_g v.$$

The g-th term tends to $e^{-\mathrm{i}k_0 g}\, v^\dagger a_g v$ as $N \to \infty$ and is bounded in absolute value by $|v^\dagger a_g v| \le |v|^2 \|a_g\|$, which is summable over g because the cells are absolutely summable. By dominated convergence for series, $Q(\mu_N)/N$ tends to $\sum_g e^{-\mathrm{i}k_0 g}\, v^\dagger a_g v = v^\dagger(\hat\gamma(k_0)+\mathrm{i}\Omega)v$, which is therefore a limit of non-negative numbers and non-negative. As v and k_0 were arbitrary, $\hat\gamma(k)+\mathrm{i}\Omega \ge 0$ for every k, and together with part (a) the criterion is an equivalence for all cell data.

Remark 3.27 (The other sign). By Lemma 3.23(b), $\hat\gamma(k) - \mathrm{i}\Omega = \overline{\hat\gamma(-k)+\mathrm{i}\Omega}$, and the complex conjugate of a positive semidefinite matrix is positive semidefinite. If $\hat\gamma(k)+\mathrm{i}\Omega \ge 0$ for every k, therefore, both $\hat\gamma(k) \pm \mathrm{i}\Omega$ are positive semidefinite at every k, and so is their average $\hat\gamma(k)$.

3.6 Saturation of the state condition

Purity of a state is not defined in these notes, and nothing in this subsection proves that a state is pure. What is proved is the equivalence of three matrix conditions on the symbol which, by analogy with finitely many modes, are candidates for the characterization of the pure translation-invariant Gaussian states.

A Gaussian state of finitely many modes is pure exactly when its covariance matrix saturates the uncertainty relation, which for one mode means $\det\gamma_B = 1$. For a translation-invariant state on the chain with one mode per site, the Fourier criterion suggests that saturation should be read at every wave number, where it can be expressed in three ways, namely that $\hat\gamma(k)+\mathrm{i}\Omega$ is singular, that $\det\hat\gamma(k) = 1$, and that $\Omega\hat\gamma(k)$ squares to -1, as $\Omega\gamma_B$ does for a pure state of finitely many modes. For a single 2×2 matrix the first is weaker than the others unless both signs of $\mathrm{i}\Omega$ are taken into account.

Lemma 3.28 (Two-by-two matrices). Lean ✓ *Let $n=1$ and let $A = \left(\begin{smallmatrix} a & c \\ \bar c & b \end{smallmatrix}\right)$ be a Hermitian 2×2 matrix with $A+\mathrm{i}\Omega \ge 0$ and $A-\mathrm{i}\Omega \ge 0$. Then $\det A \ge 1 + 2|\operatorname{Im} c|$, and the following are equivalent:*

(a) $(\Omega A)^2 = -1$;

(b) $\det A = 1$;

(c) $A + \mathrm{i}\Omega$ *and* $A - \mathrm{i}\Omega$ *are both singular.*

Proof. Since $A \pm \mathrm{i}\Omega = \left(\begin{smallmatrix} a & c \pm \mathrm{i} \\ \bar{c} \mp \mathrm{i} & b \end{smallmatrix}\right)$ and $(c \pm \mathrm{i})(\bar{c} \mp \mathrm{i}) = |c|^2 + 1 \pm 2\operatorname{Im} c$,

$$\det(A \pm \mathrm{i}\Omega) = \det A - 1 \mp 2\operatorname{Im} c, \tag{3.3}$$

and both determinants are non-negative by hypothesis, which gives the bound.

By (3.3), condition (c) says $\det A = 1 + 2\operatorname{Im} c = 1 - 2\operatorname{Im} c$, that is, $\operatorname{Im} c = 0$ and $\det A = 1$, so that (c) implies (b); conversely, $\det A = 1$ forces $\operatorname{Im} c = 0$ by the bound, and then both determinants vanish, which proves (b)⇔(c).

For (a)⇔(b), put $M = \Omega A = \left(\begin{smallmatrix} \bar{c} & b \\ -a & -c \end{smallmatrix}\right)$, so that $\operatorname{tr} M = -2\mathrm{i}\operatorname{Im} c$ and $\det M = \det A$; by the Cayley–Hamilton theorem, $M^2 = (\operatorname{tr} M)M - (\det M)1$. If (b) holds, then $\operatorname{Im} c = 0$, as just shown, so that $\operatorname{tr} M = 0$ and $M^2 = -1$. Suppose conversely that $M^2 = -1$. If $\operatorname{tr} M \neq 0$, the relation $(\operatorname{tr} M)M = (\det M - 1)1$ makes M a scalar $\lambda 1$ with $\lambda^2 = -1$, and $A = \Omega^{-1}M = -\lambda\Omega$; for $\lambda = \mathrm{i}$ this gives $A - \mathrm{i}\Omega = -2\mathrm{i}\Omega$, and for $\lambda = -\mathrm{i}$ it gives $A + \mathrm{i}\Omega = 2\mathrm{i}\Omega$, and either matrix has the eigenvalues 2 and -2, contrary to the hypothesis. Hence $\operatorname{tr} M = 0$, and $M^2 = -(\det M)1 = -1$ gives $\det A = 1$. □

Both signs are needed for (b)⇒(a): the matrix $A = \left(\begin{smallmatrix} \sqrt{2} & -\mathrm{i} \\ \mathrm{i} & \sqrt{2} \end{smallmatrix}\right)$ has $\det A = 1$ and $A + \mathrm{i}\Omega = \sqrt{2} \cdot 1 \geq 0$, but $(\Omega A)^2 = \left(\begin{smallmatrix} -3 & 2\sqrt{2}\,\mathrm{i} \\ -2\sqrt{2}\,\mathrm{i} & -3 \end{smallmatrix}\right)$. The example does not contradict the lemma, because it violates its hypothesis: $A - \mathrm{i}\Omega = \left(\begin{smallmatrix} \sqrt{2} & -2\mathrm{i} \\ 2\mathrm{i} & \sqrt{2} \end{smallmatrix}\right)$ has determinant -2 and is therefore not positive semidefinite. For a real matrix A the two hypotheses coincide, and for the symbol of cell data they coincide by Remark 3.27.

Corollary 3.29 (Saturation at every wave number)**.** Lean ✓ *Let* $n = 1$ *and let* (γ_x) *be cell data with* $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$ *for every* k*. The following are equivalent:*

(a) $\hat{\gamma}(k) + \mathrm{i}\Omega$ *is singular for every* k*;*

(b) $\det \hat{\gamma}(k) = 1$ *for every* k*;*

(c) $(\Omega\hat{\gamma}(k))^2 = -1$ *for every* k*.*

Proof. By Remark 3.27, Lemma 3.28 applies to $A = \hat{\gamma}(k)$ for every k, and it gives (b)⇔(c) at each k as well as (b)⇒(a). For (a)⇒(b), note that $\hat{\gamma}(-k) + \mathrm{i}\Omega = \overline{\hat{\gamma}(k) - \mathrm{i}\Omega}$, so that (a) at $-k$ says that $\hat{\gamma}(k) - \mathrm{i}\Omega$ is singular; both $\hat{\gamma}(k) \pm \mathrm{i}\Omega$ are then singular, and Lemma 3.28 gives $\det \hat{\gamma}(k) = 1$. □

The equivalence holds for the conditions over all k together and not wave number by wave number: at a single k, singularity of $\hat{\gamma}(k) + \mathrm{i}\Omega$ does not imply $\det \hat{\gamma}(k) = 1$ when $\operatorname{Im} \hat{\gamma}(k)_{12} > 0$, which can happen when the cells γ_x with $x \neq 0$ are not symmetric matrices. For example, the cell data $\gamma_0 = 2 \cdot 1$, $\gamma_1 = \frac{1}{2}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\gamma_{-1} = \gamma_1^{\mathsf{T}}$ and $\gamma_x = 0$ otherwise have the symbol $\hat{\gamma}(k) = \left(\begin{smallmatrix} 2 & \mathrm{i}\sin k \\ -\mathrm{i}\sin k & 2 \end{smallmatrix}\right)$, so that $\det(\hat{\gamma}(k) + \mathrm{i}\Omega) = 4 - (1 + \sin k)^2 \geq 0$ and, the diagonal being positive, $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$ for every k; at $k = \pi/2$ this matrix is singular, whereas $\det \hat{\gamma}(\pi/2) = 3$.
Both product examples satisfy the conditions of Corollary 3.29. For the vacuum, $1 + \mathrm{i}\Omega$ has the eigenvalues 0 and 2 and $(\Omega \cdot 1)^2 = \Omega^2 = -1$; the squeezed form has the constant symbol $\operatorname{diag}(1/\vartheta, \vartheta)$, whose determinant is 1.

3.7 Evolution

A symplectic rule acts on states by pullback along its automorphism, and on the data of a Gaussian state by composition with the rule, so that Gaussian states evolve into Gaussian states. On a finite region an on-site rule acts on the covariance matrix by the symplectic congruence familiar from finitely many modes, and on the chain the evolution of the cell data is a congruence of the symbol at each wave number. Until the symbol is computed, G is again an arbitrary abelian group.

Proposition 3.30 (Evolved states). Lean ✓ *Let L be a symplectic rule and ω a state.*

(a) $\omega \circ \alpha_L$ is a state with $\chi_{\omega\circ\alpha_L} = \chi_\omega \circ L$.

(b) If (ω, γ, d) is a Gaussian state, then so is $(\omega \circ \alpha_L,\ \gamma(L\cdot, L\cdot),\ d \circ L)$.

(c) If ω is translation invariant, so is $\omega \circ \alpha_L$.

Proof.

(a) Proposition 3.5(b) with $\alpha = \alpha_L$ and $Q = L$.

(b) Since L is linear, $\gamma(L\cdot, L\cdot)$ is a symmetric bilinear form and $d \circ L$ a linear form, and by (a) $\chi_{\omega\circ\alpha_L}(u) = \exp\big(\mathrm{i}d(Lu) - \frac{1}{4}\gamma(Lu, Lu)\big)$.

(c) By Proposition 2.4(c), $\alpha_L\alpha_a = \alpha_a\alpha_L$, so that $(\omega \circ \alpha_L) \circ \alpha_a = (\omega \circ \alpha_a) \circ \alpha_L = \omega \circ \alpha_L$.

□

The automorphisms of rules compose as the rules do, so that iterating an automaton on a state amounts to multiplying symbols.

Proposition 3.31 (Composition of rules). Lean ✓ *The symplectic symbols form a submonoid of the matrices over the group algebra $\mathbb{R}[G]$, and $\hat{L} \mapsto \alpha_L$ is a monoid homomorphism from it into the $*$-automorphisms of $\mathrm{CCR}(V, \sigma)$ under composition, which restricts to a group homomorphism on the units.*

Proof. The identity is symplectic, and if $\hat{L}^{\star\mathsf{T}}\Omega\hat{L} = \Omega$ and $\hat{N}^{\star\mathsf{T}}\Omega\hat{N} = \Omega$, then, since $\hat{M} \mapsto \hat{M}^{\star\mathsf{T}}$ reverses products (the involution is a ring homomorphism of the commutative ring $\mathbb{R}[G]$, and transposition reverses products), $(\hat{L}\hat{N})^{\star\mathsf{T}}\Omega\hat{L}\hat{N} = \hat{N}^{\star\mathsf{T}}(\hat{L}^{\star\mathsf{T}}\Omega\hat{L})\hat{N} = \Omega$. The homomorphism property is Proposition 2.4(b), and a monoid homomorphism maps units to units. □

By Corollary 1.11, every symplectic symbol is invertible, with a symplectic inverse, so that the submonoid of symplectic symbols is a group and the statement about units applies to all of it. On a finite region, the simplest rules are those acting at each site by the same symplectic matrix, and for them the evolution of the covariance matrix can be written down explicitly.

Corollary 3.32 (On-site evolution). Lean ✓ *Let S be a real $2n \times 2n$ matrix with $S^{\mathsf{T}}\Omega S = \Omega$, let L_S be the on-site rule $(L_S u)(x) = Su(x)$, and let O be a finite region. For every bilinear form γ,*

$$\mathrm{cov}_O\big(\gamma(L_S\cdot, L_S\cdot)\big) = T_O\ \mathrm{cov}_O(\gamma)\, T_O^{\mathsf{T}},$$

where T_O is the matrix of $u \mapsto S^{-1}u$ applied at each site of O, written in the coordinates of Definition 3.15. If γ is symmetric and satisfies the state condition on O, so does $\gamma(L_S\cdot, L_S\cdot)$.

Proof. Let S_O be the matrix of L_S on $\mathrm{loc}(O)$ in the coordinates of Lemma 3.16(b), so that $R_O(\gamma(L_S\cdot, L_S\cdot)) = S_O^{\mathsf{T}}R_O(\gamma)S_O$ and $S_O^{\mathsf{T}}\Omega_{n_O}S_O = \Omega_{n_O}$, the latter because L_S preserves σ and $R_O(\sigma) = \Omega_{n_O}$. From $S^{\mathsf{T}}\Omega S = \Omega$ and $\Omega\Omega^{\mathsf{T}} = 1$ it follows that $\Omega^{\mathsf{T}}S^{\mathsf{T}}\Omega = S^{-1}$, and site by site the same computation gives $\Omega_{n_O}^{\mathsf{T}}S_O^{\mathsf{T}}\Omega_{n_O} = S_O^{-1} = T_O$. Inserting $\Omega_{n_O}\Omega_{n_O}^{\mathsf{T}} = 1$ twice,

$$\Omega_{n_O}^{\mathsf{T}}S_O^{\mathsf{T}}R_O(\gamma)S_O\Omega_{n_O} = (\Omega_{n_O}^{\mathsf{T}}S_O^{\mathsf{T}}\Omega_{n_O})(\Omega_{n_O}^{\mathsf{T}}R_O(\gamma)\Omega_{n_O})(\Omega_{n_O}^{\mathsf{T}}S_O\Omega_{n_O}) = T_O\ \mathrm{cov}_O(\gamma)\, T_O^{\mathsf{T}},$$

which is the formula. For the last statement, L_S maps $\mathrm{loc}(O)$ into itself and preserves σ, so that for $u, w \in \mathrm{loc}(O)$ the real-pair form for γ at $L_S u, L_S w$ gives $\gamma(L_S u, L_S u) \geq 0$ and $\sigma(u, w)^2 = \sigma(L_S u, L_S w)^2 \leq \gamma(L_S u, L_S u)\gamma(L_S w, L_S w)$. □

The covariance matrix transforms with S^{-1} and not with S. The state $\omega \circ \alpha_{L_S}$ is the evolution in the Heisenberg picture, in which the Weyl operators are moved by S, and the moments of the canonical coordinates are then moved by the inverse. The two congruences agree only in special cases, for instance for a rotation acting on the vacuum, where both leave the identity unchanged, and in general they differ: for one site and one mode, $S = \mathrm{diag}(2, 1/2)$ takes the vacuum covariance 1 to $S^{-1}S^{-\mathsf{T}} = \mathrm{diag}(1/4, 4)$, whereas $SS^{\mathsf{T}} = \mathrm{diag}(4, 1/4)$; and a rotation S by an angle φ with $\sin 2\varphi \neq 0$, acting on the squeezed covariance $\mathrm{diag}(\vartheta, 1/\vartheta)$ with $\vartheta \neq 1$, gives two matrices whose off-diagonal entries have opposite signs. By Proposition 3.17, the admissibility of $T_O \operatorname{cov}_O(\gamma) T_O^{\mathsf{T}}$ is the familiar invariance of admissibility under symplectic congruences.

Example 3.33 (On-site rotations). Lean ✓ Let $n = 1$ and let R_φ be the rotation of Example 2.11; the letter R here denotes a rotation matrix and not a restricted matrix R_O. Since $R_\varphi^{\mathsf{T}} R_\varphi = 1$, the vacuum form is invariant under the on-site rule, $\gamma(L_{R_\varphi} u, L_{R_\varphi} w) = \gamma(u, w)$. By Proposition 3.30(b), the free evolution of uncoupled oscillators therefore takes a Gaussian state with the vacuum covariance to one with the same covariance and a rotated displacement; on a finite region this is Corollary 3.32 with $T_O \cdot 1 \cdot T_O^{\mathsf{T}} = 1$.

On the chain, the covariance form of translation-invariant data evolves into another such form, and its cells and symbol can be computed from those of the rule.

Theorem 3.34 (Evolution of the covariance). Lean ✓ *Let $G = \mathbb{Z}$, let L be a rule with coefficients ℓ_g, finitely many nonzero, and let (γ_x) be cell data with covariance form γ. Put*

$$\gamma'_j = \sum_{g,g'} \ell_g^{\mathsf{T}} \, \gamma_{j+g-g'} \, \ell_{g'} \qquad (j \in \mathbb{Z}).$$

Then (γ'_j) is cell data whose covariance form is $\gamma(L\cdot, L\cdot)$, and

$$\hat\gamma'(k) = \hat L(-k)^{\mathsf{T}} \, \hat\gamma(k) \, \hat L(k) = \hat L(k)^{\dagger} \, \hat\gamma(k) \, \hat L(k).$$

If L is reflection symmetric, $\ell_{-g} = \ell_g$ for all g, then $\hat L(-k) = \hat L(k)$ and $\hat\gamma'(k) = \hat L(k)^{\mathsf{T}} \hat\gamma(k) \hat L(k)$.

Proof. The cells γ'_j are finite combinations of shifts of (γ_x) and are therefore absolutely summable. They are symmetric, since $(\gamma'_j)^{\mathsf{T}} = \sum_{g,g'} \ell_{g'}^{\mathsf{T}} \gamma_{-j-g+g'} \ell_g$, which after exchanging the names g and g' is γ'_{-j}. For the form,

$$\gamma(Lu, Lw) = \sum_{x,y} \sum_{g,g'} u(x-g)^{\mathsf{T}} \ell_g^{\mathsf{T}} \gamma_{x-y} \ell_{g'} w(y-g'),$$

and the substitution $x' = x - g$, $y' = y - g'$ turns γ_{x-y} into $\gamma_{x'-y'+g-g'}$, so that the inner sum over g, g' is $\gamma'_{x'-y'}$. For the symbol, the substitution $p = j + g - g'$ gives $e^{-\mathrm{i}kj} = e^{\mathrm{i}kg} e^{-\mathrm{i}kp} e^{-\mathrm{i}kg'}$, and since the sums over g, g' are finite and the sum over p converges absolutely, the series may be rearranged into $(\sum_g \ell_g^{\mathsf{T}} e^{\mathrm{i}kg}) \hat\gamma(k) (\sum_{g'} \ell_{g'} e^{-\mathrm{i}kg'}) = \hat L(-k)^{\mathsf{T}} \hat\gamma(k) \hat L(k)$. As the ℓ_g are real, $\hat L(-k)^{\mathsf{T}} = \overline{\hat L(k)}^{\mathsf{T}} = \hat L(k)^{\dagger}$, and the last statement is immediate. □

The second form shows that the evolution is a congruence at each wave number. If L is symplectic, then $\hat L(k)^{\dagger} \Omega \hat L(k) = \Omega$ by Example 1.19, so that $\hat\gamma'(k) + \mathrm{i}\Omega = \hat L(k)^{\dagger} (\hat\gamma(k) + \mathrm{i}\Omega) \hat L(k)$, and the evolution preserves the Fourier criterion, in accordance with Proposition 3.30(b) and Theorem 3.13. The formula with $\hat L(-k)^{\mathsf{T}} = \hat L(k)^{\dagger}$ is the general form of the evolution law, and without reflection symmetry the transpose alone cannot replace the conjugate transpose: for the rule with $\ell_0 = 1$, $\ell_1 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and all other coefficients zero, the vacuum symbol is taken to $\hat L(k)^{\dagger} \hat L(k) = \left(\begin{smallmatrix} 1 & e^{-\mathrm{i}k} \\ e^{\mathrm{i}k} & 2 \end{smallmatrix}\right)$, whereas $\hat L(k)^{\mathsf{T}} \hat L(k) = \left(\begin{smallmatrix} 1 & e^{-\mathrm{i}k} \\ e^{-\mathrm{i}k} & 1 + e^{-2\mathrm{i}k} \end{smallmatrix}\right)$ is not even Hermitian.

Example 3.35 (The leapfrog applied to the vacuum). script ✓ The leapfrog F of Example 1.4 has the coefficients $\ell_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & -m^2 \end{smallmatrix}\right)$ and $\ell_{\pm 1} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, so that it is reflection symmetric, with $\hat{F}(k) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & c_k \end{smallmatrix}\right)$ and $c_k = 2\cos k - m^2$, where m is the mass. By Proposition 2.16, the rule that moves Cauchy data forward is F^{-1}, and reading $\alpha_{F^{-1}}$ as the forward time step of the lattice field is the heuristic reading that follows that proposition. The rule F^{-1} has the symbol $\hat{F}^{-1}(k) = \left(\begin{smallmatrix} c_k & -1 \\ 1 & 0 \end{smallmatrix}\right)$, that is, the coefficients $\ell_0 = \left(\begin{smallmatrix} -m^2 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\ell_{\pm 1} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, and it is again reflection symmetric. By Theorem 3.34, one forward step takes the vacuum symbol $\hat{\gamma}(k) = 1$ to

$$\hat{\gamma}'(k) = \hat{F}^{-1}(k)^{\mathsf{T}} \hat{F}^{-1}(k) = \begin{pmatrix} 1 + c_k^2 & -c_k \\ -c_k & 1 \end{pmatrix},$$

whose cells are

$$\gamma'_0 = \begin{pmatrix} 3 + m^4 & m^2 \\ m^2 & 1 \end{pmatrix}, \qquad \gamma'_{\pm 1} = \begin{pmatrix} -2m^2 & -1 \\ -1 & 0 \end{pmatrix}, \qquad \gamma'_{\pm 2} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

and zero otherwise, as the formula $\gamma'_j = \sum_g \ell_g^{\mathsf{T}} \ell_{g+j}$ for the coefficients of F^{-1} confirms directly together with $1 + c_k^2 = 3 + m^4 - 4m^2 \cos k + 2\cos 2k$. Since $\det \hat{\gamma}'(k) = 1$ for every k, the evolved covariance again satisfies the equivalent conditions of Corollary 3.29, while its correlations, absent in the vacuum, now reach to the second neighbour: a single step of range 1 creates correlations of range 2, because the covariance is quadratic in the rule. The backward step α_F gives in the same way $\hat{F}(k)^{\mathsf{T}} \hat{F}(k) = \left(\begin{smallmatrix} 1 & -c_k \\ -c_k & 1 + c_k^2 \end{smallmatrix}\right)$, the same matrix with its two diagonal entries exchanged.

3.8 Notes and references

1. *States and Gaussian states.* In [1, p. 77], states are positive, normalized linear functionals, equivalently described by their characteristic functions, and a state is Gaussian when its characteristic function is $\exp\bigl(-\frac{1}{4}\gamma(\xi,\xi) + \mathrm{i}\sum_x \xi_x^{\mathsf{T}} d_x\bigr)$, with the same factor $\frac{1}{4}$ as Definition 3.7. The thesis does not discuss the existence of Gaussian states on the lattice CCR algebra; Remarks 3.10 and 3.14 record what is involved.

2. *State condition.* Condition (a) of Definition 3.11 is the thesis's (4.5), $\gamma(\bar{\mu}, \mu) + \mathrm{i}\sigma(\bar{\mu}, \mu) \geq 0$, and Q in Subsection 3.5 is its form (4.6) [1, pp. 77–78]. The thesis obtains the necessity of the state condition "by a direct generalization of the argument leading to (2.22)"; the proof of Theorem 3.13 is a different, derivative-free argument. The thesis works throughout with $\gamma + \mathrm{i}\sigma$, which corresponds to the finite-mode condition written with $\gamma_B - \mathrm{i}\Omega$; by Proposition 3.12 and Remark 3.9 the sign is immaterial, and either sign may be called the finite-mode condition.

3. *Finite regions.* The thesis's correlation matrix of a region is the matrix of the form γ on the site vectors, ordered site by site, whereas the finite-mode covariance matrix is ordered positions first and is conjugated by Ω; the normalization, fixed by the factor $\frac{1}{4}$, is the same. The Notation section lists the conventions. Definition 3.15 and Proposition 3.17 make the translation between the two explicit; they have no counterpart in the thesis.

4. *Fourier criterion.* The Fourier convention is the thesis's (4.14), the cell indexing γ_{x-y} is that of (4.4), and Lemma 3.24 is (4.16) [1, pp. 77, 81]. Theorem 3.25 is Lemma 4.5(i) of the thesis [1, p. 82], which asserts both directions for summable cell data and argues the direction (b) with the "sharply peaked Fourier transform of a flat Gaussian" [1, p. 81]; here (b) is proved by the same idea with a rectangular window, first for finite range and then, in Remark 3.26, for all summable cell data.

5. *Purity.* Lemma 4.5(ii) of [1, p. 82] states, for one mode per site, that the correlation function "corresponds to a pure Gaussian state if and only if $\hat{\gamma}(k) + \mathrm{i}\sigma_s$ is not strictly positive for any

$k \in [-\pi, \pi]$, i.e. if $(\sigma_s \hat{\gamma}(k))^2 = -1$". Corollary 3.29 proves the equivalence of the two matrix conditions, which holds for all k together; the "i.e." is not an equivalence of the pointwise conditions, as the remark after the corollary explains. The first clause, the characterization of purity, is not proved here. The vacuum is the thesis's "coherent product state" $\hat{\gamma}_0(k) = 1$.

6. *Evolution.* The cells γ'_j of Theorem 3.34 are the thesis's (4.17a), $\gamma_t(x) = \sum_{y,z} \Gamma_t(y)^{\mathsf{T}} \gamma_0(x + y - z)\Gamma_t(z)$, and the printed symbol formula (4.17b), $\hat{\gamma}_t(k) = \overline{\hat{\Gamma}_t^{\mathsf{T}}(k)}\, \hat{\gamma}_0(k)\, \hat{\Gamma}_t(k)$ [1, p. 82], with the conjugate transpose, is the general form $\hat{\gamma}_t(k) = \hat{\Gamma}_t(k)^\dagger \hat{\gamma}_0(k) \hat{\Gamma}_t(k)$ of Theorem 3.34. For the rules of the thesis, which are reflection symmetric by (4.12) [1, p. 80] and have real symbols, the conjugate transpose is the transpose.

7. *Existence of states.* Quasi-free states of the CCR algebra, of which the Gaussian states of this chapter are the lattice instance, were introduced by Manuceau and Verbeure [15]. The theorem that a normalized function with positive semidefinite twisted kernels is the characteristic function of a unique state is due to Manuceau, Sirugue, Testard and Verbeure [11]; see also [9, §5.2]. The quantum Bochner theorem of Remark 3.4 is due to Kastler [13] and to Loupias and Miracle-Sole [14].

4 The example system and its stationary states

The automaton of Example 1.20 rotates each oscillator of the chain by a fixed angle and couples neighbouring oscillators through their momenta. It is the example on which Krüger demonstrates that a reversible Gaussian automaton can drive every finite region of the chain towards equilibrium [1], and this chapter prepares that result by describing the automaton completely at each wave number and by determining its stationary covariances. At a single wave number the automaton acts by a real 2×2 matrix of determinant one, and the analysis rests on the elementary dichotomy for such matrices. When the trace lies strictly between -2 and 2, the matrix is conjugate to a rotation, its powers stay bounded, and it preserves exactly one positive definite form up to a factor; when the absolute value of the trace exceeds two, it squeezes, and the traces ± 2, which belong to ± 1 and to the parabolic matrices, separate the two cases. Determining the parameters for which every wave number is of the first kind, the rotation angle as a function of the wave number, and the invariant form, one arrives at a complete description of the stationary reflection-symmetric covariances: their symbols are the invariant form multiplied, wave number by wave number, by a factor not less than one.

The chapter closes with the observation that the massless lattice field of the earlier chapters is the example system at the edge of the admissible parameters, so that it is not covered by the analysis.

Throughout, $G = \mathbb{Z}$ and each site carries one mode, so that cells and symbols are real or complex 2×2 matrices, and $\Omega = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Cell data, their covariance form and their symbol are those of Definition 3.22; for a rule L and cell data (γ_x) we write $L_*\gamma$ for the cell data (γ'_j) of Theorem 3.34, whose covariance form is $\gamma(L\cdot, L\cdot)$. Besides the identity $A^{\mathsf{T}}\Omega A = (\det A)\,\Omega$ of Lemma 1.12, one further identity for 2×2 matrices over a commutative ring is used repeatedly,

$$\Omega A + A^{\mathsf{T}}\Omega = (\operatorname{tr} A)\,\Omega, \tag{4.1}$$

which holds because, for $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the left side is antisymmetric with upper right entry $a + d$; so is the Cayley–Hamilton relation $A^2 = (\operatorname{tr} A)A - (\det A)1$.

4.1 Iterated evolution

The dynamics after t steps is the automaton of the t-th power of the rule, whose symbol is the t-th power of the symbol by Theorem 1.3.

Proposition 4.1 (Powers and iterates)**.** Lean ✓ *Let L, L_1, L_2 be rules and (γ_x) cell data.*

(a) $\hat{L}(k+2\pi) = \hat{L}(k)$ and $\hat{\gamma}(k+2\pi) = \hat{\gamma}(k)$ for every $k \in \mathbb{R}$, and $\widehat{L^t}(k) = \hat{L}(k)^t$ for every $t \in \mathbb{N}$.

(b) $1_*\gamma = \gamma$ and $(L_1L_2)_*\gamma = L_{2*}(L_{1*}\gamma)$; hence $(L_*)^t\gamma = (L^t)_*\gamma$ for every $t \in \mathbb{N}$, and these are again cell data.

(c) If L is symplectic, so is L^t for every t, and for a real matrix S with $S^{\mathsf{T}}\Omega S = \Omega$ the on-site rule L_S is symplectic.

Proof.

(a) The evaluation points $e^{-\mathrm{i}(k+2\pi)}$ and $e^{-\mathrm{i}k}$ coincide, and the series for $\hat{\gamma}(k + 2\pi)$ and $\hat{\gamma}(k)$ agree term by term. Evaluation at $z = e^{-\mathrm{i}k}$ is a ring homomorphism (Lemma 1.18), and so is its entrywise application to matrices, which therefore preserves powers.

(b) Cell data are determined by their covariance form, because the form takes the value $(\gamma_x)_{ij}$ on the site vectors $e_{x,i}$ and $e_{0,j}$. The rule 1 is the identity, and the form of $(L_1L_2)_*\gamma$ is $\gamma(L_1L_2\cdot, L_1L_2\cdot)$, which, since the rule of L_1L_2 is $u \mapsto L_1(L_2u)$, is the form of $L_{1*}\gamma$ evaluated at (L_2u, L_2w), that is, the form of $L_{2*}(L_{1*}\gamma)$. The statement about iterates follows by induction on t from $(L_*)^{t+1}\gamma = L_*((L^t)_*\gamma) = (L^tL)_*\gamma$, and Theorem 3.34 shows at each step that the result is cell data.

(c) The symplectic symbols form a submonoid (Proposition 3.31), which contains the powers of its elements. The involution fixes constant symbols and the constant embedding is a ring homomorphism, so that the symplecticity condition for L_S is the constant symbol of $S^{\mathsf{T}}\Omega S = \Omega$.

□

Cell data can be recovered from their symbol by the inversion formula of Fourier series, and two cell data with the same symbol therefore coincide; this is how stationarity will be read off from the symbol.

Proposition 4.2 (Fourier inversion)**.** Lean ✓ *Let (γ_x) and (γ'_x) be cell data.*

(a) For every $x \in \mathbb{Z}$, $\gamma_x = \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} e^{\mathrm{i}kx}\,\hat{\gamma}(k)\,dk$, entrywise.

(b) If $\hat{\gamma}'(k) = \hat{\gamma}(k)$ for every k, then $\gamma' = \gamma$.

Proof.

(a) Each entry $\sum_g (\gamma_g)_{ab} e^{-\mathrm{i}kg}$ of $\hat{\gamma}(k)$ is an absolutely convergent Fourier series, which may be integrated term by term against $e^{\mathrm{i}kx}$, and since $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\mathrm{i}k(x-g)}\,dk = \delta_{g,x}$ only the term $g = x$ remains, as in the proof of Lemma 3.24.

(b) By (a), both families are the same integrals of the same function.

□

Lemma 4.3 (Reflection-symmetric cells)**.** Lean ✓ *Let (γ_x) be cell data with $\gamma_{-x} = \gamma_x$ for every x. Then $\hat{\gamma}(-k) = \hat{\gamma}(k)$, and $\hat{\gamma}(k)$ is a real symmetric matrix for every k.*

Proof. The substitution $x \mapsto -x$ gives $\hat{\gamma}(-k) = \sum_x \gamma_{-x} e^{-\mathrm{i}kx} = \hat{\gamma}(k)$, and by Lemma 3.23(b) $\overline{\hat{\gamma}(k)} = \hat{\gamma}(-k) = \hat{\gamma}(k)$, so that every entry is real. A Hermitian matrix with real entries is symmetric. □

4.2 The example system

We now fix the rule of Example 1.20. Its coefficient at the origin rotates each oscillator by an angle φ, and the coefficients at ± 1 couple each site to its neighbours through the momentum row of that rotation, with strength λ.

Definition 4.4 (The example system)**.** For $\varphi, \lambda \in \mathbb{R}$, the *example system* is the nearest-neighbour rule $L = L_{\varphi,\lambda}$ of Example 1.20, with coefficients

$$\ell_0 = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad \ell_1 = \ell_{-1} = \ell_{\pm} = \lambda \begin{pmatrix} \sin\varphi & \cos\varphi \\ 0 & 0 \end{pmatrix},$$

and all other coefficients zero. We write $\tau(k) = \cos\varphi + \lambda \cos k \sin\varphi$.

Proposition 4.5 (The symbol)**.** Lean ✓ *For all $\varphi, \lambda, k \in \mathbb{R}$, the Fourier symbol of $L_{\varphi,\lambda}$ is the real matrix*

$$\hat{L}(k) = \ell_0 + 2\cos k\,\ell_{\pm} = \begin{pmatrix} \cos\varphi + 2\lambda\cos k\sin\varphi & -\sin\varphi + 2\lambda\cos k\cos\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix},$$

and

(a) $\hat{L}(-k) = \hat{L}(k)$, and the rule is reflection symmetric, $\ell_{-g} = \ell_g$ for all g;

(b) $\det \hat{L}(k) = 1$, $\operatorname{tr} \hat{L}(k) = 2\tau(k)$ *and* $\hat{L}(k)^{\mathsf{T}} \Omega \hat{L}(k) = \Omega$*;*

(c) the rule is symplectic;

(d) for cell data (γ_x)*, the evolved cell data have the symbol* $\widehat{L_*\gamma}(k) = \hat{L}(k)^{\mathsf{T}}\, \hat{\gamma}(k)\, \hat{L}(k)$*.*

Proof. The symbol is $\ell_\pm e^{\mathrm{i}k} + \ell_0 + \ell_\pm e^{-\mathrm{i}k} = \ell_0 + 2\cos k\, \ell_\pm$, which is real.

(a) The cosine is even, and the only nonzero coefficients are ℓ_0 and the equal coefficients at ± 1.

(b) The determinant is $(\cos\varphi + 2\lambda \cos k \sin\varphi)\cos\varphi - \sin\varphi(-\sin\varphi + 2\lambda\cos k\cos\varphi) = \cos^2\varphi + \sin^2\varphi$, the trace is read off, and the last identity is Lemma 1.12 with $\det \hat{L}(k) = 1$.

(c) This is Example 1.20.

(d) This is Theorem 3.34, whose reflection-symmetric form applies because L is reflection symmetric by (a); since $\hat{L}(k)$ is real, its transpose is its conjugate transpose.

□

In the regime the symbol is not symmetric, its off-diagonal entries being $-\sin\varphi + 2\lambda\cos k\cos\varphi$ and $\sin\varphi$, which would be equal only if $\lambda\cos k\cos\varphi = \sin\varphi$, contrary to Lemma 4.10; so the transpose in (d) cannot be omitted.

Example 4.6 (The running example)**.** script ✓ For $\varphi = \pi/2$ the on-site rotation is by a quarter turn, $\tau(k) = \lambda\cos k$, and

$$\hat{L}(k) = \begin{pmatrix} 2\lambda\cos k & -1 \\ 1 & 0 \end{pmatrix}.$$

The rule maps the label $u(x) = (u_q, u_p)$ of a Weyl operator to $(-u_p(x) + \lambda(u_q(x-1) + u_q(x+1)),\, u_q(x))$; under the heuristic reading of Proposition 2.16, the field itself then moves by L^{-1}. For $\lambda = 1$ the rule is the inverse leapfrog of the massless lattice field, as the last section shows.

4.3 The regime of rotations

A real 2×2 matrix of determinant one with trace in $(-2, 2)$ has the eigenvalues $e^{\pm\mathrm{i}\theta}$ with $\theta \in (0, \pi)$ and is conjugate to a rotation by θ, while a matrix of determinant one whose trace has absolute value greater than two has real eigenvalues inverse to each other and stretches one direction while compressing another. Since $\operatorname{tr} \hat{L}(k) = 2\tau(k)$, every wave number is a rotation exactly when $|\tau| < 1$ everywhere.

Lemma 4.7 (Maximum of $|\tau|$)**.** Lean ✓ *For all* φ, λ, k*,* $|\tau(k)| \le |\cos\varphi| + |\lambda\sin\varphi|$*, with equality for some* k*.*

Proof. The inequality is the triangle inequality with $|\cos k| \le 1$. Equality holds at $k = 0$ if $\cos\varphi$ and $\lambda\sin\varphi$ are not of opposite signs, and at $k = \pi$ otherwise, since then $\cos\varphi$ and $-\lambda\sin\varphi$ are not of opposite signs. □

Besides the rotation of every wave number, the convergence of Section 5 needs the chain to be coupled, since without coupling a squeezed initial state simply rotates for ever at each site.

Definition 4.8 (The regime)**.** The pair (φ, λ) lies in the *regime* if

$$|\cos\varphi| + |\lambda\sin\varphi| < 1 \qquad \text{and} \qquad \lambda\sin\varphi \neq 0.$$

The first condition already forces $\sin\varphi \neq 0$ (Lemma 4.10 below), so that the second only excludes the uncoupled chain $\lambda = 0$. Every numbered statement below that assumes the regime remains true under the first condition alone; the second is used in the remarks after Propositions 4.12 and 4.14 and in Example 4.22.

Proposition 4.9 (Forms of the regime). Lean ✓ *Let $\varphi, \lambda \in \mathbb{R}$.*

(a) $|\cos\varphi| + |\lambda \sin\varphi| < 1$ *if and only if* $|\operatorname{tr} \hat{L}(k)| < 2$ *for every* k.

(b) If $\sin\varphi \neq 0$, *then* $|\cos\varphi| + |\lambda \sin\varphi| < 1$ *if and only if* $|\lambda| < f_{\mathrm{crit}}$, *where*

$$f_{\mathrm{crit}} = \frac{1 - |\cos\varphi|}{|\sin\varphi|}.$$

(c) $f_{\mathrm{crit}} = \tan(\varphi/2)$ *for* $0 < \varphi \leq \pi/2$, *and* $f_{\mathrm{crit}} = \cot(\varphi/2)$ *for* $\pi/2 < \varphi < \pi$.

Proof.

(a) Since $|\operatorname{tr} \hat{L}(k)| = 2|\tau(k)|$, Lemma 4.7 shows that the maximum over k of $|\operatorname{tr} \hat{L}(k)|$ is $2(|\cos\varphi| + |\lambda \sin\varphi|)$ and is attained, which is the claim.

(b) Divide $|\lambda|\,|\sin\varphi| < 1 - |\cos\varphi|$ by $|\sin\varphi| > 0$.

(c) For $0 < \varphi < \pi$ write $\varphi = 2x$ with $\sin x, \cos x > 0$, so that $\sin\varphi = 2\sin x \cos x > 0$ while $1 - \cos\varphi = 2\sin^2 x$ and $1 + \cos\varphi = 2\cos^2 x$. If $\varphi \leq \pi/2$, then $\cos\varphi \geq 0$ and $f_{\mathrm{crit}} = (1 - \cos\varphi)/\sin\varphi = \sin x/\cos x$, and if $\varphi > \pi/2$, then $\cos\varphi < 0$ and $f_{\mathrm{crit}} = (1 + \cos\varphi)/\sin\varphi = \cos x/\sin x$.

□

For negative φ the same computation, with $|\sin\varphi| = -\sin\varphi$, gives $f_{\mathrm{crit}} = |\tan(\varphi/2)|$ for $-\pi/2 \leq \varphi < 0$ and $|\cot(\varphi/2)|$ for $-\pi < \varphi < -\pi/2$. In the running example $f_{\mathrm{crit}} = 1$, and the regime is $0 < |\lambda| < 1$.
Whether the invariant form of Section 4.6 is written with the projector P_k or with its conjugate depends on the sign of $\sin\varphi - \lambda \cos k \cos\varphi$ (Proposition 4.16(d)), and the regime fixes this sign. The bound it needs goes beyond the triangle inequality, because the regime bounds $|\lambda \sin\varphi|$ while the sign depends on $|\lambda \cos\varphi|$.

Lemma 4.10 (A sign). Lean ✓ *If* $|\cos\varphi| + |\lambda \sin\varphi| < 1$, *then* $\sin\varphi \neq 0$, $|\lambda \cos\varphi| < |\sin\varphi|$, *and* $\sin\varphi - \lambda \cos k \cos\varphi$ *has the sign of* $\sin\varphi$ *for every* k.

Proof. Write $a = |\cos\varphi|$ and $b = |\sin\varphi|$. Since $a < 1$ and $a^2 + b^2 = 1$, we have $b > 0$. Multiplying $|\lambda| b < 1 - a$ by $a \geq 0$ gives $|\lambda| b\, a \leq (1 - a)a$, and since $1 - a > 0$ and $a < 1 + a$, also $(1 - a)a < (1 - a)(1 + a) = b^2$. Hence $|\lambda| a\, b < b^2$, and division by b gives $|\lambda \cos\varphi| < b$. Then $|\lambda \cos k \cos\varphi| \leq |\lambda \cos\varphi| < |\sin\varphi|$, so that $\sin\varphi - \lambda \cos k \cos\varphi$ differs from $\sin\varphi$ by less than $|\sin\varphi|$ and therefore has the sign of $\sin\varphi$. □

4.4 The rotation angle

In the regime each $\hat{L}(k)$ is a rotation in suitable coordinates, and its angle, which depends on k unless $\lambda = 0$, is determined by the trace.

Definition 4.11 (Rotation angle). For (φ, λ) in the regime, $\theta(k) = \arccos \tau(k) = \arccos(\cos\varphi + \lambda \cos k \sin\varphi)$.

Proposition 4.12 (The angle). Lean ✓ *Let* (φ, λ) *lie in the regime. Then* $|\tau(k)| < 1$, $\cos\theta(k) = \tau(k)$, $\operatorname{tr} \hat{L}(k) = 2\cos\theta(k)$, $\theta(k) \in (0, \pi)$ *and* $\sin\theta(k) > 0$ *for every* k, *and*

(a) $\theta(-k) = \theta(k)$, *and* θ *is continuous;*

(b) θ *is differentiable, with* $\theta'(k) = \lambda \sin\varphi \sin k/\sin\theta(k)$;

(c) on $[0, \pi]$, θ *is strictly increasing if* $\lambda \sin\varphi > 0$ *and strictly decreasing if* $\lambda \sin\varphi < 0$.

Proof. By Lemma 4.7 and the regime, $|\tau(k)| < 1$. On $[-1,1]$ the arccosine inverts the cosine and maps $(-1,1)$ into $(0,\pi)$, where the sine is positive, and the trace is Proposition 4.5(b).

(a) τ is even and continuous, and the arccosine is continuous.

(b) The arccosine has the derivative $-1/\sqrt{1-x^2}$ at $|x| < 1$, and $\sqrt{1-\tau(k)^2} = \sin\theta(k)$; with $\tau'(k) = -\lambda \sin\varphi \sin k$ the chain rule gives the formula.

(c) The cosine is strictly decreasing on $[0,\pi]$, so that τ is strictly decreasing there if $\lambda \sin\varphi > 0$ and strictly increasing if $\lambda \sin\varphi < 0$, and its values lie in $(-1,1)$, where the arccosine is strictly decreasing.

□

Although (c) follows without derivatives, (b) adds, through $\lambda \sin\varphi \neq 0$, that θ' vanishes on $[0,\pi]$ only at the end points, so that θ is a change of variables on each half of the circle, which is what the convergence argument will use. In the running example, $\theta(k) = \arccos(\lambda \cos k)$ and $\sin\theta(k) = \sqrt{1-\lambda^2\cos^2 k}$.

4.5 Powers of the symbol

After t steps the symbol is $\hat{L}(k)^t$, and a matrix of determinant one has powers of a simple closed form, since by the Cayley–Hamilton relation each power is a combination of the matrix and the identity.

Lemma 4.13 (Powers). Lean ✓ *Let A be a real 2×2 matrix with $\det A = 1$ and $\operatorname{tr} A = 2\cos a$, where $\sin a \neq 0$. Then for every $t \in \mathbb{N}$,*

$$A^t = \frac{\sin(ta)}{\sin a}\, A - \frac{\sin((t-1)a)}{\sin a}\, 1.$$

In the regime this holds for $A = \hat{L}(k)$ and $a = \theta(k)$.

Proof. Put $U_t = \sin(ta)/\sin a$, so that $U_0 = 0$ and $U_{-1} = -1$, and the formula $A^t = U_t A - U_{t-1}1$ holds for $t = 0$. If it holds for t, then by $A^2 = 2\cos a\, A - 1$,

$$A^{t+1} = U_t A^2 - U_{t-1}A = (2\cos a\, U_t - U_{t-1})A - U_t 1,$$

and $2\cos a\, U_t - U_{t-1} = U_{t+1}$ is the identity $\sin((t+1)a) + \sin((t-1)a) = 2\sin(ta)\cos a$. The last statement follows from Propositions 4.5(b) and 4.12. □

The same powers can be written through the spectral projectors of $\hat{L}(k)$, in terms of which Proposition 4.16(e) expresses the invariant form. They are built from the real matrix

$$N(k) = \frac{\cos\theta(k)\, 1 - \hat{L}(k)}{\sin\theta(k)},$$

defined in the regime, which squares to -1 because, by the Cayley–Hamilton relation, $(\cos\theta\, 1 - \hat{L})^2 = \cos^2\theta\, 1 - 2\cos\theta\, \hat{L} + \hat{L}^2 = -\sin^2\theta\, 1$. The letter N also denotes neighbourhoods in these notes; with the argument k it always means this matrix.

Proposition 4.14 (Spectral projectors). Lean ✓ *Let (φ,λ) lie in the regime, let $P_k = \frac{1}{2}(1 + \mathrm{i}N(k))$, a complex 2×2 matrix, and let $\bar{P}_k$ be its complex conjugate. Then $P_k + \bar{P}_k = 1$, and*

(a) $P_k^2 = P_k$ and $P_k\bar{P}_k = \bar{P}_k P_k = 0$;

(b) $\hat{L}(k) = e^{\mathrm{i}\theta(k)}P_k + e^{-\mathrm{i}\theta(k)}\bar{P}_k$, and $\hat{L}(k)P_k = e^{\mathrm{i}\theta(k)}P_k$;

(c) $\hat{L}(k)^t = e^{\mathrm{i}t\theta(k)}P_k + e^{-\mathrm{i}t\theta(k)}\bar{P}_k$ for every $t \in \mathbb{N}$.

Proof. Since $N = N(k)$ is real, $\bar{P}_k = \frac{1}{2}(1 - \mathrm{i}N)$, and $P_k + \bar{P}_k = 1$ uses nothing else.

(a) With $N^2 = -1$, $P_k^2 = \frac{1}{4}(1+2\mathrm{i}N-N^2) = \frac{1}{2}(1+\mathrm{i}N) = P_k$, and $P_k\bar{P}_k = \bar{P}_kP_k = \frac{1}{4}(1+N^2) = 0$, the two factors being polynomials in N.

(b) $e^{\mathrm{i}\theta}P_k + e^{-\mathrm{i}\theta}\bar{P}_k = \cos\theta\, 1 - \sin\theta\, N = \hat{L}(k)$ by the definition of N, and multiplying on the right by P_k gives the second identity by (a).

(c) By induction on t, the cross terms of the product with (b) vanishing by (a).

□

Expanding (c) gives $\hat{L}(k)^t = \cos(t\theta)1 - \sin(t\theta)N(k)$, which is Lemma 4.13 again, and it is this real form that the convergence argument will use. The projector P_k is Hermitian exactly when $N(k)$ is antisymmetric, which in the regime happens only at $\cos k = 0$, where the diagonal of $\tau(k)1 - \hat{L}(k)$, namely $(-\lambda\cos k\sin\varphi, \lambda\cos k\sin\varphi)$, vanishes; at every other wave number the two projectors are not orthogonal.

4.6 The invariant form

A rotation preserves the identity form, and a matrix conjugate to a rotation preserves the transported form. For $\hat{L}(k)$ this form can be written down directly: since $\hat{L}^{\mathsf{T}}\Omega\hat{L} = \Omega$, the matrix $\Omega(a1 - \hat{L})$ is preserved for every scalar a, and by (4.1) the choice $a = \cos\theta(k) = \frac{1}{2}\operatorname{tr}\hat{L}(k)$ makes it symmetric. Normalized to determinant one and oriented to be positive definite, it is the form of the following definition.

Definition 4.15 (Invariant form)**.** For (φ, λ) in the regime,

$$\hat{\varepsilon}(k) = -\frac{\operatorname{sgn}(\sin\varphi)}{\sin\theta(k)}\,\Omega\big(\cos\theta(k)\,1 - \hat{L}(k)\big) = -\operatorname{sgn}(\sin\varphi)\,\Omega N(k).$$

Since $\cos\theta = \tau$, the entries of $\hat{L}(k)$ give

$$\hat{\varepsilon}(k) = \frac{\operatorname{sgn}(\sin\varphi)}{\sin\theta(k)}\begin{pmatrix} \sin\varphi & -\lambda\cos k\sin\varphi \\ -\lambda\cos k\sin\varphi & \sin\varphi - 2\lambda\cos k\cos\varphi \end{pmatrix}, \tag{4.2}$$

and

$$\sin^2\theta(k) = 1 - \tau(k)^2 = \sin^2\varphi - 2\lambda\cos k\cos\varphi\sin\varphi - \lambda^2\cos^2 k\sin^2\varphi \tag{4.3}$$

is the determinant of the matrix in (4.2). Without the factor $\operatorname{sgn}(\sin\varphi)$ the matrix would be negative definite for $\sin\varphi < 0$.

Proposition 4.16 (Properties of the invariant form)**.** Lean ✓ *Let (φ, λ) lie in the regime. Then $\hat{\varepsilon}(-k) = \hat{\varepsilon}(k)$ and $\hat{L}(k)^{\mathsf{T}}\hat{\varepsilon}(k)\hat{L}(k) = \hat{\varepsilon}(k)$ for every k, and*

(a) $\hat{\varepsilon}(k)$ is real symmetric, and $\hat{\varepsilon}$ is continuous;

(b) $\det\hat{\varepsilon}(k) = 1$ and $(\Omega\hat{\varepsilon}(k))^2 = -1$;

(c) $\hat{\varepsilon}(k)$ is positive definite, with $\hat{\varepsilon}(k)_{11} = |\sin\varphi|/\sin\theta(k)$;

(d) $\operatorname{tr}(\mathrm{i}\Omega P_k) = (\sin\varphi - \lambda\cos k\cos\varphi)/\sin\theta(k)$, a real number with the sign of $\sin\varphi$;

(e) $\hat{\varepsilon}(k) = \operatorname{sgn}(\sin\varphi)\,\mathrm{i}\Omega(P_k - \bar{P}_k)$, and for $\sin\varphi > 0$, $\hat{\varepsilon}(k) + \mathrm{i}\Omega = 2\mathrm{i}\Omega P_k$.

Proof. Evenness follows from that of θ and $\hat{L}$. For the invariance, with $a = \cos\theta(k)$ and $\hat{L} = \hat{L}(k)$, the identity $\hat{L}^{\mathsf{T}}\Omega\hat{L} = \Omega$ gives $\hat{L}^{\mathsf{T}}\Omega(a1 - \hat{L})\hat{L} = a\Omega - \Omega\hat{L} = \Omega(a1 - \hat{L})$, and $\hat{\varepsilon}(k)$ is a scalar multiple of $\Omega(a1 - \hat{L})$.

(a) Symmetry is visible in (4.2), and continuity holds because θ and $\hat{L}$ are continuous and $\sin\theta$ does not vanish.

(b) By (4.2) and (4.3), $\det\hat{\varepsilon} = \operatorname{sgn}(\sin\varphi)^2\sin^2\theta/\sin^2\theta = 1$, since $\sin\varphi \neq 0$ by Lemma 4.10. Further, $\Omega\hat{\varepsilon} = -\operatorname{sgn}(\sin\varphi)\Omega^2 N = \operatorname{sgn}(\sin\varphi)N$, whose square is $N^2 = -1$.

(c) The $(1,1)$ entry is $\operatorname{sgn}(\sin\varphi)\sin\varphi/\sin\theta = |\sin\varphi|/\sin\theta > 0$, and a real symmetric 2×2 matrix E with $E_{11} > 0$ and $\det E > 0$ is positive definite, because $E_{11}\, v^{\mathsf{T}}Ev = (E_{11}v_1 + E_{12}v_2)^2 + (\det E)v_2^2$.

(d) Since $\mathrm{i}\Omega P_k = \frac{\mathrm{i}}{2}\Omega - \frac{1}{2}\Omega N$ and $\operatorname{tr}\Omega = 0$, the trace equals $-\frac{1}{2}\operatorname{tr}(\Omega N)$, and the trace of $\Omega(\tau 1 - \hat{L})$ is $-2\sin\varphi + 2\lambda\cos k\cos\varphi$; the sign is that of Lemma 4.10.

(e) Since $P_k - \bar{P}_k = \mathrm{i}N$, we have $\mathrm{i}\Omega(P_k - \bar{P}_k) = -\Omega N$, and for $\sin\varphi > 0$ the relation $\bar{P}_k = 1 - P_k$ gives $\hat{\varepsilon} + \mathrm{i}\Omega = \mathrm{i}\Omega(P_k - \bar{P}_k + 1) = 2\mathrm{i}\Omega P_k$.

□

In the running example,

$$\hat{\varepsilon}(k) = \frac{1}{\sqrt{1-\lambda^2\cos^2 k}}\begin{pmatrix} 1 & -\lambda\cos k \\ -\lambda\cos k & 1 \end{pmatrix}.$$

The entries of $\hat{\varepsilon}$ are real-analytic, 2π-periodic functions of k, since $1 - \tau(k)^2$ is a trigonometric polynomial that does not vanish; their Fourier coefficients therefore decay exponentially, and $\hat{\varepsilon}$ is the symbol of reflection-symmetric cell data, as is $\eta\hat{\varepsilon}$ for every real-analytic, even, 2π-periodic η. By Corollary 3.29, (b) then says that these cell data saturate the state condition at every wave number.
A factor multiplying $\hat{\varepsilon}(k)$ adds noise to the mode k, and the state condition becomes a threshold on the factor.

Proposition 4.17 (Threshold). Lean ✓ *Let E be a real symmetric positive definite 2×2 matrix with $\det E = 1$, and $\eta \in \mathbb{R}$. Then $\eta E + \mathrm{i}\Omega \geq 0$ if and only if $\eta \geq 1$. In particular, in the regime, $\eta\,\hat{\varepsilon}(k) + \mathrm{i}\Omega \geq 0$ if and only if $\eta \geq 1$.*

Proof. Write $e_1 = E_{11} > 0$ and $e_2 = E_{12}$. If $\eta E + \mathrm{i}\Omega \geq 0$, its $(1,1)$ entry ηe_1 is non-negative, so that $\eta \geq 0$, and its determinant, which by (3.3) is $\eta^2 \det E - 1 = \eta^2 - 1$, is non-negative; hence $\eta \geq 1$. Conversely, for $w_\pm = (\sqrt{e_1}, (e_2 \mp \mathrm{i})/\sqrt{e_1})$ a direct computation, using $e_1E_{22} - e_2^2 = 1$, gives $E \pm \mathrm{i}\Omega = w_\pm w_\pm^\dagger$, and

$$\eta E + \mathrm{i}\Omega = \tfrac{\eta+1}{2}(E + \mathrm{i}\Omega) + \tfrac{\eta-1}{2}(E - \mathrm{i}\Omega)$$

is a combination of positive semidefinite matrices with non-negative coefficients when $\eta \geq 1$. The last statement is the case $E = \hat{\varepsilon}(k)$, by Proposition 4.16. □

4.7 Invariant forms of a rotation

To show that every reflection-symmetric stationary covariance is built from $\hat{\varepsilon}$, one needs that $\hat{\varepsilon}(k)$ spans the symmetric forms preserved by $\hat{L}(k)$, and this follows from the description of the real 2×2 matrices commuting with a given one.

Lemma 4.18 (Commutant). Lean ✓ *Let A be a real 2×2 matrix that is not a multiple of the identity. Every real matrix X with $XA = AX$ is of the form $X = p1 + qA$ with $p, q \in \mathbb{R}$.*

Proof. Subtracting multiples of the identity, we may take $A_0 = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right) \neq 0$ and $X_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ with $X_0A_0 = A_0X_0$. The commutator $X_0A_0 - A_0X_0$ has the entries $yc - bz$, $2(xb - ya)$, $2(za - xc)$ and $bz - yc$, so that (x,y,z) and (a,b,c) have vanishing cross product, and $(x,y,z) = q(a,b,c)$ because $(a,b,c) \neq 0$. □

Proposition 4.19 (Invariant forms). Lean ✓ *Let A be a real 2×2 matrix with $\det A = 1$ that is not a multiple of the identity.*

(a) A real symmetric matrix S with $A^{\mathsf T}SA = S$ is of the form $S = q\,\Omega(A - \frac{1}{2}(\operatorname{tr}A)1)$ with $q \in \mathbb{R}$.

(b) If $|\operatorname{tr}A| < 2$ and $E \neq 0$ is a real symmetric matrix with $A^{\mathsf T}EA = E$, then every such S is $S = \eta E$ for a unique $\eta \in \mathbb{R}$.

Proof.

(a) By Lemma 1.12, $A^{\mathsf T}\Omega = \Omega A^{-1}$, so that $A^{\mathsf T}SA = S$ says that $X = \Omega^{-1}S$ commutes with A. By Lemma 4.18, $S = \Omega(p1 + qA)$. Symmetry of S, with $\Omega^{\mathsf T} = -\Omega$ and (4.1), becomes $(2p + q\operatorname{tr}A)\Omega = 0$, so that $p = -\frac{1}{2}q\operatorname{tr}A$.

(b) By (a), both S and E are multiples of $\Omega(A - \frac{1}{2}\operatorname{tr}A\,1)$, the multiple for E being nonzero, and uniqueness holds because $E \neq 0$.

□

The proof of (b) uses only (a), so that the hypothesis $|\operatorname{tr}A| < 2$ is not needed there; it matters for the converse. Conversely, $\Omega(A - \frac{1}{2}\operatorname{tr}A\,1)$ is symmetric and invariant by (4.1), and its determinant is $1 - \frac{1}{4}(\operatorname{tr}A)^2$, positive exactly in the elliptic case. If the forms are not required to be symmetric, the invariant ones span a plane, which also contains Ω. This is why reflection symmetry, which makes symbols real, enters the theorem below: the cell data with symbol $2\hat\varepsilon(k) + \mathrm{i}\sin k\,\Omega$ are stationary and satisfy the state condition, since $2\hat\varepsilon(k) + \mathrm{i}s\Omega = s\big((2/s)\hat\varepsilon(k) + \mathrm{i}\Omega\big) \geq 0$ for $s = 1 + \sin k > 0$ by Proposition 4.17, and trivially for $s = 0$, but their symbol is not of the form $\eta\hat\varepsilon$.

Corollary 4.20 (Coefficient). Lean ✓ *In the regime, put $h_k(S) = -\frac{1}{2}\operatorname{tr}(\Omega\hat\varepsilon(k)\Omega S)$ for a real matrix S. Then $h_k(\eta\,\hat\varepsilon(k)) = \eta$, and every real symmetric S with $\hat L(k)^{\mathsf T}S\hat L(k) = S$ equals $h_k(S)\,\hat\varepsilon(k)$.*

Proof. By Proposition 4.16(b), $h_k(\eta\hat\varepsilon) = -\frac{\eta}{2}\operatorname{tr}((\Omega\hat\varepsilon)^2) = -\frac{\eta}{2}\operatorname{tr}(-1) = \eta$. Since $|\operatorname{tr}\hat L(k)| < 2$ and $\hat\varepsilon(k) \neq 0$ is symmetric and invariant, Proposition 4.19(b) gives $S = \eta\hat\varepsilon(k)$ for some η, and then $\eta = h_k(S)$. □

4.8 The stationary states

A Gaussian state (ω, γ, d) is stationary under the automaton when its covariance and its displacement are both preserved, $\gamma(L\cdot, L\cdot) = \gamma$ and $d \circ L = d$. The theorem concerns the covariance; the displacement is settled in the remark after it. By the preceding results, reflection-symmetric cell data are preserved exactly when their symbol is preserved at every wave number, a preserved real symbol is a multiple of $\hat\varepsilon(k)$ at each k, and the state condition then says that the multiple is at least one.

Theorem 4.21 (Stationary covariances). Lean ✓ *Let (φ, λ) lie in the regime, and let (γ_x) be cell data with $\gamma_{-x} = \gamma_x$ for every x. The following are equivalent:*

(i) $L_*\gamma = \gamma$, and $\hat\gamma(k) + \mathrm{i}\Omega \geq 0$ for every k;

(ii) there is a continuous function $\eta\colon \mathbb{R} \to \mathbb{R}$ with $\eta(-k) = \eta(k) \geq 1$ and $\hat\gamma(k) = \eta(k)\,\hat\varepsilon(k)$ for every k.

Proof. (ii)⇒(i). By Proposition 4.5(d) and the invariance of $\hat\varepsilon$, the evolved cell data have the symbol $\hat L^{\mathsf T}\eta\hat\varepsilon\hat L = \eta\hat\varepsilon = \hat\gamma$, and by Proposition 4.2(b) they equal γ. The state condition is Proposition 4.17.

(i)⇒(ii). By Lemma 4.3, $S(k) = \hat\gamma(k)$ is real symmetric, even in k, and continuous by Lemma 3.23. Taking symbols in $L_*\gamma = \gamma$ gives $\hat L(k)^{\mathsf T}S(k)\hat L(k) = S(k)$, so that $S(k) = \eta(k)\hat\varepsilon(k)$ with $\eta(k) = h_k(S(k))$ by Corollary 4.20. This η is continuous and even, being a trace of products of continuous even functions, and $\eta(k) \geq 1$ by Proposition 4.17. □

The proof of (ii)⇒(i) uses neither the continuity and evenness of η nor reflection symmetry, but these are automatic: if $\hat\gamma = \eta\hat\varepsilon$ for cell data with $\eta \geq 1$, then $\eta = \hat\gamma_{11}\sin\theta/|\sin\varphi|$ is continuous, the symbol is real, and $\hat\gamma(-k) = \overline{\hat\gamma(k)} = \hat\gamma(k)$ makes η even and, by Proposition 4.2(b), the cells reflection symmetric. A stationary covariance of this form has $\det\hat\gamma(k) = \eta(k)^2$. For $\eta \equiv 1$ this is one at every wave number, the saturated covariance, while $\eta > 1$ adds noise to the corresponding modes; $\eta(k)$ measures the excess noise of the mode k, as $\coth(\beta\omega/2)$ does for a thermal oscillator. For the displacement, a translation-invariant $d(u) = \sum_x \delta^{\mathsf T}u(x)$ satisfies $d\circ L = d$ exactly when $\hat L(0)^{\mathsf T}\delta = \delta$, and since $\hat L(0)$ is conjugate to a rotation by $\theta(0) \in (0,\pi)$, it has no eigenvalue one, which forces $\delta = 0$.

Example 4.22 (Stationary covariances of the running example)**.** script ✓ For $\varphi = \pi/2$ and $0 < |\lambda| < 1$, the stationary reflection-symmetric cell data satisfying the Fourier criterion are exactly those with

$$\hat\gamma(k) = \frac{\eta(k)}{\sqrt{1-\lambda^2\cos^2 k}}\begin{pmatrix} 1 & -\lambda\cos k \\ -\lambda\cos k & 1\end{pmatrix}, \qquad \eta(-k) = \eta(k) \geq 1,$$

where η is continuous and such that this is the symbol of cell data, for instance real-analytic and 2π-periodic. The vacuum, $\hat\gamma \equiv 1$, is reflection symmetric and satisfies the Fourier criterion, so that by Theorem 4.21 it would be stationary only if its symbol had this form; its off-diagonal entry vanishes, so it is not stationary, and whether it approaches one of these states is the question of convergence.

4.9 The massless field at the edge of the regime

The lattice Klein–Gordon field of Example 3.35 is a nearest-neighbour rule, and one may ask whether every one of its wave numbers is a rotation. None of its masses makes every wave number elliptic, and only the massless field is an example system, which lies on the boundary of the regime.

Proposition 4.23 (The leapfrog and the regime)**.** Lean ✓ *Let $c_k = 2\cos k - m^2$ for a real m^2.*

(a) The symbols $\hat F$ and $\hat F^{-1}$ of the leapfrog and its inverse (Example 3.35) have $\operatorname{tr}\hat F(k) = \operatorname{tr}\hat F^{-1}(k) = c_k$, and $|c_{k_0}| \geq 2$ for $k_0 = \pi$ if $m^2 \geq 0$ and $k_0 = 0$ if $m^2 < 0$.

(b) The inverse leapfrog at $m^2 = 0$ is the example system with $\varphi = \pi/2$ and $\lambda = 1$.

(c) The pair $(\pi/2, 1)$ is not in the regime: $|\cos\frac{\pi}{2}| + |\sin\frac{\pi}{2}| = 1$, and $\lambda = 1 = f_{\mathrm{crit}}(\pi/2)$.

Proof.

(a) Both traces are c_k by Example 3.35, and $|c_\pi| = 2 + m^2$ for $m^2 \geq 0$, while $|c_0| = 2 - m^2 > 2$ for $m^2 < 0$.

(b) At $m^2 = 0$ the inverse leapfrog has $\ell_0 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0\end{smallmatrix}\right)$ and $\ell_{\pm1} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0\end{smallmatrix}\right)$, which are the coefficients of Definition 4.4 for $\varphi = \pi/2$ and $\lambda = 1$.

(c) $\cos\frac{\pi}{2} = 0$ and $\sin\frac{\pi}{2} = 1$.

□

At $\lambda = 1$ the running example has $\operatorname{tr}\hat L(0) = 2$ and $\operatorname{tr}\hat L(\pi) = -2$, and neither $\hat L(0)$ nor $\hat L(\pi)$ is ±1, so that the strict inequality of the regime fails exactly at the two parabolic modes $k = 0$ and $k = \pi$. For $m^2 \neq 0$, part (a) gives a mode with $|\operatorname{tr}| > 2$, which squeezes.

4.10 Notes and references

1. *The example system.* Definition 4.4 and Proposition 4.5 are (4.18) of [1], where the coupling is written f and the symbol $\hat{\Gamma}(k) = \Gamma_0 + 2f \cos k\, \Gamma_\pm$ [1, p. 82]. The evolved symbol is (4.17b), printed with the conjugate transpose, and for this real symbol the conjugate transpose and the transpose coincide.

2. *The regime.* The trace formula and the angle are (4.19), and f_{crit} with its half-angle forms is (4.20) [1, p. 83], printed with absolute values, as here. The thesis states the regime with $|f| \leq f_{\mathrm{crit}}$ and restricts to the nondegenerate case, which we read as the strict inequality of Definition 4.8 [1, pp. 83–84]. The thesis excludes $\sin\varphi = 0$ as degenerate [1, p. 88] but not $\lambda = 0$, and its Proposition 4.7 does not hold as stated for $\lambda = 0$, since an initial squeezed state then rotates for ever. Lemma 4.10 makes precise the appeal to "the condition on f from (4.20)" behind the sign statement of [1, p. 88]. Figure 4.3 of the thesis [1, p. 84] ($f = 0.4$, $\varphi = \pi/4$) shows the angle increasing on $[0, \pi]$, in accordance with Proposition 4.12(c).

3. *Projectors.* Proposition 4.14 is Lemma 4.6 and (4.24) of [1, pp. 84–85]; the thesis proves idempotence through eigenvectors (its footnote 8), here through $N(k)^2 = -1$.

4. *Stationary states.* The thesis writes the invariant form as $\hat{\varepsilon}(k) = \mathrm{i}\sigma_s(P_k - \bar{P}_k)$ for $0 < \varphi < \pi$ and $\mathrm{i}\sigma_s(\bar{P}_k - P_k)$ for $-\pi < \varphi < 0$, with the real expression (4.28) for the first case [1, pp. 87, 89]; Definition 4.15 is one formula for both, by Proposition 4.16(e). Theorem 4.21 is Theorem 4.8 of the thesis, with the factor written g there and η here, and the rotation angle written $\alpha(k)$ there and $\theta(k)$ here, since α denotes automorphisms in these notes. The thesis proves that every $g\hat{\varepsilon}$ with $g \geq 1$ is stationary and that the limit of its dynamics has this form, and it states that *all* stationary reflection-symmetric states are of this form; the completeness half, (i)⇒(ii), rests on Proposition 4.19, which is not in the thesis. The state condition is taken in its Fourier form; by Theorem 3.25 and Remark 3.26 it is equivalent to the state condition of the covariance form.

5. *The massless field.* Proposition 4.23 is not in the thesis. It shows that no mass makes every wave number of the lattice Klein–Gordon field of Example 3.35 elliptic, and that the massless field is the example system on the boundary of the regime of the thesis's convergence result.

5 Convergence to the stationary states

A reversible dynamics does not bring the state of the chain to a limit in norm unless the state is stationary, since the distance between the states at consecutive times does not change with time. What it can do is spread the correlations of any finite region over ever larger distances, so that every expectation value settles to a limit. For the example system in the regime this happens for every Gaussian initial state with zero displacement whose covariance is given by cell data. For reflection-symmetric data the symbol of the limit is, wave number by wave number, of the form found in Theorem 4.21. After t steps the symbol $\hat{L}(k)^t$ is conjugate to the rotation by the angle $t\theta(k)$, so that the evolved symbol of the covariance consists of a part independent of t and parts proportional to $\cos(2t\theta(k))$ and $\sin(2t\theta(k))$. The cells are the Fourier coefficients of this symbol. Because $\theta(k)$ varies with k, the Fourier coefficients of the oscillating parts tend to zero, and the limit is the average of the symbol over the rotation, which is invariant under $S \mapsto \hat{L}(k)^{\mathsf{T}} S \hat{L}(k)$.

As in the previous section, (φ, λ) lies in the regime of Definition 4.8, and $\theta(k)$ is the rotation angle of Definition 4.11. The real matrix $N(k)$ is defined by $\hat{L}(k) = \cos\theta(k)\,1 - \sin\theta(k)\,N(k)$ and satisfies $N(k)^2 = -1$, and $\hat{\varepsilon}(k)$ is the invariant form of Definition 4.15.

5.1 Oscillatory integrals

The Riemann–Lebesgue lemma states that $\int f(u) e^{iru}\,du$ tends to zero as $r \to \infty$ for every integrable f. The integrals that arise here carry the phase $r\theta(k)$, with θ the rotation angle, instead of ru. Since θ is strictly monotone on each half of the circle, a change of variables brings them to the standard form.

Lemma 5.1 (Oscillatory integrals along a monotone phase)**.** Lean ✓ *Let $a \le b$, let g be strictly monotone on $[a,b]$ and differentiable on (a,b) with $g' \neq 0$ there, and let f be integrable on $[a,b]$. Then*

$$\int_a^b f(k)\, e^{\mathrm{i}rg(k)}\,dk \longrightarrow 0 \qquad (r \to +\infty).$$

Proof. The endpoints have measure zero, so that the integral may be taken over the open interval (a,b), on which g is injective with a nonvanishing derivative. Its image $I = g((a,b))$ is measurable, and g has an inverse $g^{-1}\colon I \to (a,b)$. The change-of-variables formula for an injective differentiable map gives

$$\int_I G(u)\,du = \int_a^b |g'(k)|\, G(g(k))\,dk$$

for every measurable function G, with G integrable on I exactly when $|g'|\,(G \circ g)$ is integrable on (a,b). For $G(u) = H(u)e^{iru}$ with $H = (f/|g'|) \circ g^{-1}$ the right side is the integral in question, and $|g'|\,(H \circ g) = f$ is integrable, so that H is integrable on I. Hence

$$\int_a^b f(k)\, e^{\mathrm{i}rg(k)}\,dk = \int_{\mathbb{R}} \mathbf{1}_I(u) H(u)\, e^{\mathrm{i}ru}\,du,$$

the integral of the integrable function $\mathbf{1}_I H$ against e^{iru}, which tends to zero by the Riemann–Lebesgue lemma. □

When g' tends to zero at an endpoint, as it will for the rotation angle θ at $k = 0$ and $k = \pm\pi$, the function H may be unbounded there. The proof requires only its integrability, which the change of variables supplies, so that no separate estimate near the critical points of g is needed.

Corollary 5.2 (Oscillatory integrals of the example)**.** Lean ✓ *Let (φ, λ) lie in the regime, and let $f\colon \mathbb{R} \to \mathbb{C}$ be continuous. Then, for either choice of sign,*

$$\int_{-\pi}^{\pi} f(k)\, e^{\pm 2\mathrm{i}t\theta(k)}\,dk \longrightarrow 0 \qquad (t \to \infty,\ t \in \mathbb{N}).$$

Proof. Split the integral at $k = 0$. By Proposition 4.12, θ is differentiable with $\theta'(k) = \lambda \sin\varphi \sin k / \sin\theta(k)$, which vanishes nowhere on $(0, \pi)$ or on $(-\pi, 0)$, because $\lambda \sin\varphi \neq 0$ in the regime; on $[0, \pi]$ it is strictly monotone, and on $[-\pi, 0]$ it is strictly monotone in the opposite sense, being even. Lemma 5.1 with $r = 2t$, applied on each half to θ for the upper sign and to $-\theta$ for the lower, gives the claim. □

In this chapter the condition $\lambda \sin\varphi \neq 0$ of the regime enters only through this corollary. For $\lambda = 0$ the symbol $\hat{L}(k)$ is the same rotation matrix at every wave number, and the phase does not vary with k.

5.2 The limit of the covariance

The evolved cell data $(L_*)^t\gamma$ have the symbol $(\hat{L}(k)^t)^{\mathsf{T}}\hat{\gamma}(k)\hat{L}(k)^t$, by Proposition 4.1 and the reflection symmetry of $\hat{L}$ (Proposition 4.5). The power $\hat{L}^t$ is the rotation $\cos(t\theta)\,1 - \sin(t\theta)\,N$, and the quadratic expression in it separates into a part that does not depend on t and parts at twice the angle.

Lemma 5.3 (Splitting of the evolved symbol)**.** Lean ✓ *Let (φ, λ) lie in the regime, and let A be a complex 2×2 matrix. At each wave number k, with $\theta = \theta(k)$ and $N = N(k)$, and for every $t \in \mathbb{N}$,*

$$(\hat{L}(k)^t)^{\mathsf{T}} A\, \hat{L}(k)^t = C + \cos(2t\theta)\, A_c + \sin(2t\theta)\, A_s,$$

where $C = \frac{1}{2}(A + N^{\mathsf{T}}AN)$, $A_c = \frac{1}{2}(A - N^{\mathsf{T}}AN)$ and $A_s = -\frac{1}{2}(N^{\mathsf{T}}A + AN)$.

Proof. By Proposition 4.14(c) and $P_k - \bar{P}_k = \mathrm{i}N$, the power is $\hat{L}^t = \cos(t\theta)\,1 - \sin(t\theta)\,N$, so that

$$(\hat{L}^t)^{\mathsf{T}} A \hat{L}^t = \cos^2(t\theta)\, A - \cos(t\theta)\sin(t\theta)\,(N^{\mathsf{T}}A + AN) + \sin^2(t\theta)\, N^{\mathsf{T}}AN,$$

and the double-angle formulas $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $\cos x \sin x = \frac{1}{2}\sin 2x$ give the stated form. □

The part C is the average of $(\hat{L}^t)^{\mathsf{T}} A \hat{L}^t$ over a full rotation.

Definition 5.4 (Limit symbol and limit cells)**.** Let (φ, λ) lie in the regime. For cell data (γ_x) with symbol $\hat{\gamma}$, the *limit symbol* is

$$C(k) = \tfrac{1}{2}(\hat{\gamma}(k) + N(k)^{\mathsf{T}}\hat{\gamma}(k)N(k)),$$

and the *limit cells* are its Fourier coefficients,

$$\gamma_{\infty,x} = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\mathrm{i}kx}\, C(k)\, dk \qquad (x \in \mathbb{Z}).$$

Lemma 5.5 (The limit cells)**.** Lean ✓ *Let (φ, λ) lie in the regime. For cell data (γ_x), every $\gamma_{\infty,x}$ is a real matrix, and $\gamma_{\infty,-x} = \gamma_{\infty,x}^{\mathsf{T}}$.*

Proof. The matrix $N(k)$ is real and even in k, and $\hat{\gamma}(-k) = \overline{\hat{\gamma}(k)}$ by Lemma 3.23, so that $C(-k) = \overline{C(k)}$. Conjugating the defining integral and substituting $k \mapsto -k$ gives $\overline{\gamma_{\infty,x}} = \frac{1}{2\pi}\int e^{-\mathrm{i}kx} C(-k)\, dk = \gamma_{\infty,x}$. Since $\hat{\gamma}(k)$ is Hermitian, so is $C(k)$, and the same substitution gives $\gamma_{\infty,x}^{\mathsf{T}} = \frac{1}{2\pi}\int e^{\mathrm{i}kx}\, \overline{C(k)}\, dk = \frac{1}{2\pi}\int e^{-\mathrm{i}kx} C(k)\, dk = \gamma_{\infty,-x}$. □

The proof uses of N only that $N(k)$ is real and even in k.

Remark 5.6 (Summability of the limit cells). Let (φ, λ) lie in the regime and let (γ_x) be cell data. Then $(\gamma_{\infty,x})$ are cell data, and their symbol is C.

Proof. The 2π-periodic functions $F(k) = \sum_x c_x e^{-\mathrm{i}kx}$ with $\sum_x |c_x| < \infty$ form an algebra under pointwise multiplication, since the coefficients of a product are the convolution of the coefficients, and the convolution of two summable sequences is summable. The entries of $\hat{\gamma}$ belong to it by the definition of cell data. The function τ is a trigonometric polynomial with $|\tau| < 1$, so that $1 - \tau^2$ is a positive trigonometric polynomial; its square root $\sin\theta$ and the reciprocal $1/\sin\theta$ extend holomorphically to a strip around the real axis, and the entries of $N = (\tau\, 1 - \hat{L})/\sin\theta$ are therefore 2π-periodic functions holomorphic in a strip, whose Fourier coefficients decay exponentially. Hence the entries of C lie in the algebra, $C(k) = \sum_x c_x e^{-\mathrm{i}kx}$ with summable matrix coefficients c_x, and integrating term by term gives $\gamma_{\infty,x} = c_x$. By Lemma 5.5 these are real with $\gamma_{\infty,-x} = \gamma_{\infty,x}^{\mathsf{T}}$. □

Theorem 5.7 (Convergence of the covariance)**.** Lean ✓ *Let (φ, λ) lie in the regime and let (γ_x) be cell data. Then for every $x \in \mathbb{Z}$,*

$$\lim_{t\to\infty} \big((L_*)^t \gamma\big)_x = \gamma_{\infty,x}.$$

Proof. The evolved data $(L_*)^t\gamma$ are again cell data by Proposition 4.1(b), so that by Fourier inversion (Proposition 4.2) and Lemma 5.3 with $A = \hat{\gamma}(k)$,

$$\big((L_*)^t \gamma\big)_x = \gamma_{\infty,x} + \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\mathrm{i}kx}\big(\cos(2t\theta(k))A_c(k) + \sin(2t\theta(k))A_s(k)\big)\,dk.$$

The symbol $\hat{\gamma}$ is continuous by Lemma 3.23, and N is continuous in the regime, since θ is and $\sin\theta > 0$; hence the entries of A_c and A_s are continuous. Writing the cosine and the sine through $e^{\pm 2\mathrm{i}t\theta(k)}$ expresses each entry of the integral as a combination of four integrals of the form of Corollary 5.2, with $f(k)$ equal to $e^{\mathrm{i}kx}$ times an entry of $A_c(k)$ or of $A_s(k)$, and each of them tends to zero. □

For $\lambda = 0$ every wave number is rotated by the same matrix, and the covariance of a state that is not invariant under this rotation need not converge.

Proposition 5.8 (No convergence without coupling)**.** Lean ✓ *For $\varphi = \pi/2$, $\lambda = 0$ and the cell data $\gamma_0 = \mathrm{diag}(2, \frac{1}{2})$, $\gamma_x = 0$ for $x \neq 0$, the entry $\big((L_*)^t\gamma\big)_{0,11}$ equals 2 for even t and $\frac{1}{2}$ for odd t; in particular it does not converge.*

Proof. At $\lambda = 0$ the symbol is the constant matrix $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and $\hat{\gamma} \equiv \gamma_0$, so that the evolved symbol is the constant $(R^t)^{\mathsf{T}}\gamma_0 R^t$ and, by Fourier inversion, equals the cell at the origin. Since $R^2 = -1$, this matrix is γ_0 for even t and $R^{\mathsf{T}}\gamma_0 R = \mathrm{diag}(\frac{1}{2}, 2)$ for odd t. □

The hypothesis $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$ below holds at every k whenever the covariance form of the cell data satisfies the state condition, by Remark 3.26. In particular, by Theorem 3.13, it holds for the covariance of a Gaussian state.

Proposition 5.9 (The limit is stationary)**.** Lean ✓ *Let (φ, λ) lie in the regime, let (γ_x) be cell data and $k \in \mathbb{R}$.*

(a) $\hat{L}(k)^{\mathsf{T}} C(k) \hat{L}(k) = C(k)$.

(b) If $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$, *then* $C(k) + \mathrm{i}\Omega \geq 0$.

(c) If $\gamma_{-x} = \gamma_x$ *for every x and* $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$, *then* $C(k) = \eta\,\hat{\varepsilon}(k)$ *for some real* $\eta \geq 1$.

Proof.

(a) Write $A = \hat{\gamma}(k)$ and $\theta = \theta(k)$, so that $\hat{L} = \cos\theta\, 1 - \sin\theta\, N$. Since $(N^{\mathsf{T}})^2 = (N^2)^{\mathsf{T}} = -1$, one has $N^{\mathsf{T}} C N = \frac{1}{2}(N^{\mathsf{T}} A N + A) = C$ and $N^{\mathsf{T}} C + C N = \frac{1}{2}(N^{\mathsf{T}} A - A N + A N - N^{\mathsf{T}} A) = 0$, and therefore $\hat{L}^{\mathsf{T}} C \hat{L} = \cos^2\theta\, C - \cos\theta \sin\theta\,(N^{\mathsf{T}} C + C N) + \sin^2\theta\, N^{\mathsf{T}} C N = C$.

(b) N is real with $N^2 = -1$, hence $\operatorname{tr} N = 0$ and $\det N = 1$, and $N^{\mathsf{T}}\Omega N = (\det N)\,\Omega = \Omega$ by Lemma 1.12. Therefore $C + \mathrm{i}\Omega = \frac{1}{2}\big((\hat\gamma + \mathrm{i}\Omega) + N^{\mathsf{T}}(\hat\gamma + \mathrm{i}\Omega)N\big)$, and $N^{\mathsf{T}} = N^*$ because N is real, so that both terms are positive semidefinite.

(c) By Lemma 4.3, $\hat\gamma(k)$ is real symmetric, hence so is $C(k)$. By (a) and Corollary 4.20, $C(k) = \eta\,\hat\varepsilon(k)$ with $\eta = h_k(C(k))$, and $\eta \geq 1$ by (b) and Proposition 4.17.

□

Example 5.10 (The vacuum in the running example)**.** script ✓ For $\varphi = \pi/2$ and $0 < |\lambda| < 1$, the vacuum of Example 3.20 has $\hat\gamma \equiv 1$ and, by Example 4.22, is not stationary. With $\sin\theta(k) = \sqrt{1 - \lambda^2\cos^2 k}$ one finds $N(k) = (\sin\theta(k))^{-1}\left(\begin{smallmatrix} -\lambda\cos k & 1 \\ -1 & \lambda\cos k \end{smallmatrix}\right)$, hence

$$N^{\mathsf{T}}N = \frac{1}{\sin^2\theta}\begin{pmatrix} 1+\lambda^2\cos^2 k & -2\lambda\cos k \\ -2\lambda\cos k & 1+\lambda^2\cos^2 k \end{pmatrix}, \qquad C(k) = \frac{1}{\sin^2\theta(k)}\begin{pmatrix} 1 & -\lambda\cos k \\ -\lambda\cos k & 1 \end{pmatrix}.$$

This is $\eta(k)\hat\varepsilon(k)$ with $\eta(k) = (1 - \lambda^2\cos^2 k)^{-1/2} \geq 1$, with excess noise wherever $\cos k \neq 0$. By Remark 5.6 the limit cells are cell data with symbol C; they are reflection symmetric, since C is real and even, and η is continuous and even with $\eta \geq 1$, so that by Theorem 4.21 the cells of the vacuum converge to those of a stationary covariance. Here the entries of C are themselves holomorphic in a strip, and the limit cells decay exponentially.

5.3 The limit state

The convergence of the covariance cells yields the convergence of the characteristic functions, since the covariance form on a vector of finite support involves only finitely many cells. To pass from characteristic functions to arbitrary observables one uses that the Weyl operators span a dense subspace, together with a bound on states that is uniform in the state.

Let (γ_x) be cell data with covariance form γ, and let ω be a given Gaussian state with covariance form γ and displacement zero. Its evolution by t steps, $\omega_t = \omega \circ \alpha_{L^t}$, is by Proposition 3.30 a Gaussian state with covariance form $\gamma(L^t\cdot, L^t\cdot)$. The limit cells are cell data by Remark 5.6, and we denote their covariance form by γ_∞.

Lemma 5.11 (Norm of a state)**.** Lean ✓ *For every state ω on $\mathrm{CCR}(V,\sigma)$ and every $a \in \mathrm{CCR}(V,\sigma)$, $|\omega(a)| \leq \|a\|$.*

Proof. The form $(a,b) \mapsto \omega(a^*b)$ is positive semidefinite, and the Cauchy–Schwarz inequality gives $|\omega(a)|^2 = |\omega(1^*a)|^2 \leq \omega(1)\,\omega(a^*a) = \omega(a^*a)$. The element a^*a is positive in the C*-algebra $\mathrm{CCR}(V,\sigma)$, with spectrum in $[0, \|a^*a\|] = [0, \|a\|^2]$, so that $\|a\|^2\,1 - a^*a$ has non-negative spectrum and is the square of its self-adjoint square root; positivity of ω then gives $\omega(a^*a) \leq \|a\|^2$. □

Theorem 5.12 (Convergence of the states)**.** Lean ✓ *Let (φ,λ) lie in the regime, and let ω, ω_t and γ_∞ be as above.*

(a) For every $u \in V$, $\chi_{\omega_t}(u) \to \exp\!\big(-\frac{1}{4}\gamma_\infty(u,u)\big)$.

(b) There is a Gaussian state ω_∞ with covariance form γ_∞ and displacement zero such that $\omega_t(a) \to \omega_\infty(a)$ for every $a \in \mathrm{CCR}(V,\sigma)$, that is, $\omega_t \to \omega_\infty$ in the weak-$*$ topology; ω_∞ is the only state with this characteristic function.

Proof.

(a) Since the displacement vanishes, $\chi_{\omega_t}(u) = \exp\!\big(-\frac{1}{4}\gamma(L^tu, L^tu)\big)$, and by Theorem 3.34 and Proposition 4.1, $\gamma(L^tu, L^tu) = \sum_{x,y} u(x)^{\mathsf{T}}\big((L_*)^t\gamma\big)_{x-y}\,u(y)$. The sum is finite, each cell converges by Theorem 5.7, and the exponential is continuous.

(b) On a finite combination $b = \sum_j z_j W(u_j)$ the sequence $\omega_t(b) = \sum_j z_j \chi_{\omega_t}(u_j)$ converges by (a). The span of the Weyl operators is dense in $\mathrm{CCR}(V, \sigma)$, which is its closure; for arbitrary a and $\zeta > 0$ choose such a b with $\|a - b\| < \zeta$. By Lemma 5.11, $|\omega_t(a) - \omega_{t'}(a)| \leq 2\zeta + |\omega_t(b) - \omega_{t'}(b)|$, so that $(\omega_t(a))_t$ is a Cauchy sequence. Its limit $\omega_\infty(a)$ is linear in a and satisfies $|\omega_\infty(a)| \leq \|a\|$, so that it is continuous, and $\omega_\infty(1) = 1$ and $\omega_\infty(a^* a) \geq 0$ hold in the limit because the non-negative reals form a closed subset of $\mathbb{C}$. By (a), $\chi_{\omega_\infty}(u) = \exp(-\frac{1}{4}\gamma_\infty(u, u))$, and γ_∞ is symmetric by Lemma 5.5; hence ω_∞ is Gaussian with covariance form γ_∞ and displacement zero, and it is unique by Proposition 3.5(a).

□

Since ω is a Gaussian state, $\hat{\gamma}(k) + \mathrm{i}\Omega \geq 0$ for every k by Theorem 3.13 and Remark 3.26. By Proposition 5.9 the symbol C of the limit therefore satisfies $\hat{L}(k)^{\mathsf{T}} C(k) \hat{L}(k) = C(k)$ and $C(k) + \mathrm{i}\Omega \geq 0$ at every wave number, and by Remark 5.6 and Fourier inversion the limit cells are stationary. For reflection-symmetric data $C(k) = \eta(k)\hat{\varepsilon}(k)$ with $\eta(k) \geq 1$, so that the limit is one of the stationary covariances of Theorem 4.21.

5.4 Notes and references

1. *Convergence.* Theorem 5.7 is Proposition 4.7 of [1, p. 87]. The thesis assumes a pure initial state of finite correlation length. It treats the oscillatory integrals by a stationary-phase expansion around the extrema of the rotation angle, written as an equality, without showing that the contributions near the extrema vanish [1, pp. 85–86]. Here only summability of the initial cells is assumed, and the change of variables of Lemma 5.1 reduces the decay to the Riemann–Lebesgue lemma. The thesis does not state the condition $\lambda \neq 0$ explicitly; Proposition 5.8 shows that the statement fails without it. The thesis remarks that the first moments do not converge unless they vanish [1, p. 86]. This holds for translation-invariant displacements, which the dynamics rotates at the single wave number $k = 0$. For a displacement supported on finitely many sites one expects decay to zero by the same dephasing, now with the phases $e^{\pm \mathrm{i}t\theta(k)}$, but this is not proved here.

2. *The limit.* Proposition 5.9(c) identifies the limit symbol with the form of Theorem 4.8 of [1], wave number by wave number. The thesis concludes that the limit is a stationary state in the sense of cell data, on the grounds that $\hat{\varepsilon}$ is continuous [1, p. 89]. Continuity of a symbol does not by itself imply summability of its Fourier coefficients. The conclusion holds nevertheless, by the real-analyticity of N and the algebra property used in Remark 5.6.

3. *The limit state.* Theorem 5.12 is the first part of Theorem 4.9 of [1], the convergence of expectation values. The second part, convergence of the reduced density matrices of finite regions in trace norm, is not treated in these notes. Lemma 5.11 is the standard bound $\|\omega\| = \omega(1)$ for positive functionals on a unital C*-algebra.

6 Gaussian channels

The automata of the previous chapters are reversible: each step is an automorphism of the CCR algebra and can be undone by its inverse. A lattice of oscillators coupled to an environment loses this property. Its dynamics is still a physical operation on the observables, that is, a unital completely positive map, but the map need not have a completely positive inverse, and in general it adds noise to the states. When the environment is Gaussian and the coupling quadratic, such a map is expected to send Weyl operators to multiples of Weyl operators, as the dilations later in the chapter indicate: a Weyl operator $W(u)$ goes to a multiple of $W(Lu)$, where the linear map L need no longer preserve σ. The multiple consists of a phase $e^{\mathrm{i}\mu(u)}$, which only displaces the states, and a damping factor $e^{-\nu(u,u)/4}$ with a quadratic form ν, the noise, which compensates the loss of symplecticity. This chapter develops these Gaussian channels over an arbitrary real vector space with an alternating form. The lattice enters only at the end, where the definition of a quantum cellular automaton is extended to completely positive dynamics and two notions of locality are introduced; Gaussian automata on the chain are treated in Chapter 7.

Throughout, V is a real vector space with an alternating bilinear form σ, no longer only the lattice phase space, and $\mathrm{CCR}(V,\sigma)$ is the C*-algebra generated by the Weyl operators $W(u)$ on $\ell^2(V)$, constructed exactly as in Chapter 2, with $W(u)W(w) = e^{-\mathrm{i}\sigma(u,w)/2}W(u+w)$ and $W(u)^* = W(-u)$. The letters V, σ and L thus have a more general meaning in this chapter than in the table of notation, the lattice data being one instance. States, characteristic functions and Gaussian states on $\mathrm{CCR}(V,\sigma)$ are defined as in Chapter 3, whose definitions, and the propositions cited from it below, use no property of the lattice. For bilinear forms γ and β on V we write $\gamma + \mathrm{i}\beta \geq 0$ if the matrix $\big[\gamma(u_j,u_k) + \mathrm{i}\beta(u_j,u_k)\big]_{j,k}$ is positive semidefinite for every finite family $u_1,\dots,u_m$ in V; for $\beta = \sigma$ this is the state condition of Chapter 3.

6.1 Completely positive maps

A linear map T between C*-algebras is *positive* if it maps positive elements to positive elements, and *completely positive* if, for every k, the map applied entrywise to $k \times k$ matrices over the first algebra is positive. Positivity refers to the order of a C*-algebra, in which the positive elements are the self-adjoint elements with spectrum in $[0,\infty)$; by a basic theorem of the subject these are exactly the elements of the form x^*x, and finite sums of such are again of this form [9] (pinpoint to be checked). The CCR algebra is a closed subalgebra of the bounded operators on a Hilbert space, and we first check that the order it inherits from there is this intrinsic one. Applied also to the algebra of $k \times k$ matrices over $\mathrm{CCR}(V,\sigma)$, a closed subalgebra of the bounded operators on the k-fold direct sum of the Hilbert space, this makes complete positivity of maps of the CCR algebra unambiguous.

Proposition 6.1 (Positivity in closed subalgebras). Lean ✓ *Let A be a C*-algebra and $B \subseteq A$ a norm-closed $*$-subalgebra containing the unit of A. An element $x \in B$ is positive in A if and only if $x = s^*s$ for some $s \in B$. In particular an element of $\mathrm{CCR}(V,\sigma)$ is positive in the C*-algebra $\mathrm{CCR}(V,\sigma)$ if and only if it is a positive operator.*

Proof. If $x = s^*s$, then x is positive in A by the theorem just quoted. Conversely, let $x \in B$ be positive in A, so that x is self-adjoint with spectrum in a bounded interval $[0,c]$. By the Weierstrass theorem there are real polynomials p_j converging to the square root uniformly on $[0,c]$, and the continuous functional calculus of A gives $p_j(x) \to s = \sqrt{x}$ in norm. Each $p_j(x)$ lies in B, which contains the unit and is closed, so $s \in B$; it is self-adjoint, and $s^*s = x$. For the last statement take for A the bounded operators on $\ell^2(V)$ and $B = \mathrm{CCR}(V,\sigma)$: an element is positive in A exactly when it is a positive operator, and by the theorem quoted above, applied in B, it is positive in B exactly when it has the form s^*s with $s \in B$, which by the first statement is the same condition. □

Proposition 6.2 (Composition of completely positive maps)**.** Lean ✓ *The identity map is completely positive, and the composite of two completely positive maps is completely positive.*

Proof. Applying $g \circ f$ entrywise to a matrix is the same as applying f entrywise and then g, and each step preserves positivity. □

A positive map between C*-algebras is automatically bounded, and every $*$-homomorphism is completely positive [2]. Both facts are used below: the first to extend identities from Weyl operators to the whole algebra, the second to regard automorphisms as channels.

6.2 Gaussian channels and the product formula

A Gaussian channel is specified by the data described at the beginning of the chapter, a linear map, a noise form and a displacement, together with a completely positive map that realizes them on Weyl operators. The data are subject to a positivity condition, in which the noise compensates the failure of the linear map to preserve σ.

Definition 6.3 (Gaussian channel)**.** A *Gaussian channel* on $\mathrm{CCR}(V,\sigma)$ consists of a linear map $L\colon V \to V$, a symmetric bilinear form ν on V (the *noise*), a linear form μ on V (the *displacement*) and a completely positive map T of $\mathrm{CCR}(V,\sigma)$ into itself, such that

$$T(W(u)) = e^{\mathrm{i}\mu(u) - \nu(u,u)/4}\, W(Lu) \qquad (u \in V) \tag{6.1}$$

and

$$\nu + \mathrm{i}\big(\sigma - \sigma(L\cdot, L\cdot)\big) \geq 0. \tag{6.2}$$

The *dissipation form* of the channel is

$$\Delta_T(u,w) = \nu(u,w) + \mathrm{i}\big(\sigma(u,w) - \sigma(Lu, Lw)\big).$$

Condition (6.2) says that the dissipation form is positive semidefinite. In the literature the maps T of this kind are called quasi-free completely positive maps. For finitely many modes it is known that (6.2) is necessary and sufficient for a completely positive map with (6.1) to exist [3] (scope of the theorem there to be checked); in these notes neither direction is proved, and both the map and the condition are part of the definition. Since T is linear and bounded, it is determined by (6.1), as the following lemma shows.

Lemma 6.4 (Unit and uniqueness)**.** Lean ✓ *A Gaussian channel satisfies $T(1) = 1$. Two bounded linear maps from $\mathrm{CCR}(V,\sigma)$ into a normed space that agree on all Weyl operators are equal.*

Proof. Since $W(0) = 1$, (6.1) at $u = 0$ gives $T(1) = 1$. The linear span of the Weyl operators is a $*$-algebra, because a product of Weyl operators is a multiple of a Weyl operator, and it is dense in $\mathrm{CCR}(V,\sigma)$ by definition of the latter; since two bounded linear maps that agree on a set agree on the closure of its linear span, maps agreeing on all Weyl operators are equal. □

A $*$-automorphism maps products to products. A Gaussian channel does so only up to a scalar for products of two Weyl operators, and the scalar is governed by the dissipation form.

Proposition 6.5 (Product formula)**.** Lean ✓ *For all $u, w \in V$,*

$$T\big(W(u)W(w)\big) = e^{-\Delta_T(u,w)/2}\, T\big(W(u)\big)\, T\big(W(w)\big).$$

Proof. By the Weyl relation and (6.1), the left-hand side equals

$$e^{-\mathrm{i}\sigma(u,w)/2}\, e^{\mathrm{i}\mu(u+w) - \nu(u+w,u+w)/4}\, W(Lu + Lw),$$

while the product on the right equals

$$e^{\mathrm{i}\mu(u) + \mathrm{i}\mu(w) - \nu(u,u)/4 - \nu(w,w)/4}\, e^{-\mathrm{i}\sigma(Lu,Lw)/2}\, W(Lu + Lw).$$

Since μ is linear and ν symmetric, $\nu(u+w,u+w)=\nu(u,u)+2\nu(u,w)+\nu(w,w)$, and the ratio of the two scalars is

$$\exp\Big(-\tfrac{1}{2}\nu(u,w)-\tfrac{\mathrm{i}}{2}\sigma(u,w)+\tfrac{\mathrm{i}}{2}\sigma(Lu,Lw)\Big)=e^{-\Delta_T(u,w)/2}.$$

□

Example 6.6 (The attenuator). script ✓ Let $V=\mathbb{R}^2$ be one mode, $\sigma(u,w)=u^{\mathsf{T}}\Omega w$, and fix $0<\zeta<1$. The data $L=\zeta\cdot 1$, $\nu(u,w)=(1-\zeta^2)\,u^{\mathsf{T}}w$ and $\mu=0$ satisfy (6.2), because the dissipation form is $(1-\zeta^2)(u^{\mathsf{T}}w+\mathrm{i}u^{\mathsf{T}}\Omega w)$, a positive multiple of the vacuum's form $1+\mathrm{i}\Omega$, whose eigenvalues are 0 and 2. This is the attenuator of quantum optics, which transmits the fraction ζ^2 of the intensity and fills the loss with vacuum noise; the beam splitter of Example 6.14 indicates where a completely positive map realizing it comes from. We use it as the running example of the chapter. Statements about the attenuator below concern the Gaussian channel with these data, which is unique by Lemma 6.4 and whose existence is not proved in these notes.

6.3 Special cases and reversible channels

Condition (6.2) simplifies in three cases. Without noise it forces L to preserve σ; for L preserving σ it only asks the noise to be non-negative; and at the opposite extreme, $L=0$, it is the state condition. In the last case every $T(W(u))$ is a multiple of the unit, so that the functionals $\omega\circ T$ agree on Weyl operators for all states ω, and hence, by Lemma 6.4, the channel replaces every state by one fixed state (a state by the argument in the proof of Proposition 6.10), with characteristic function $e^{\mathrm{i}\mu(u)-\nu(u,u)/4}$.

Proposition 6.7 (Special cases). Lean ✓ *Let L be linear and ν symmetric.*

(a) If $0+\mathrm{i}(\sigma-\sigma(L\cdot,L\cdot))\geq 0$, then $\sigma(Lu,Lw)=\sigma(u,w)$ for all u,w.

(b) If L preserves σ, then (6.2) holds if and only if $\nu(u,u)\geq 0$ for all u.

(c) If $L=0$, then (6.2) is the state condition $\nu+\mathrm{i}\sigma\geq 0$.

Proof. (a) Put $a=\sigma(u,w)-\sigma(Lu,Lw)$. For the pair u,w the matrix of the hypothesis is $\left(\begin{smallmatrix}0 & \mathrm{i}a\\ -\mathrm{i}a & 0\end{smallmatrix}\right)$, whose determinant $-a^2$ must be non-negative; hence $a=0$.

(b) The matrices of (6.2) are the Gram matrices $[\nu(u_j,u_k)]$ of the real symmetric form ν, which are positive semidefinite for all families exactly when $\nu(u,u)\geq 0$ for all u.

(c) For $L=0$ the form $\sigma(L\cdot,L\cdot)$ vanishes, so (6.2) reads $\nu+\mathrm{i}\sigma\geq 0$.

□

Proposition 6.8 (Reversible channels). Lean ✓ *Let L be a linear bijection of V preserving σ. With $\nu=0$ and $\mu=0$, the automorphism α_L with $\alpha_L(W(u))=W(Lu)$ (Lemma 2.1, whose proof uses only that L is a bijection preserving σ) is a Gaussian channel. Conversely, a Gaussian channel with $\nu=0$, $\mu=0$ and bijective L is α_L.*

Proof. As a $*$-automorphism, α_L is completely positive, (6.1) holds, and the matrices of (6.2) vanish. Conversely, L preserves σ by Proposition 6.7(a), so α_L is defined; the channel and α_L are bounded and agree on Weyl operators, hence coincide by Lemma 6.4. □

6.4 Concatenation and the action on states

Applying two channels in succession is again a channel. In the Heisenberg picture the first channel to act on a state is the last to act on an observable, so that $T_2 \circ T_1$ applies T_1 to the observable first, while on states T_2 acts first, $\omega \mapsto \omega \circ T_2 \mapsto \omega \circ T_2 \circ T_1$.

Proposition 6.9 (Concatenation)**.** Lean ✓ *If T_1 and T_2 are Gaussian channels, then $T_2 \circ T_1$ is a Gaussian channel with linear map L_2L_1, noise $\nu_1 + \nu_2(L_1\cdot, L_1\cdot)$ and displacement $\mu_1 + \mu_2 \circ L_1$. Its dissipation form is $\Delta_{T_1}(u, w) + \Delta_{T_2}(L_1u, L_1w)$.*

Proof. The composite is completely positive by Proposition 6.2, and applying (6.1) twice gives

$$T_2(T_1(W(u))) = e^{\mathrm{i}\mu_1(u) - \nu_1(u,u)/4}\, e^{\mathrm{i}\mu_2(L_1u) - \nu_2(L_1u, L_1u)/4}\, W(L_2L_1u).$$

The formula for the dissipation form follows from the decomposition

$$\sigma(u, w) - \sigma(L_2L_1u, L_2L_1w) = \big(\sigma(u, w) - \sigma(L_1u, L_1w)\big) + \big(\sigma(L_1u, L_1w) - \sigma(L_2L_1u, L_2L_1w)\big).$$

Consequently the matrix of (6.2) for a family (u_j) is the sum of the matrix of the first channel for (u_j) and that of the second for (L_1u_j), and a sum of positive semidefinite matrices is positive semidefinite. □

In the Schrödinger picture a channel acts on states, sending ω to $\omega \circ T$. The following proposition shows that Gaussian states go to Gaussian states, with the covariance transformed by L and increased by the noise.

Proposition 6.10 (Action on Gaussian states)**.** Lean ✓ *Let ω be a Gaussian state with covariance γ and displacement d. Then $\omega \circ T$ is a Gaussian state with covariance $\gamma(L\cdot, L\cdot) + \nu$ and displacement $d \circ L + \mu$.*

Proof. The functional $\omega \circ T$ is bounded, and it takes the value 1 at the unit because $T(1) = 1$. For every b in the CCR algebra the element $T(b^*b)$ is positive, since T is, and therefore of the form s^*s, so that $\omega(T(b^*b)) = \omega(s^*s) \geq 0$; thus $\omega \circ T$ is a state. Its characteristic function is

$$\omega(T(W(u))) = e^{\mathrm{i}\mu(u) - \nu(u,u)/4}\, e^{\mathrm{i}d(Lu) - \gamma(Lu, Lu)/4},$$

which is the Gaussian characteristic function with the symmetric form $\gamma(L\cdot, L\cdot) + \nu$ and the linear form $d \circ L + \mu$. □

In the running example, the attenuator carries a Gaussian state with the vacuum covariance, $\gamma(u, w) = u^{\mathsf{T}}w$ and $d = 0$, to the state with covariance

$$\zeta^2 u^{\mathsf{T}}w + (1 - \zeta^2)\, u^{\mathsf{T}}w = u^{\mathsf{T}}w,$$

so that it leaves the vacuum covariance invariant, as it should, since it only mixes the mode with vacuum noise. A Gaussian state with the thermal covariance $\gamma(u, w) = c\, u^{\mathsf{T}}w$, $c > 1$, goes to the state with covariance $(\zeta^2 c + 1 - \zeta^2)\, u^{\mathsf{T}}w$, so that the excess $c - 1$ over the vacuum is multiplied by ζ^2.

6.5 Finitely many modes

For n modes, $V = \mathbb{R}^{2n}$ with $\sigma(u, w) = u^{\mathsf{T}}\Omega w$, Gaussian states are described by covariance matrices, and the covariance matrix γ_B of a covariance form γ is $\Omega^{\mathsf{T}}\gamma\Omega$ (Chapter 3); it is admissible if $\gamma_B^{\mathsf{T}} = \gamma_B$ and $\gamma_B - \mathrm{i}\Omega \geq 0$. In this language the data (L, ν) of a Gaussian channel become a pair of matrices, while the displacement is carried separately.

Definition 6.11 (Channel matrices)**.** Real $2n \times 2n$ matrices (X, Y) are *channel matrices* if $Y^{\mathsf{T}} = Y$ and $Y - \mathrm{i}\Omega + \mathrm{i}X\Omega X^{\mathsf{T}} \geq 0$. They act on covariance matrices by $\gamma_B \mapsto X\gamma_B X^{\mathsf{T}} + Y$.

Proposition 6.12 (Properties of channel matrices). Lean ✓

(a) Channel matrices map admissible covariance matrices to admissible ones.

(b) Let L, ν and γ also denote the real matrices of a linear map and of two bilinear forms on $\mathbb{R}^{2n}$, and put $X = \Omega^{\mathsf{T}} L^{\mathsf{T}} \Omega$ and $Y = \Omega^{\mathsf{T}} \nu \Omega$. Then

$$\Omega^{\mathsf{T}}(L^{\mathsf{T}}\gamma L + \nu)\Omega = X(\Omega^{\mathsf{T}}\gamma\Omega)X^{\mathsf{T}} + Y.$$

(c) For symmetric ν, the matrices (X, Y) of (b) are channel matrices if and only if L and ν satisfy (6.2).

Proof. (a) The output $X\gamma_B X^{\mathsf{T}} + Y$ is symmetric, and

$$X\gamma_B X^{\mathsf{T}} + Y - \mathrm{i}\Omega = X(\gamma_B - \mathrm{i}\Omega)X^{\mathsf{T}} + \left(Y - \mathrm{i}\Omega + \mathrm{i}X\Omega X^{\mathsf{T}}\right)$$

is a sum of positive semidefinite matrices.

(b) Insert $\Omega\Omega^{\mathsf{T}} = 1$ between the factors.

(c) The matrices of (6.2) are $U^* H U$ with U real and $H = \nu + \mathrm{i}(\Omega - L^{\mathsf{T}}\Omega L)$, and taking for U the standard basis shows that (6.2) holds if and only if $H \geq 0$. Using $\Omega^{\mathsf{T}} = -\Omega$ and $\Omega\Omega^{\mathsf{T}} = 1$ one computes

$$Y - \mathrm{i}\Omega + \mathrm{i}X\Omega X^{\mathsf{T}} = \Omega^{\mathsf{T}} H^{\mathsf{T}} \Omega;$$

congruence by the orthogonal matrix Ω and transposition preserve positive semidefiniteness of Hermitian matrices, and Y is symmetric exactly when ν is.

□

For the attenuator, $X = \zeta \cdot 1$ and $Y = (1 - \zeta^2)1$, and the condition reads $(1 - \zeta^2)(1 - \mathrm{i}\Omega) \geq 0$. If channel matrices (X, Y) arise from L and ν as in (b) and $Y = 0$, then $\nu = 0$, and by (c) and Proposition 6.7(a) the map L preserves σ. Since σ is nondegenerate on $\mathbb{R}^{2n}$, such an L is injective, hence bijective; then $X = L^{-1}$, the inverse by which covariance matrices transform (Corollary 3.32), and by Proposition 6.8 the Gaussian channel with these data and $\mu = 0$ exists and is α_L.

6.6 Dilations

The attenuator arises from a beam splitter: the mode is mixed with an auxiliary mode prepared in the vacuum, and the auxiliary mode is then discarded. In general, let a second real vector space V' with an alternating form σ' describe an ancilla, and let S be a linear map of $V \times V'$ preserving the form

$$(\sigma \oplus \sigma')((u, u'), (w, w')) = \sigma(u, w) + \sigma'(u', w');$$

for bilinear forms γ on V and γ' on V', the form $\gamma \oplus \gamma'$ is defined in the same way. When S is bijective, the reversible dynamics α_S of system and ancilla, followed by discarding an ancilla prepared in a Gaussian state with covariance γ', should act on covariances through the two components of the image of a system vector, $S(u, 0) = (S_{11}u, S_{21}u)$. The following proposition establishes the corresponding statements for the data; the construction at the level of the algebra is not carried out in these notes.

Proposition 6.13 (Dilation data). Lean ✓ *Let S preserve $\sigma \oplus \sigma'$, and let γ' be a symmetric bilinear form on V' with $\gamma' + \mathrm{i}\sigma' \geq 0$. Then:*

(a) for all $u, w \in V$,

$$\sigma(S_{11}u, S_{11}w) + \sigma'(S_{21}u, S_{21}w) = \sigma(u, w);$$

(b) the data $L = S_{11}$ *and* $\nu = \gamma'(S_{21}\cdot, S_{21}\cdot)$ *satisfy* (6.2)*;*

(c) for every form γ *on* V*,*

$$(\gamma \oplus \gamma')(S(u,0), S(w,0)) = \gamma(S_{11}u, S_{11}w) + \gamma'(S_{21}u, S_{21}w).$$

Proof. (a) This is the preservation of $\sigma \oplus \sigma'$ at $(u,0)$ and $(w,0)$.

(b) By (a),

$$\nu + \mathrm{i}(\sigma - \sigma(S_{11}\cdot, S_{11}\cdot)) = (\gamma' + \mathrm{i}\sigma')(S_{21}\cdot, S_{21}\cdot),$$

so the matrices of (6.2) for a family (u_j) are those of $\gamma' + \mathrm{i}\sigma'$ for $(S_{21}u_j)$.

(c) This is the definition of $\gamma \oplus \gamma'$ at $S(u,0) = (S_{11}u, S_{21}u)$.

□

Part (c) computes the covariance that the system would have after the dilation, $\gamma(S_{11}\cdot, S_{11}\cdot) + \gamma'(S_{21}\cdot, S_{21}\cdot)$. By Proposition 6.10 this is the output covariance of the Gaussian channel with $L = S_{11}$ and $\nu = \gamma'(S_{21}\cdot, S_{21}\cdot)$, whose existence is not proved here, and part (b) says that these data always satisfy the condition (6.2). An ancilla state with displacement d' would contribute the displacement $\mu = d' \circ S_{21}$.

Example 6.14 (The beam splitter)**.** script ✓ For one system and one ancilla mode, let S act on positions and on momenta by the same rotation

$$\begin{pmatrix} \zeta & \sqrt{1-\zeta^2} \\ -\sqrt{1-\zeta^2} & \zeta \end{pmatrix},$$

mixing system and ancilla. Since the same orthogonal matrix acts on positions and on momenta, S preserves $\sigma \oplus \sigma'$, and $S_{11} = \zeta \cdot 1$, $S_{21} = -\sqrt{1-\zeta^2} \cdot 1$. With the ancilla in the vacuum, $\gamma'(u', w') = u'^{\mathsf{T}} w'$, the noise is $(1-\zeta^2) u^{\mathsf{T}} w$, which is the noise of the attenuator of Example 6.6. At the level of the algebra, α_S followed by the vacuum state on the ancilla is expected to give a completely positive map with these data; this step is not carried out in these notes.

6.7 Quantum cellular automata and locality

We return to the lattice. Let G be an abelian group and $\mathcal{A}$ a net on G: each region $O \subseteq G$ is assigned a closed $*$-subalgebra $\mathcal{A}(O)$ of a C*-algebra A, monotonically in O, and a translation action α of G by $*$-automorphisms of A carries $\mathcal{A}(O)$ to $\mathcal{A}(O+a)$ (Chapter 2). A reversible automaton is a $*$-automorphism commuting with the translations, together with a finite neighbourhood by which it moves observables at most. Dropping invertibility, we arrive at the general definition.

Definition 6.15 (Quantum cellular automaton)**.** A *quantum cellular automaton* on $\mathcal{A}$ is a unital completely positive map $T\colon A \to A$ together with a finite set $N \subseteq G$, its *neighbourhood*, such that $T\alpha_a = \alpha_a T$ for all $a \in G$ and $T(\mathcal{A}(O)) \subseteq \mathcal{A}(O+N)$ for every region O.

The last condition bounds how far a single step can spread an observable. A second notion of locality asks instead that observables in sufficiently separated regions be processed independently. We state both for an arbitrary linear map.

Definition 6.16 (Notions of locality)**.** A linear map T of A

(I) has *finite propagation with neighbourhood* N if $T(\mathcal{A}(O)) \subseteq \mathcal{A}(O+N)$ for all regions O;

(II) *factorizes with respect to* $N' \subseteq G$, where $0 \in N' = -N'$, if $T(b_1 b_2) = T(b_1)T(b_2)$ whenever $b_1 \in \mathcal{A}(O_1)$, $b_2 \in \mathcal{A}(O_2)$ and $(O_1 + N') \cap O_2 = \emptyset$ (so that, since $0 \in N'$, the regions are disjoint, and, since $N' = -N'$, the separation hypothesis is symmetric in O_1 and O_2).

Finite propagation with the neighbourhood N is part of the definition of an automaton (T, N). Factorization says that the image of a product of observables in sufficiently separated regions is the product of their images. For the CCR net of Theorem 2.3 the local algebras of disjoint regions commute, and the product $b_1 b_2$ takes the place of the observable $b_1 \otimes b_2$ of the composite region. A $*$-automorphism factorizes with respect to every N', since it is multiplicative; for a channel the product formula of Proposition 6.5 shows what factorization demands of the dissipation form.

Proposition 6.17 (Reversible automata)**.** Lean ✓ *If (T, N) is a reversible quantum cellular automaton, then (T, N) is a quantum cellular automaton.*

Proof. A $*$-automorphism is unital and completely positive, and the remaining conditions are the same in both definitions. □

6.8 Notes and references

(1) *Definition.* Definition 6.3 is (4.32) of [1, p. 92], with the displacement of the standard definition of a quasi-free completely positive map added; the thesis's linear map and noise are Γ and g. Condition (6.2) is (4.34). That it characterizes complete positivity is Theorem 2.6 of [1], taken from Demoen, Vanheuverzwijn and Verbeure [3] (scope of the theorem there to be checked); in these notes it is assumed through the definition. The displacement appears already in (2.38) of [1]. For finitely many modes, the channel matrices of Definition 6.11 and their condition are equivalent, by conjugation with Ω, to the phase-space description of Holevo and Werner [4] (normalization to be checked). The term "dissipation form" for Δ_T is the thesis's [1, p. 93].

(2) *The product formula and states.* Proposition 6.5 is (4.35) of [1, p. 92]. The thesis prints the output characteristic function (4.33) with $+g(\xi, \xi)/4$; Proposition 6.10 shows that the sign is $-$, in agreement with the covariance update after (2.41) of [1]. Granted that the attenuator and the vacuum exist, which is not proved in these notes, the sign $+$ cannot be right: the attenuator would then carry the vacuum to a functional with $|\chi(u)| = e^{(1-2\zeta^2)u^{\mathsf{T}}u/4}$, which exceeds 1 for $u \neq 0$ and $\zeta^2 < \frac{1}{2}$, whereas the characteristic function of a state is bounded by 1 (Proposition 3.3(b)). The two special solutions of (4.34) noted on p. 92 of the thesis are Proposition 6.7(b) and (c).

(3) *Dilations.* Proposition 6.13(c) is the reduction (4.40) of [1, p. 98] for an arbitrary ancilla, and part (b) goes beyond the thesis, which checks only that the resulting noise is real and symmetric. For finitely many modes, every Gaussian channel has a Gaussian dilation with a pure Gaussian environment [5] (theorem to be identified and checked).

(4) *Automata and locality.* Definition 6.15 corresponds to Definition 4.1 of [1, p. 73], with the quasi-local algebra there replaced by a net, as for Definition 2.6; the thesis also asks $0 \in N$, but by monotonicity of the net, enlarging a neighbourhood is always allowed, so the condition is not needed here. Notions (I) and (II) of Definition 6.16 are the first two of the four notions of locality of [1, p. 93]; the thesis states (II) for $A_1 \otimes A_2$ on disjoint finite regions, with $M = -M$ (and $0 \in M$ without loss of generality), which corresponds to the product $b_1 b_2$ here. For a bounded map of the CCR net, asking the condition for finite regions only is equivalent: every element of the $*$-algebra generated by the Weyl operators of a region lies in the local algebra of a finite subregion, these elements are dense in the local algebra, and the product is continuous. In finite dimensions, notion (I) is causality preservation (Definition 1 of [6]) and notion (II) the factorization condition (6) of [6], which defines their class of factorizing channels (Definition 5 there) (pinpoints to be checked).

7 Irreversible Gaussian automata on the chain

Chapter 6 developed Gaussian channels over an arbitrary real vector space with an alternating form and extended the definition of a quantum cellular automaton to completely positive dynamics. We now return to the chain and ask which Gaussian channels commute with the translations, and how the two notions of locality of Definition 6.16 read on their data. For a channel whose linear map is a rule and whose noise is given by summable cells, locality becomes a statement about where these two families vanish: finite propagation concerns the rule alone, while factorization constrains the noise and, through the symplectic form pulled back by the rule, the rule as well. The thesis asks for factorization because it would let the action on a single site determine the whole automaton [1, p. 91] (pinpoint to be checked), and it asks that the class of automata so obtained be closed under concatenation; the chapter shows how far both wishes are met.

Throughout, $G = \mathbb{Z}$, the phase space V and the form σ are those of the lattice, and a rule L has coefficients ℓ_m, finitely many of them nonzero, with range $N_L = \{m : \ell_m \neq 0\}$ (Corollary 1.5). Cell data (ν_x) and their covariance form and symbol are as in Definition 3.22. The existence of a completely positive map with prescribed data is not proved in these notes (Chapter 6). Every statement below therefore concerns the channels with given data and holds vacuously if there is none, and a conclusion about a class of channels, such as the failure of closure under concatenation, presupposes that the channels in question exist; the only channels constructed here are the automorphisms of symplectic rules.

7.1 Lattice channels

Definition 7.1 (Lattice channel)**.** A Gaussian channel T on $\mathrm{CCR}(V, \sigma)$ is a *lattice channel* with rule L and noise cells (ν_x) if its linear map is L, (ν_x) are cell data whose covariance form is its noise ν, and its displacement is translation invariant, $\mu(\kappa_{a*}u) = \mu(u)$ for all $a \in \mathbb{Z}$ and $u \in V$.

The automaton α_L of a symplectic rule (Theorem 2.8) is a lattice channel with zero noise cells and zero displacement, by Proposition 6.8. The running example of the chapter is irreversible.

Example 7.2 (The lossy chain)**.** script ✓ Let $n = 1$, let L be the example system of Definition 4.4, fix $0 < \zeta < 1$, and consider the data ζL, with coefficients $\zeta\ell_m$, and the on-site noise cells $\nu_0 = (1 - \zeta^2) \cdot 1$, $\nu_x = 0$ for $x \neq 0$. These are the data that Proposition 6.9 assigns to $\alpha_L \circ T_1$, where T_1 has rule $\zeta \cdot 1$ and on-site noise $(1-\zeta^2)\cdot 1$, the attenuator of Example 6.6 at every site. In the Schrödinger picture this is one step of the example system followed by the loss of the fraction $1 - \zeta^2$ of the intensity at every site into vacuum.

Locality is a statement about local algebras, and a channel is known only through its values on Weyl operators. The following lemma translates between the two.

Lemma 7.3 (Weyl operators in local algebras)**.** Lean ✓ *Let $O \subseteq \mathbb{Z}$.*

(a) $\mathcal{A}(O)$ is the closure of the linear span of the Weyl operators $W(u)$ with $\operatorname{supp} u \subseteq O$.

(b) $W(u) \in \mathcal{A}(O)$ if and only if $\operatorname{supp} u \subseteq O$.

Proof.

(a) The span in question is a $*$-algebra, since a product of Weyl operators is a multiple of the Weyl operator of the sum, $\operatorname{supp}(u+w) \subseteq \operatorname{supp} u \cup \operatorname{supp} w$ and $W(u)^* = W(-u)$. Its closure is therefore a closed $*$-algebra containing the generators of $\mathcal{A}(O)$ and contained in $\mathcal{A}(O)$, hence equal to $\mathcal{A}(O)$.

(b) One direction is contained in (a). For the other, suppose that $\operatorname{supp} u \not\subseteq O$. The Weyl operator $W(v)$ maps the point mass δ_0 in $\ell^2(V)$ to a multiple of modulus one of the point

mass δ_{-v}, so that the continuous linear functional $\phi(a) = \langle \delta_{-u}, a\,\delta_0 \rangle$ satisfies $\phi(W(v)) \neq 0$ exactly for $v = u$. It vanishes on every generator of $\mathcal{A}(O)$, and hence on $\mathcal{A}(O)$ by (a), but not on $W(u)$.

□

Proposition 7.4 (Lattice channels are automata)**.** Lean ✓ *A lattice channel T with rule L commutes with every translation, $\alpha_a \circ T = T \circ \alpha_a$, and (T, N_L) is a quantum cellular automaton.*

Proof. The channel is unital by Lemma 6.4 and completely positive by definition. Both $\alpha_a \circ T$ and $T \circ \alpha_a$ are bounded and linear. On a Weyl operator the first gives $e^{\mathrm{i}\mu(u)-\nu(u,u)/4} W(\kappa_{a*} L u)$ and the second $e^{\mathrm{i}\mu(\kappa_{a*}u)-\nu(\kappa_{a*}u,\kappa_{a*}u)/4} W(L\kappa_{a*}u)$; these agree, because the rule commutes with translations, the covariance form of cell data is translation invariant (Lemma 3.23(a)) and so is the displacement. Lemma 6.4 gives equality. For locality, $T(W(u))$ is a multiple of $W(Lu)$ and $\operatorname{supp} Lu \subseteq \operatorname{supp} u + N_L$ by Corollary 1.5, so that T maps the Weyl operators of $\mathcal{A}(O)$ into the closed subalgebra $\mathcal{A}(O + N_L)$; by continuity and Lemma 7.3(a) it maps $\mathcal{A}(O)$ there. □

7.2 Pull-back cells and locality

Factorization involves the product formula of Proposition 6.5, and hence the dissipation form $\nu + \mathrm{i}(\sigma - \sigma(L\cdot, L\cdot))$. Its three terms are translation-invariant forms, and we describe each by its cells. For the noise these are the given cells; for σ they are Ω at the origin and zero elsewhere; and for $\sigma(L\cdot, L\cdot)$ they are obtained, as in the evolution of a covariance in Theorem 3.34, by pulling the cells of σ back along the rule.

Definition 7.5 (Pull-back cells)**.** For a rule L and a family (ν_x) of real $2n \times 2n$ matrices, the *pull-back cells* are

$$(L_*\nu)_j = \sum_{m,m'} \ell_m^{\mathsf{T}}\, \nu_{j+m-m'}\, \ell_{m'} \qquad (j \in \mathbb{Z}),$$

the sum running over the finitely many m, m' with $\ell_m, \ell_{m'} \neq 0$. We write $(L_*\sigma)_j = \sum_m \ell_m^{\mathsf{T}} \Omega\, \ell_{m+j}$ for the pull-back cells of the on-site cells $\Omega\delta_{x,0}$ of σ, and $\delta_{\bullet,0}\,\nu_0$ for the on-site cells equal to a matrix ν_0 at the origin and to zero elsewhere. These are the cells $L_*\gamma$ of Chapter 4, written there for cell data; the subscript records that forms are pulled back along L, an operation often written L^* in geometry. The *form* of an arbitrary family (ν_x) is $\nu(u, w) = \sum_{x,y} u(x)^{\mathsf{T}} \nu_{x-y} w(y)$, a finite sum; it determines the family, since $\nu(e_{x,i}, e_{0,j}) = (\nu_x)_{ij}$.

Lemma 7.6 (Forms of pull-back cells)**.** Lean ✓ *Let L be a rule.*

(a) For every family (ν_x), the form $\sum_{x,y} u(x)^{\mathsf{T}} (L_*\nu)_{x-y} w(y)$ is $\nu(Lu, Lw)$, where ν is the form of the cells (ν_x); if (ν_x) are cell data, so are $(L_*\nu)$.

(b) $\sigma(Lu, Lw) = \sum_{x,y} u(x)^{\mathsf{T}} (L_*\sigma)_{x-y} w(y)$, and $(L_*\sigma)_j$ is the coefficient of z^j in $\hat{L}^{\star\mathsf{T}} \Omega \hat{L}$, where $\hat{L}^\star$ replaces z by z^{-1}.

(c) If (ν_x) are cell data, the symbol of $L_*\nu$ is $\widehat{L_*\nu}(k) = \hat{L}(k)^\dagger\, \hat{\nu}(k)\, \hat{L}(k)$.

Proof.

(a) The cells $(L_*\nu)$ are the cells γ' of Theorem 3.34, and the computation of the form in its proof uses neither summability nor symmetry of the cells; the second statement is part of that theorem.

(b) The first statement is (a) for the on-site cells of σ, whose form is σ. In $\hat{L}^{\star\mathsf{T}} \Omega \hat{L} = \sum_{m,m'} \ell_m^{\mathsf{T}} \Omega\, \ell_{m'} z^{m'-m}$ the coefficient of z^j collects the terms with $m' = m + j$.

(c) This is the formula for the symbol in Theorem 3.34.

□

Theorem 7.7 (Locality on the chain). Lean ✓ *Let T be a lattice channel with rule L and noise cells (ν_x).*

(I) For a finite set $N \subseteq \mathbb{Z}$, T has finite propagation with neighbourhood N if and only if $N_L \subseteq N$.

(II) For a finite set $N' \subseteq \mathbb{Z}$ with $0 \in N' = -N'$, T factorizes with respect to N' if and only if $\nu_x = 0$ and $(L_*\sigma)_x = 0$ for every $x \notin N'$.

Proof.

(I) If $N_L \subseteq N$, then T has finite propagation with neighbourhood N_L by Proposition 7.4, and hence with N. Conversely, for the site vector $u = e_{0,i}$, $T(W(u))$ is a nonzero multiple of $W(Lu)$ and lies in $\mathcal{A}(N)$, so that $\operatorname{supp} Lu \subseteq N$ by Lemma 7.3(b). Since $(Lu)(m)$ is the i-th column of ℓ_m, every m for which this column is nonzero lies in N, and taking all i gives $N_L \subseteq N$.

(II) Suppose first that the cells vanish outside N', and let u, w have supports with $(\operatorname{supp} u + N') \cap \operatorname{supp} w = \emptyset$, so that $x - y \notin N'$ for $x \in \operatorname{supp} u$ and $y \in \operatorname{supp} w$. By Lemma 7.6, $\nu(u, w) = 0$ and $\sigma(Lu, Lw) = 0$, and $\sigma(u, w) = 0$ because the supports are disjoint; the dissipation form vanishes on (u, w), and $T(W(u)W(w)) = T(W(u))T(W(w))$ by Proposition 6.5. For regions O_1, O_2 with $(O_1 + N') \cap O_2 = \emptyset$, the maps $(b_1, b_2) \mapsto T(b_1 b_2)$ and $(b_1, b_2) \mapsto T(b_1)T(b_2)$ are bilinear and continuous on $\mathcal{A}(O_1) \times \mathcal{A}(O_2)$ and agree on pairs of Weyl operators, hence everywhere by Lemma 7.3(a). Conversely, suppose that T factorizes with respect to N', let $x \notin N'$, and let $u = e_{x,i}$ and $w = e_{0,j}$ be site vectors. The regions $\{x\}$ and $\{0\}$ are separated by N', and for every real $t \neq 0$ factorization together with the product formula gives $e^{-\Delta_T(tu,w)/2}\, T(W(tu))\, T(W(w)) = T(W(tu))\, T(W(w))$, where the product on the right is a nonzero multiple of a unitary. Hence $e^{-t\Delta_T(u,w)/2} = 1$ for every $t \neq 0$; choosing t so small that $|t\Delta_T(u, w)/2| < 2\pi$ forces $\Delta_T(u, w) = 0$. Its real part is $\nu(u, w) = (\nu_x)_{ij}$, and its imaginary part is $\sigma(u, w) - \sigma(Lu, Lw) = -((L_*\sigma)_x)_{ij}$, because $\sigma(u, w) = 0$ for $x \neq 0$.

□

For $N' = \{0\}$ the condition on the rule is that $(L_*\sigma)_x = \sum_m \ell_m^{\mathsf{T}} \Omega\, \ell_{m+x}$ vanish for $x \neq 0$. These are the coefficient equations (1.1) of a symplectic rule for Ω at every index other than 0, without the equation at the index 0; the rule need not be symplectic, and $\hat{L}^{\star\mathsf{T}} \Omega \hat{L}$ need only be a constant matrix. The two notions of locality are independent of each other in the following sense.

Proposition 7.8 (Independence of the two notions). Lean ✓ *Let $n \geq 1$.*

(a) There is a Gaussian channel that factorizes with respect to $\{0\}$ and does not have finite propagation with neighbourhood $\{0\}$.

(b) For every finite set $N' \subseteq \mathbb{Z}$ with $0 \in N' = -N'$ there are cell data (ν_x) of finite range such that the rule 0 and the noise ν satisfy (6.2), and no lattice channel with these data factorizes with respect to N'.

Proof.

(a) The shift, the rule with coefficient $\ell_1 = 1$ and all others zero, is symplectic, and its automorphism is a lattice channel. Since $\hat{L}^{\star\mathsf{T}} \Omega \hat{L} = z^{-1} \Omega z = \Omega$, the pull-back cells of σ vanish off the origin, and Theorem 7.7(II) gives factorization with respect to $\{0\}$; since $N_L = \{1\}$, part (I) excludes finite propagation with neighbourhood $\{0\}$.

(b) Let $x_0 = 1 + \max_{m \in N'} |m|$, and let $\nu_0 = 2 \cdot 1$, $\nu_{\pm x_0} = \frac{1}{2} \cdot 1$ and $\nu_x = 0$ otherwise. For the rule 0, condition (6.2) is the state condition $\nu + \mathrm{i}\sigma \geq 0$ (Proposition 6.7(c)), which holds by Theorem 3.25(a), since $\hat{\nu}(k) + \mathrm{i}\Omega = (1 + \cos kx_0) \cdot 1 + (1 + \mathrm{i}\Omega)$ is a sum of positive semidefinite matrices. Since $x_0 \notin N'$ and $\nu_{x_0} \neq 0$, Theorem 7.7(II) excludes factorization with respect to N' for every lattice channel with these data.

□

Every lattice channel with the rule 0 has finite propagation with neighbourhood $\{0\}$ by Theorem 7.7(I), since $N_L = \emptyset$ for $L = 0$. Part (b) therefore shows that finite propagation does not imply factorization, and part (a) that factorization does not imply finite propagation with a prescribed neighbourhood.

7.3 Concatenation

Applying two lattice channels in succession gives a lattice channel whose rule is the product of the two rules and whose noise collects the first noise and the pull-back of the second. The ranges of the resulting families are controlled by those of the factors.

Proposition 7.9 (Concatenation of lattice channels)**.** Lean ✓ *If T_1 and T_2 are lattice channels with rules L_1, L_2 and noise cells $(\nu_{1,x})$, $(\nu_{2,x})$, then $T_2 \circ T_1$ is a lattice channel with rule $L_2 L_1$ and noise cells $\nu_1 + L_{1*}\nu_2$.*

Proof. By Proposition 6.9, $T_2 \circ T_1$ is a Gaussian channel with linear map $L_2 L_1$, noise $\nu_1 + \nu_2(L_1 \cdot, L_1 \cdot)$ and displacement $\mu_1 + \mu_2 \circ L_1$. The noise is the form of the cells $\nu_1 + L_{1*}\nu_2$ by Lemma 7.6(a), which are cell data, and the displacement is translation invariant because L_1 commutes with translations. □

Proposition 7.10 (Ranges under concatenation)**.** Lean ✓ *Let L, L_1, L_2 be rules.*

(a) $N_{L_2 L_1} \subseteq N_{L_1} + N_{L_2}$.

(b) If (ν_x) vanishes outside a finite set M, then $(L_*\nu)_j = 0$ for $j \notin M + (N_L - N_L)$.

(c) $(L_*\sigma)_j = 0$ *for* $j \notin N_L - N_L$.

(d) Let N_1, N_2 be finite, N_2 nonempty, $N_{L_i} \subseteq N_i$, and let $(\nu_{i,x})$ vanish outside $N_i - N_i$ for $i = 1, 2$. Then $N_{L_2 L_1} \subseteq N_1 + N_2$, and $\nu_1 + L_{1*}\nu_2$ vanishes outside $(N_1 + N_2) - (N_1 + N_2)$.

(e) For on-site noises $\nu_{i,x} = \delta_{x,0}\nu_{i,0}$, the noise cells $\nu_1 + L_{1*}\nu_2$ vanish off the origin if and only if $(L_{1*}\nu_2)_x = 0$ for every $x \neq 0$.

Proof.

(a) The coefficient of z^j in $\hat{L}_2 \hat{L}_1$ is a sum of products of coefficients of z^b in $\hat{L}_2$ and of z^a in $\hat{L}_1$ with $a + b = j$, and is zero unless such $a \in N_{L_1}$ and $b \in N_{L_2}$ exist.

(b) A term of $(L_*\nu)_j$ is nonzero only if $j + m - m' \in M$ for some $m, m' \in N_L$, that is, $j \in M + (N_L - N_L)$.

(c) This is (b) for the on-site cells of σ, with $M = \{0\}$.

(d) The first statement follows from (a). Choose $c \in N_2$. If $x = a - b$ with $a, b \in N_1$, then $x = (a + c) - (b + c) \in (N_1 + N_2) - (N_1 + N_2)$, so that ν_1 vanishes outside this set. By (b), the cells $L_{1*}\nu_2$ vanish outside $(N_2 - N_2) + (N_{L_1} - N_{L_1})$, and an element $(a - b) + (a' - b')$ of this set, with $a, b \in N_2$ and $a', b' \in N_{L_1} \subseteq N_1$, is $(a' + a) - (b' + b)$.

(e) The first summand vanishes off the origin.

□

The inclusion in (a) can be strict, since coefficients may cancel: the rules with symbols $(1-z)\cdot 1$ and $(1+z)\cdot 1$ have ranges $\{0,1\}$, and their product $(1-z^2)\cdot 1$ has range $\{0,2\}$, a proper subset of $\{0,1,2\}$.
Part (d) says that channels whose noise is correlated only within the differences of their neighbourhoods form a class closed under concatenation. The channels that factorize with respect to $\{0\}$, whose noise and whose pull-back of σ are both on site, do not, as the next subsection shows for the noise; part (e) says what closure would require of it.
The pull-back of σ can also leave the site under concatenation. Since a family is determined by its form, Lemma 7.6(a) applied twice gives $(L_2L_1)_*\nu = L_{1*}(L_{2*}\nu)$ for every family. If $L_{1*}\sigma$ and $L_{2*}\sigma$ are on site and $n=1$, the on-site cell of $L_{2*}\sigma$ is antisymmetric and hence a multiple $c\,\Omega$, and then $(L_2L_1)_*\sigma = c\,L_{1*}\sigma$ is again on site. For $n=2$ this fails. With coordinates (q_1,q_2,p_1,p_2), let L_1 have the coefficients $\ell_0 = P_2$ and $\ell_1 = P_1$, the projections onto the second and the first mode, so that it shifts the first mode and fixes the second, and let L_2 act at every site by the matrix $S = 1 + E_{12}$, that is, $q_1 \mapsto q_1 + q_2$. Both $L_{1*}\sigma$ and $L_{2*}\sigma$ are on site, the latter with the cell $S^{\mathsf T}\Omega S = \Omega + E_{23} - E_{32}$, but the cell of $(L_2L_1)_*\sigma$ at $j=1$ is $\ell_0^{\mathsf T}(S^{\mathsf T}\Omega S)\ell_1 = P_2(E_{23}-E_{32})P_1 = E_{23} \neq 0$.

7.4 The example system

For the example system, the cell at distance two of the pull-back of on-site noise can be computed in closed form, and it decides when two steps with on-site noise again have on-site noise.

Theorem 7.11 (Two steps with on-site noise)**.** Lean ✓ *Let $n=1$, let L be the example system of Definition 4.4, with $r = (\sin\varphi, \cos\varphi)^{\mathsf T}$ the transposed second row of ℓ_0, and let ν_0 be a real 2×2 matrix. Then*

$$(L_*\nu)_2 = (\nu_0)_{11}\,\lambda^2\, rr^{\mathsf T}$$

for the on-site cells $\nu_x = \delta_{x,0}\nu_0$. If ν_0 is symmetric and positive semidefinite, then $(L_*\nu)_x = 0$ for every $x \neq 0$ if and only if $\lambda = 0$ or $(\nu_0)_{11} = 0$.

Proof. For on-site cells, $(L_*\nu)_j = \sum_{m'-m=j} \ell_m^{\mathsf T}\nu_0\ell_{m'}$. For $j=2$ only $(m,m') = (-1,1)$ occurs, and with $\ell_{\pm 1} = \lambda e_1 r^{\mathsf T}$,

$$\ell_{-1}^{\mathsf T}\nu_0\ell_1 = \lambda^2\, r\,(e_1^{\mathsf T}\nu_0 e_1)\, r^{\mathsf T} = \lambda^2(\nu_0)_{11}\, rr^{\mathsf T}.$$

If the cells vanish off the origin, the trace of this cell, $\lambda^2(\nu_0)_{11}|r|^2 = \lambda^2(\nu_0)_{11}$, vanishes. Conversely, if $\lambda = 0$ only $m = m' = 0$ contributes. If $(\nu_0)_{11} = 0$ and ν_0 is symmetric and positive semidefinite, then $\nu_0 e_1 = 0$, since such a matrix with a zero diagonal entry has a zero row and column; hence $\nu_0\ell_{\pm 1} = 0 = \ell_{\pm 1}^{\mathsf T}\nu_0$, and every term with $j \neq 0$ contains some $\ell_{\pm 1}$. □

By the congruence $\gamma_B = \Omega^{\mathsf T}\gamma\,\Omega$ of Chapter 3, which exchanges the diagonal entries of a 2×2 form, the entry $(\nu_0)_{11}$ of the noise form is the momentum entry of the matrix that the noise adds to the covariance matrix of each site, that is, twice the variance it adds to the momentum. Pulled back by one step of the rule, this part of the noise correlates sites two apart through the two coupling factors in $\ell_{-1}^{\mathsf T}\nu_0\ell_1$, while noise with $(\nu_0)_{11} = 0$, such as $\nu_0 = \operatorname{diag}(0,b)$, which adds to the position alone, is not read by the coupling $\ell_{\pm 1} = \lambda e_1 r^{\mathsf T}$ and stays on site. For the lossy chain the consequence is the following.

Proposition 7.12 (The lossy chain)**.** Lean ✓ *Let $0 < \zeta < 1$, and let ζL and the on-site cells $(1-\zeta^2)\cdot 1$ be the data of Example 7.2.*

(a) The data satisfy (6.2)*.*

(b) $((\zeta L)_*\sigma)_j = 0$ for every $j \neq 0$.

(c) The rule ζL is not symplectic.

(d) If $\lambda \neq 0$, the noise cells of two steps, $\nu + (\zeta L)_* \nu$, are nonzero at $j = 2$.

Proof.

(a) Since L is symplectic, $\sigma(\zeta L u, \zeta L w) = \zeta^2 \sigma(u, w)$, and the matrices of (6.2) are $(1-\zeta^2)$ times those of $\gamma + \mathrm{i}\sigma$, with γ the vacuum form of the cells $\delta_{x,0} \cdot 1$; the vacuum form satisfies the state condition (Example 3.20).

(b) By Lemma 7.6(b), $((\zeta L)_* \sigma)_j$ is the coefficient of z^j in $\zeta^2 \hat{L}^{\star \mathsf{T}} \Omega \hat{L} = \zeta^2 \Omega$, since the example system is symplectic (Example 1.20).

(c) $(\zeta \hat{L})^{\star \mathsf{T}} \Omega (\zeta \hat{L}) = \zeta^2 \Omega \neq \Omega$, so that ζL is not symplectic by Theorem 1.7.

(d) The on-site first summand does not contribute at $j = 2$. The pull-back cells of the rule ζL are ζ^2 times those of L, so that by Theorem 7.11 with $\nu_0 = (1-\zeta^2) \cdot 1$ the second summand at $j = 2$ is $\zeta^2(1-\zeta^2)\lambda^2 r r^{\mathsf{T}} \neq 0$.

□

By Theorem 7.7, every lattice channel with the data of the lossy chain therefore has finite propagation with neighbourhood $\{-1, 0, 1\}$ and factorizes with respect to $\{0\}$, although its rule is not symplectic, whereas two steps of it, a lattice channel by Proposition 7.9, no longer factorize with respect to $\{0\}$ when $\lambda \neq 0$.

7.5 The Fourier form of the condition and dilations

For translation-invariant data, condition (6.2) reduces, as the state condition does in Theorem 3.25, to one Hermitian matrix at each wave number.

Theorem 7.13 (Fourier form of the condition)**.** Lean ✓ *Let (ν_x) be cell data of finite range and L a rule. Then L and ν satisfy* (6.2) *if and only if*

$$\hat{\nu}(k) + \mathrm{i}\Omega - \mathrm{i}\,\hat{L}(k)^\dagger \Omega \hat{L}(k) \geq 0 \qquad \text{for every } k \in \mathbb{R}.$$

Proof. By Lemma 7.6, the Hermitian form of (6.2) on complex combinations of vectors, $h(f) = \sum_{x,y} f(x)^\dagger a_{x-y} f(y)$ for finitely supported $f \colon \mathbb{Z} \to \mathbb{C}^{2n}$, has the cells $a_j = \nu_j + \mathrm{i}(\Omega \delta_{j,0} - (L_* \sigma)_j)$, of finite range, with $a_{-j} = a_j^\dagger$, and $\sum_j a_j e^{-\mathrm{i}kj} = \hat{\nu}(k) + \mathrm{i}\Omega - \mathrm{i}\hat{L}(k)^\dagger \Omega \hat{L}(k)$ by Lemma 7.6(b) and the computation of the symbol in Theorem 3.34. The proof of Lemma 3.24 and Theorem 3.25 uses of the cells $a_g = \gamma_g + \mathrm{i}\delta_{g,0}\Omega$ only that they are summable and satisfy $a_{-g} = a_g^\dagger$, and the necessity part only their finite range; it applies verbatim to the cells a_j and gives both directions. □

The finite range is assumed only for simplicity, since the cells a_j of summable noise are summable and Remark 3.26 extends the necessity to them. At each wave number, the matrix of the theorem is the matrix $\nu + \mathrm{i}(\Omega - L^{\mathsf{T}} \Omega L)$ whose positivity is the condition for finitely many modes (proof of Proposition 6.12(c)), with the real matrices L and ν replaced by the complex matrices $\hat{L}(k)$ and $\hat{\nu}(k)$ and the transpose by the conjugate transpose, so that the condition on the chain is the finite-mode condition of Chapter 6 read at every wave number.

For the lossy chain, $\hat{\nu}(k) = (1-\zeta^2) \cdot 1$ and $(\zeta \hat{L}(k))^\dagger \Omega (\zeta \hat{L}(k)) = \zeta^2 \Omega$, since the example system is symplectic, so that the matrix of the theorem is $(1-\zeta^2)(1+\mathrm{i}\Omega)$ at every wave number, which is positive semidefinite, in agreement with Proposition 7.12(a).

It remains to consider dilations on the chain. Let a translation-invariant ancilla with n' modes per site be coupled to the chain by a rule S on $n + n'$ modes per site that preserves the form of $\Omega_n \oplus \Omega_{n'}$. As in Proposition 6.13, $S(u, 0) = (S_{11} u, S_{21} u)$, where the symbol $\hat{S}_{11}$ of the system block is a $2n \times 2n$ and the symbol $\hat{S}_{21}$ of the block from system to ancilla a $2n' \times 2n$ matrix of Laurent polynomials; the pull-back cells $S_{21*}\gamma$ of cells of the ancilla are defined as in Definition 7.5.

Theorem 7.14 (Dilations on the chain). Lean ✓ *Let $\hat{S}^{\star\mathsf{T}}(\Omega_n \oplus \Omega_{n'})\hat{S} = \Omega_n \oplus \Omega_{n'}$, and let γ_0 be a real $2n' \times 2n'$ matrix with $\gamma_0 + \mathrm{i}\Omega_{n'} \geq 0$, which forces γ_0 to be symmetric. Then the rule S_{11} and the noise with cells $\nu = S_{21*}(\delta_{\bullet,0}\gamma_0)$ satisfy (6.2); $\hat{\nu}(k) = \hat{S}_{21}(k)^\dagger \gamma_0 \hat{S}_{21}(k)$ for every k; $N_{S_{11}} \subseteq N_S$; and $\nu_j = 0$ for $j \notin N_S - N_S$.*

Proof. The block of the system in $\hat{S}^{\star\mathsf{T}}(\Omega_n \oplus \Omega_{n'})\hat{S} = \Omega_n \oplus \Omega_{n'}$ reads $\hat{S}_{11}^{\star\mathsf{T}}\Omega_n\hat{S}_{11} + \hat{S}_{21}^{\star\mathsf{T}}\Omega_{n'}\hat{S}_{21} = \Omega_n$, which by the computation of Lemma 7.6(b) says that $\sigma(S_{11}u, S_{11}w) + \sigma'(S_{21}u, S_{21}w) = \sigma(u, w)$, with σ' the form of the ancilla; this is Proposition 6.13(a) for the translation-invariant maps S_{11} and S_{21}, and Proposition 6.13(b) gives (6.2), since the on-site form of $\gamma_0 + \mathrm{i}\Omega_{n'}$ is positive semidefinite. The symbol is computed as in Theorem 3.34 for the rectangular matrix $\hat{S}_{21}$, and the ranges follow as in Proposition 7.10(b), because the blocks of $\hat{S}$ have ranges within N_S. □

For the lossy chain, let B be the beam splitter of Example 6.14 at every site, with one ancilla mode per site, and let $S = (L \oplus 1) \circ B$, which preserves $\Omega_1 \oplus \Omega_1$ since both factors do. On a system vector, $S(u, 0) = (\zeta Lu, -\sqrt{1-\zeta^2}\, u)$, so that $S_{11} = \zeta L$ and $S_{21} = -\sqrt{1-\zeta^2} \cdot 1$, and with the ancilla in the vacuum, $\gamma_0 = 1$, the theorem gives the on-site noise $(1-\zeta^2)\cdot 1$ of the lossy chain. The composite $\alpha_{L\oplus 1} \circ \alpha_B$ acts on states first by $\alpha_{L\oplus 1}$, so that this is one step of the example system followed by the beam splitter, as in Example 7.2. The other order, $S = B \circ (L \oplus 1)$, gives $S_{21} = -\sqrt{1-\zeta^2}\, L$ and the noise cells $(1-\zeta^2)L_*(\delta_{\bullet,0} \cdot 1)$, which by the computation of Theorem 7.11 are nonzero at $j = 2$ when $\lambda \neq 0$.

7.6 Notes and references

(1) *Lattice channels.* Definition 7.1 is the translation-invariant setting of [1, p. 92], whose rule and noise are $\Gamma(x)$ and $g(x)$, the rule there of finite range. The notes ask for summable noise cells, as in Chapter 3; Theorem 7.13 assumes finite range, which the data of Proposition 7.8(b) have.

(2) *Locality.* Theorem 7.7 is Lemma 4.11 of [1, p. 95], which is stated with "if"; its proof argues the converse for notion (II), and the scaling step that turns $e^{-\Delta_T/2} = 1$ into $\Delta_T = 0$ is not written there. The thesis's condition (4.36), $\sum_x \Gamma_x^+ \Gamma_{\Delta+x} = 0$ with $\Gamma^+ = -\sigma_s \Gamma^{\mathsf{T}} \sigma_s$, is $-\Omega(L_*\sigma)_\Delta = 0$. The thesis's witness for "(II) does not imply (I)" in the proof of Lemma 4.10(v) [1, pp. 94–95], the completely depolarizing channel, maps every observable to a multiple of the unit and so has finite propagation with neighbourhood $\{0\}$. Since every automaton has finite propagation for some finite neighbourhood, what can be meant is that factorization gives no bound on the neighbourhood, which Proposition 7.8(a) shows for the neighbourhood $\{0\}$ and the same argument with the shift by $1 + \max_{m\in N} |m|$ for any nonempty finite N. Part (b) makes explicit the thesis's witness for the converse, noise of range beyond the given set.

(3) *Concatenation.* Proposition 7.9 and Proposition 7.10 are the discussion of [1, p. 96], where the ranges of the composite are written as $N_1 + N_2$ and $M_1 \cup (M_2 + N_1 - N_1)$; only the inclusions hold, by the example after the proof for the first and by Theorem 7.11 for the second, where the noise $\mathrm{diag}(0, b)$ has an on-site pull-back although $M_2 + N_L - N_L = \{-2, \ldots, 2\}$. The nonempty N_2 in (d) is supplied in the thesis by its convention $0 \in N$. Part (e) is (4.39).

(4) *The example system.* The thesis states that the noise of two steps does not fulfil (4.39) whenever the coupling is nonzero [1, p. 97]; by Theorem 7.11 this holds for isotropic noise $c \cdot 1$, $c > 0$, and fails for noise with $(\nu_0)_{11} = 0$, which adds variance to the position quadrature alone, such as $\nu_0 = \mathrm{diag}(0, b)$, $b > 0$, with the symplectic rule L, for which these data satisfy (6.2) by Proposition 6.7(b). The remark after Lemma 4.11 of [1, p. 95] asserts that for one mode per site a rule of both types need not be symplectic; the lossy chain is an explicit instance, with noise that satisfies (6.2).

(5) *Dilations.* Theorem 7.14 is Lemma 4.12 of [1, p. 97] at the level of covariances, with the reduction (4.40) of [1, p. 98]; that the data satisfy the condition goes beyond the thesis, which checks that the noise is real and symmetric. The thesis writes the symbol of the noise with transposes, $\hat{C}^{\mathsf{T}}(k)\hat{\gamma}'\hat{C}(k)$; the conjugate transpose of Theorem 7.14 agrees with it for reflection-symmetric blocks, as in the note on (4.17) in Chapter 3. The range statements correspond to Lemma 4.10(iii)–(iv) of [1, p. 94]. The construction of the channel on the algebra is not carried out here. In finite dimensions without translation invariance, factorization does not imply that a channel is a local dilation [6] (pinpoint to be checked).

(6) *Not treated here.* Section 4.3 of the thesis also defines two further notions of locality, (III) by local Kraus operators and (IV) by local dilations, relates the four notions in Lemma 4.10 [1, pp. 93–94], and leaves open whether every irreversible Gaussian automaton is of type (IV) [1, p. 98] (pinpoint to be checked).

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Lean declarations

The linked Lean ✓ tag marks a result whose Lean proof has been checked and whose correspondence with the statement here has been reviewed. Definitions are listed for navigation; only marked statements were reviewed. Files marked (in BosonicLean) belong to the library BosonicLean, on which this one depends, and the declarations listed under them are defined there.

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